



Module Manual

Master of Science (M.Sc.)

Mechatronics

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Program description

Content

The consecutive international master program "Mechatronics" extends the education in engineering, mathematics and natural science of the bachelor studies. It provides systematic, scientific and autonomous problem solving capabilities needed in industry and research.

The program covers the methods of computation, design and implementation of mechatronic systems. Students specialize individually and develop the ability to work in the interfaces of the interconnected sub-disciplines. Based on personal interest, students are able to adapt their study programs within a broad catalogue of elective courses.

Career prospects

The consecutive international Master course "Mechatronics" prepares graduates for a wide range of job profiles in mechatronics engineering.

Graduates can work in multiple disciplines directly, e.g. System Design, Intelligent Systems or Robotics.

Additionally graduates have a multifaceted knowledge of methods for interdisciplinary topics.

Graduates may decide for direct entry into companies or to take up academic careers, e.g. Ph.D. studies, in universities or other research institutions. In companies they can take up jobs as specialists or subsequently qualify for demanding management tasks in the technical area (e.g. project, group, or team leader; R&D director).

The program is designed to be universal and allows graduates to work in a variety of different industrial sectors and with different projects.

Learning target

Knowledge

- Graduates are able to present specific and interdisciplinary knowledge of mechatronics and related fields with a broad theoretical and methodological foundation.
- They can explain and discuss advanced and research-related methods of mechatronics.
- Graduates are familiar with development and simulation methods for software, control engineering, mechatronic, mechanical and robotic systems and can compare different methods with each other.
- Graduates will be able to present the state of the art and research and provide an overview of applications in industry and research.
- Graduates will be able to adapt the methods they have learned to specific applications.

Skills

- Graduates are proficient in the application of highly sophisticated methods and procedures in mechatronics and related fields (e.g. development and simulation methods for software, control engineering, mechatronic, mechanical and robotic systems).
- Graduates are able to break down both scientific and more complex practical problems, evaluate and analyze the results and select and apply suitable solution strategies on this basis. Depending on the elective subjects, the focus may be on development methodology (e.g. of embedded or haptic systems), machine learning and artificial intelligence (e.g. image processing, autonomous agents), or robotics (e.g. applied humanoid robotics, industrial, construction, underwater or medical robotics).
- They can examine and assess current and future technologies, methods and scientific developments and incorporate technical, economic and ecological constraints into problem solutions.
- Graduates are able to work independently on research questions using suitable existing methods, develop new methods, document their chosen solutions and present them to an expert audience.

Social competence

- Graduates are able to present the procedure and results of their work in English in a written and oral form that is understandable to the target audience.
- They can communicate with experts and laypersons about advanced content and problems in mechatronics. They can respond appropriately to questions, additions and comments.
- They are able to deal with problems arising in mechatronics, its sub-disciplines and related disciplines. In particular, they are able to put themselves in the mindset of other disciplines and communicate with representatives of these fields.
- Graduates are able to work in groups. They can define, distribute and integrate subtasks. They can make time arrangements and interact socially. They have the ability and willingness to assume leadership responsibility.

Independence

- Graduates are able to expand their technical knowledge, obtain the necessary specialist information and place it in the context of their knowledge.
- They are able to realistically assess their existing skills and weaknesses as well as the consequences of their actions and work through deficits independently.
- They can also systematically reflect on the non-technical effects of engineering activities and evaluate their actions in a socio-economic context.
- They can learn complex topics in a self-organized and self-motivated manner, develop research areas and find new problems (lifelong learning).

Program structure

The course is designed modularly and is based on the university-wide standardized course structure with uniform module sizes (multiples of six credit points (CP)).

The program combines the disciplines of mechanical and electrical engineering and supports concentration in interdisciplinary fields of system design and system implementation.

All modules in the first semester are mandatory. This helps especially students from abroad to familiarize themselves with the university and culture.

Afterwards the students can broadly personalize their studies due to the high number and variety of elective courses.

In the common core skills, students take the following modules:

- Finite element analysis and Vibration theory (12 CP)
- Theory and design of control systems and Design and implementation of software systems (12 CP)
- Robotics (6 CP)
- Complementary courses business and management (catalogue) (6 CP)
- Nontechnical elective complementary courses (catalogue) (6 CP).

Students specialize by selecting modules amounting to 36 CP, which can be chosen from a module catalog containing modules with a size of six credits.

Instead, open modules can be attend to the maximum extent of twelve credit points, in which smaller specialized courses can be combined, individually.

Students write a master thesis and one additional research project.

- Research project (12 CP)
- Master thesis (30 CP)

Core Qualification

Module M0523: Business & Management

Module Responsible	Prof. Matthias Meyer
Admission Requirements	Successful completion of the modul "Foundations of Management"
Recommended Previous Knowledge	None
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence <i>Knowledge</i> • Students are able to find their way around selected special areas of management within the scope of business management. • Students are able to explain basic theories, categories, and models in selected special areas of business management. • Students are able to interrelate technical and management knowledge. <i>Skills</i> • Students are able to apply basic methods in selected areas of business management. • Students are able to explain and give reasons for decision proposals on practical issues in areas of business management. Personal Competence <i>Social Competence</i> • Students are able to communicate in small interdisciplinary groups and to jointly develop solutions for complex problems <i>Autonomy</i> • Students are capable of acquiring necessary knowledge independently by means of research and preparation of material.	
Workload in Hours	Depends on choice of courses
Credit points	6

Courses

Information regarding lectures and courses can be found in the corresponding module handbook published separately.

Module M0524: Non-technical Courses for Master	
Module Responsible	Dagmar Richter
Admission Requirements	None
Recommended Previous Knowledge	None
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence <i>Knowledge</i>	The Nontechnical Academic Programms (NTA) imparts skills that, in view of the TUHH's training profile, professional engineering studies require but are not able to cover fully. Self-reliance, self-management, collaboration and professional and personnel management competences. The department implements these training objectives in its teaching architecture, in its teaching and learning arrangements, in teaching areas and by means of teaching offerings in which students can qualify by opting for specific competences and a competence level at the Bachelor's or Master's level. The teaching offerings are pooled in two different catalogues for nontechnical complementary courses. The Learning Architecture consists of a cross-disciplinarily study offering. The centrally designed teaching offering ensures that courses in the nontechnical academic programms follow the specific profiling of TUHH degree courses. The learning architecture demands and trains independent educational planning as regards the individual development of competences. It also provides orientation knowledge in the form of "profiles". The subjects that can be studied in parallel throughout the student's entire study program - if need be, it can be studied in one to two semesters. In view of the adaptation problems that individuals commonly face in their first semesters after making the transition from school to university and in order to encourage individually planned semesters abroad, there is no obligation to study these subjects in one or two specific semesters during the course of studies. Teaching and Learning Arrangements provide for students, separated into B.Sc. and M.Sc., to learn with and from each other across semesters. The challenge of dealing with interdisciplinarity and a variety of stages of learning in courses are part of the learning architecture and are deliberately encouraged in specific courses. Fields of Teaching are based on research findings from the academic disciplines cultural studies, social studies, arts, historical studies, communication studies, migration studies and sustainability research, and from engineering didactics. In addition, from the winter semester 2014/15 students on all Bachelor's courses will have the opportunity to learn about business management and start-ups in a goal-oriented way. The fields of teaching are augmented by soft skills offers and a foreign language offer. Here, the focus is on encouraging goal-oriented communication skills, e.g. the skills required by outgoing engineers in international and intercultural situations. The Competence Level of the courses offered in this area is different as regards the basic training objective in the Bachelor's and Master's fields. These differences are reflected in the practical examples used, in content topics that refer to different professional application contexts, and in the higher scientific and theoretical level of abstraction in the B.Sc. This is also reflected in the different quality of soft skills, which relate to the different team positions and different group leadership functions of Bachelor's and Master's graduates in their future working life. Specialized Competence (Knowledge) Students can • explain specialized areas in context of the relevant non-technical disciplines, • outline basic theories, categories, terminology, models, concepts or artistic techniques in the disciplines represented in the learning area, • different specialist disciplines relate to their own discipline and differentiate it as well as make connections, • sketch the basic outlines of how scientific disciplines, paradigms, models, instruments, methods and forms of representation in the specialized sciences are subject to individual and socio-cultural interpretation and historicity, • Can communicate in a foreign language in a manner appropriate to the subject.
<i>Skills</i>	Professional Competence (Skills) In selected sub-areas students can • apply basic and specific methods of the said scientific disciplines, • question a specific technical phenomena, models, theories from the viewpoint of another, aforementioned specialist discipline, • to handle simple and advanced questions in aforementioned scientific disciplines in a successful manner, • justify their decisions on forms of organization and application in practical questions in contexts that go beyond the technical relationship to the subject.
Personal Competence <i>Social Competence</i>	Personal Competences (Social Skills)

<i>Autonomy</i>	Students will be able • to learn to collaborate in different manner, • to present and analyze problems in the abovementioned fields in a partner or group situation in a manner appropriate to the addressees, • to express themselves competently, in a culturally appropriate and gender-sensitive manner in the language of the country (as far as this study-focus would be chosen), • to explain nontechnical items to auditorium with technical background knowledge. Personal Competences (Self-reliance) Students are able in selected areas • to reflect on their own profession and professionalism in the context of real-life fields of application • to organize themselves and their own learning processes • to reflect and decide questions in front of a broad education background • to communicate a nontechnical item in a competent way in written form or verbally • to organize themselves as an entrepreneurial subject country (as far as this study-focus would be chosen)
Workload in Hours	Depends on choice of courses
Credit points	6

Courses
Information regarding lectures and courses can be found in the corresponding module handbook published separately.

Module M0563: Robotics				
Courses				
Title	Typ		Hrs/wk	CP
Robotics: Modelling and Control (L0168)	Integrated Lecture		4	4
Robotics: Modelling and Control (L1305)	Project-/problem-based Learning		2	2
Module Responsible	Dr. Martin Gomse			
Admission Requirements	None			
Recommended Previous Knowledge	Fundamentals of electrical engineering Broad knowledge of mechanics Fundamentals of control theory			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	Students are able to describe fundamental properties of robots and solution approaches for multiple problems in robotics. Students are able to derive and solve equations of motion for various manipulators. Students can generate trajectories in various coordinate systems. Students can design linear and partially nonlinear controllers for robotic manipulators.			
<i>Knowledge</i>				
<i>Skills</i>				
Personal Competence				
<i>Social Competence</i>	Students are able to work goal-oriented in small mixed groups.			
<i>Autonomy</i>	Students are able to recognize and improve knowledge deficits independently. With instructor assistance, students are able to evaluate their own knowledge level and define a further course of study.			
Workload in Hours	Independent Study Time 96, Study Time in Lecture 84			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	Yes	None	Subject theoretical and practical work	Teilnahme an PBL-Einheiten sowie Erreichen des Gesamtziels und der jeweiligen Session-Ziele
Examination	Written exam			
Examination duration and scale	120 min			
Assignment for the Following Curricula	Aircraft Systems Engineering: Core Qualification: Elective Compulsory International Management and Engineering: Specialisation II. Product Development and Production: Elective Compulsory International Management and Engineering: Specialisation II. Mechatronics: Elective Compulsory Aeronautics: Core Qualification: Elective Compulsory Mechanical Engineering - Product Development and Production: Specialisation Product Development: Elective Compulsory Mechanical Engineering - Product Development and Production: Specialisation Production: Elective Compulsory Mechanical Engineering - Product Development and Production: Specialisation Materials: Elective Compulsory Mechanical Engineering and Management: Core Qualification: Compulsory Mechatronics: Core Qualification: Compulsory Product Development, Materials and Production: Specialisation Product Development: Elective Compulsory Product Development, Materials and Production: Specialisation Production: Elective Compulsory Product Development, Materials and Production: Specialisation Materials: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Product Development and Production: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory			

Course L0168: Robotics: Modelling and Control	
Typ	Integrated Lecture
Hrs/wk	4
CP	4
Workload in Hours	Independent Study Time 64, Study Time in Lecture 56
Lecturer	Dr. Martin Gomse
Language	EN
Cycle	WiSe
Content	Fundamental kinematics of rigid body systems Newton-Euler equations for manipulators Trajectory generation Linear and nonlinear control of robots
Literature	Craig, John J.: Introduction to Robotics Mechanics and Control, Third Edition, Prentice Hall. ISBN 0201-54361-3 Spong, Mark W.; Hutchinson, Seth; Vidyasagar, M. : Robot Modeling and Control. WILEY. ISBN 0-471-64990-2

Course L1305: Robotics: Modelling and Control	
Typ	Project-/problem-based Learning
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Dr. Martin Gomse
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M0808: Finite Elements Methods				
Courses				
Title		Typ	Hrs/wk	CP
Finite Element Methods (L0291)		Lecture	2	3
Finite Element Methods (L0804)		Recitation Section (large)	2	3
Module Responsible	Prof. Benedikt Kriegesmann			
Admission Requirements	None			
Recommended Previous Knowledge	Mechanics I (Statics, Mechanics of Materials) and Mechanics II (Hydrostatics, Kinematics, Dynamics) Mathematics I, II, III (in particular differential equations)			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence				
<i>Knowledge</i>	The students possess an in-depth knowledge regarding the derivation of the finite element method and are able to give an overview of the theoretical and methodical basis of the method.			
<i>Skills</i>	The students are capable to handle engineering problems by formulating suitable finite elements, assembling the corresponding system matrices, and solving the resulting system of equations.			
Personal Competence				
<i>Social Competence</i>	Students can work in small groups on specific problems to arrive at joint solutions.			
<i>Autonomy</i>	The students are able to independently solve challenging computational problems and develop own finite element routines. Problems can be identified and the results are critically scrutinized.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	None			
Examination	Written exam			
Examination duration and scale	90 min			
Assignment for the Following Curricula	Computational Engineering: Core Qualification: Elective Compulsory Energy Systems: Core Qualification: Elective Compulsory Aircraft Systems Engineering: Core Qualification: Elective Compulsory International Management and Engineering: Specialisation II. Mechatronics: Elective Compulsory International Management and Engineering: Specialisation II. Product Development and Production: Elective Compulsory Aeronautics: Core Qualification: Elective Compulsory Mechanical Engineering - Product Development and Production: Core Qualification: Compulsory Mechatronics: Core Qualification: Compulsory Biomedical Engineering: Specialisation Management and Business Administration: Elective Compulsory Biomedical Engineering: Specialisation Medical Technology and Control Theory: Elective Compulsory Biomedical Engineering: Specialisation Artificial Organs and Regenerative Medicine: Elective Compulsory Biomedical Engineering: Specialisation Implants and Endoprotheses: Elective Compulsory Product Development, Materials and Production: Core Qualification: Compulsory Technomathematics: Specialisation III. Engineering Science: Elective Compulsory Theoretical Mechanical Engineering: Core Qualification: Compulsory			

Course L0291: Finite Element Methods	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Benedikt Kriegesmann
Language	EN
Cycle	WiSe
Content	- General overview on modern engineering - Displacement method - Hybrid formulation - Isoparametric elements - Numerical integration - Solving systems of equations (statics, dynamics) - Eigenvalue problems - Non-linear systems - Applications - Programming of elements (Matlab, hands-on sessions) - Applications
Literature	Bathe, K.-J. (2000): Finite-Elemente-Methoden. Springer Verlag, Berlin

Course L0804: Finite Element Methods	
Typ	Recitation Section (large)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Benedikt Kriegesmann
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M0846: Control Systems Theory and Design			
Courses			
Title	Typ	Hrs/wk	CP
Control Systems Theory and Design (L0656)	Lecture	2	4
Control Systems Theory and Design (L0657)	Recitation Section (small)	2	2
Module Responsible	Prof. Timm Faulwasser		
Admission Requirements	None		
Recommended Previous Knowledge	Introduction to Control Systems or Grundlagen der Regelungstechnik		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence <i>Knowledge</i> The students can explain the system properties controllability and observability, and their relationship to state feedback and state estimation, respectively. They can explain the significance of a minimal realization. They can explain observer-based state feedback and how it can be used to achieve tracking and disturbance rejection. They can extend all of the above to multi-input multi-output systems. They can explain the z-transform and its relationship with the Laplace Transform; they can explain state space models and transfer function models of discrete-time systems. They can explain the experimental identification of ARX models of dynamic systems. They know concepts for two-degree of freedom designs and flatness-based feedforward control. They know concepts for Lyapunov-based stability analysis, passivity and dissipativity analysis of LTI systems and in very basic form also for nonlinear systems. They can compute Lyapunov functions and storage functions for LTI systems. They know how the stability of equilibria of nonlinear systems can be analyzed using the Jacobi linearization. They know the basic of optimal control of linear systems, they know the Bellman equation also for nonlinear systems. They know the concepts of optimal trajectory tracking designs for LTI systems. <i>Skills</i> • Students can transform transfer function models into state space models and vice versa • They can assess controllability and observability and construct minimal realisations • They can design LQR controllers for multivariable plants, they can perform a 2 DoF design for LTI systems • They can carry out a controller design both in continuous-time and discrete-time domain, and decide which is appropriate for a given sampling rate • They can identify state space models of dynamic systems from experimental data • They can carry out all these tasks using standard software tools (Matlab Control Toolbox, System Identification Toolbox, Simulink) • They can analyze the stability of equilibria of nonlinear systems, they can compute Lyapunov and storage functions for LTI systems. Personal Competence <i>Social Competence</i> Students can work in small groups on specific problems to arrive at joint solutions. <i>Autonomy</i> Students can obtain information from provided sources (lecture notes, software documentation, experiment guides) and use it when solving given problems. They can assess their knowledge in weekly on-line tests and thereby control their learning progress.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56		
Credit points	6		
Course achievement	None		
Examination	Written exam		
Examination duration and scale	120 min		
Assignment for the Following Curricula	Electrical Engineering and Information Technology: Core Qualification: Compulsory Electrical Engineering: Core Qualification: Compulsory Energy Systems: Core Qualification: Elective Compulsory Energy Systems: Core Qualification: Elective Compulsory Energy Systems: Specialisation Energy Systems: Elective Compulsory Aircraft Systems Engineering: Core Qualification: Elective Compulsory Aeronautics: Core Qualification: Elective Compulsory Mechanical Engineering - Product Development and Production: Core Qualification: Elective Compulsory Mechanical Engineering and Management: Specialisation Mechatronics: Elective Compulsory Mechatronics: Core Qualification: Compulsory Biomedical Engineering: Specialisation Artificial Organs and Regenerative Medicine: Elective Compulsory Biomedical Engineering: Specialisation Implants and Endoprostheses: Elective Compulsory Biomedical Engineering: Specialisation Medical Technology and Control Theory: Compulsory Biomedical Engineering: Specialisation Management and Business Administration: Elective Compulsory Product Development, Materials and Production: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Core Qualification: Compulsory		

Course L0656: Control Systems Theory and Design	
Typ	Lecture
Hrs/wk	2
CP	4
Workload in Hours	Independent Study Time 92, Study Time in Lecture 28
Lecturer	Prof. Timm Faulwasser
Language	EN
Cycle	WiSe
Content	State space methods (single-input single-output) • State space models and transfer functions, state feedback • Coordinate basis, similarity transformations • Solutions of state equations, matrix exponentials, Caley-Hamilton Theorem • Controllability and pole placement • State estimation, observability, Kalman decomposition • Observer-based state feedback control, reference tracking • Transmission zeros • Optimal pole placement, symmetric root locus Multi-input multi-output systems • Transfer function matrices, state space models of multivariable systems, Gilbert realization • Poles and zeros of multivariable systems, minimal realization • Closed-loop stability • Pole placement for multivariable systems, LQR design, Kalman filter Digital Control • Discrete-time systems: difference equations and z-transform • Discrete-time state space models, sampled data systems, poles and zeros • Frequency response of sampled data systems, choice of sampling rate System identification and model order reduction • Least squares estimation, ARX models, persistent excitation • Identification of state space models, subspace identification • Balanced realization and model order reduction Case study • Modelling and multivariable control of a process evaporator using Matlab and Simulink Software tools • Matlab/Simulink
Literature	• Werner, H., Lecture Notes „Control Systems Theory and Design“ • T. Kailath "Linear Systems", Prentice Hall, 1980 • K.J. Astrom, B. Wittenmark "Computer Controlled Systems" Prentice Hall, 1997 • L. Ljung "System Identification - Theory for the User", Prentice Hall, 1999

Course L0657: Control Systems Theory and Design	
Typ	Recitation Section (small)
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Timm Faulwasser
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M1222: Design and Implementation of Software Systems				
Courses				
Title			Typ	Hrs/wk
Design and Implementation of Software Systems (L1657)			Lecture	2
Design and Implementation of Software Systems (L1658)			Project-/problem-based Learning	3
Module Responsible	Prof. Bernd-Christian Renner			
Admission Requirements	None			
Recommended Previous Knowledge	- Imperativ programming languages (C, Pascal, Fortran or similar) - Simple data types (integer, double, char, boolean), arrays, if-then-else, for, while, procedure and function calls			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence <i>Knowledge</i>	- Students know the importance of software engineering, especially in the context of embedded systems, and the relevant requirements and metrics. - They know different paradigms of programming language processing and various relevant terms - They know the language constructs and concepts of procedural and object-oriented programming, including the concrete realizations in Java (i.e., inheritance, polymorphism, late-binding, etc.) - They have extensive knowledge of software modeling using UML. - They understand the difference between an ideal world and the limitations of a real, mechatronic system and can implement algorithms that work despite these limitations (e.g., motor slip, sensor noise, environmental influence).			
<i>Skills</i>	- Students can design and implement software for mechatronic systems. They can design and implement a challenging software system and its components for a mechatronic system based on a textual specification. - They can practically apply concepts of object-oriented software development and describe the software using UML. - They can also argue in a system-oriented manner with a view to the combination of hardware and software, including interface problems and the non-ideal behavior of real mechatronic systems. - They are proficient in using an integrated development environment (IDE) and a typical development tool for source code management and versioning (Git).			
Personal Competence <i>Social Competence</i>	Students are able to work goal-oriented in small mixed groups, learning and broadening teamwork abilities and define and distribute tasks within the team.			
<i>Autonomy</i>	Students are able to solve individually exercises related to this lecture with instructional direction. Students are able to plan, execute and summarize a mechatronic experiment.			
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	No	10 %	Attestation	
Examination	Written exam			
Examination duration and scale	90 min			
Assignment for the Following Curricula	Mechatronics: Core Qualification: Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory			

Course L1657: Design and Implementation of Software Systems	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Bernd-Christian Renner
Language	EN
Cycle	WiSe
Content	The lecture covers the disciplined and engineering approach to creating software for mechatronic systems. Methods and tools for software modeling are discussed. The Java language is used for the practical examples. Content: • Introduction to software techniques • Procedural Programming • Object oriented software design • Java • Software Engineering (Software Development and Life Cycles , Software Modelling in UML) In the practical module part, students learn the specification and implementation of software modules for a mechatronic, embedded system (LEGO Mindstorms). • They implement individual software modules based on a specification. • They test their implementation using simulation and real hardware. • They model the software using UML and specify larger software to solve a larger task. • They use the created modules and specifications to implement the entire software system for a mechatronic system. • They develop countermeasures for limitations of the system and environmental influence (e.g., lighting conditions, motor slip and speed). • They document the source code and present the solution.
Literature	• "The Pragmatic Programmer: From Journeyman to Master" Andrew Hunt, David Thomas, Ward Cunningham • "Core LEGO MINDSTORMS Programming: Unleash the Power of the Java Platform" Brian Bagnall Prentice Hall PTR, 1st edition (March, 2002) ISBN 0130093645 • "Objects First with Java: A Practical Introduction using BlueJ" David J. Barnes & Michael Kölling Prentice Hall/ Pearson Education; 2003, ISBN 0-13-044929-6

Course L1658: Design and Implementation of Software Systems	
Typ	Project-/problem-based Learning
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Lecturer	Prof. Bernd-Christian Renner
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M0751: Vibration Theory			
Courses			
Title Vibration Theory (L0701)	Typ Integrated Lecture	Hrs/wk 4	CP 6
Module Responsible	Prof. Norbert Hoffmann		
Admission Requirements	None		
Recommended Previous Knowledge	• Calculus • Linear Algebra • Engineering Mechanics		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence <i>Knowledge</i> • Students are able to denote terms and concepts of Vibration Theory and develop them further. • Students know methods of modeling and simulation for free, driven, self-excited and parameter driven vibrations. • Students know about concepts of linear and nonlinear vibration problems. • Students know basic tasks of vibration problems of discrete and continuous systems. <i>Skills</i> • Students are able to denote methods of Vibration Theory and develop them further. • Students are able to apply and expand methods of modeling and simulation for free, forced, self-excited and parameter driven vibrations. • Students are able to solve linear and nonlinear vibration problems. Personal Competence <i>Social Competence</i> • Students can analyze vibration problems, work on them, and reach working results also in teams or groups. • Students are able to document the results of vibration studies also in groups. <i>Autonomy</i> • Students are able to individually analyze and solve vibration problems. • Students are able to approach individually research tasks in Vibration Theory.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56		
Credit points	6		
Course achievement	None		
Examination	Written exam		
Examination duration and scale	2 Hours		
Assignment for the Following Curricula	Energy Systems: Core Qualification: Elective Compulsory International Management and Engineering: Specialisation II. Mechatronics: Elective Compulsory Mechanical Engineering - Product Development and Production: Core Qualification: Compulsory Mechanical Engineering and Management: Specialisation Mechatronics: Elective Compulsory Mechatronics: Core Qualification: Compulsory Biomedical Engineering: Specialisation Artificial Organs and Regenerative Medicine: Elective Compulsory Biomedical Engineering: Specialisation Implants and Endoprostheses: Elective Compulsory Biomedical Engineering: Specialisation Medical Technology and Control Theory: Elective Compulsory Biomedical Engineering: Specialisation Management and Business Administration: Elective Compulsory Product Development, Materials and Production: Core Qualification: Compulsory Naval Architecture and Ocean Engineering: Core Qualification: Elective Compulsory Naval Architecture and Ocean Engineering: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Core Qualification: Elective Compulsory		

Course L0701: Vibration Theory	
Typ	Integrated Lecture
Hrs/wk	4
CP	6
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56
Lecturer	Prof. Norbert Hoffmann
Language	DE/EN
Cycle	WiSe
Content	Linear and Nonlinear Single and Multiple Degree of Freedom Vibrations • Free vibration • Self-excited vibration • Parameter driven vibration • Forced vibration • Multi degree of freedom vibration • Continuum vibration • Irregular vibration
Literature	German - K. Magnus, K. Popp, W. Sextro: Schwingungen. Physikalische Grundlagen und mathematische Behandlung von Schwingungen. English - K. Magnus: Vibrations.

Module M1223: Selected Topics of Mechatronics (Alternative A: 12 LP)			
Courses			
Title	Typ	Hrs/wk	CP
Applied Automation (L1592)	Project-/problem-based Learning	3	3
Advanced Training Course SE-ZERT (L2739)	Project-/problem-based Learning	2	3
Development Management for Mechatronics (L1512)	Lecture	2	3
Fatigue & Damage Tolerance (L0310)	Lecture	2	3
Industry 4.0 for Engineers (L2012)	Lecture	2	3
Microcontroller Circuits: Implementation in Hardware and Software (L0087)	Seminar	2	2
Microsystems Technology (L0724)	Lecture	2	4
Model-Based Systems Engineering (MBSE) with SysML/UML (L1551)	Project-/problem-based Learning	3	3
Sustainable Industrial Production (L2863)	Lecture	2	3
Process Measurement Engineering (L1077)	Lecture	2	3
Process Measurement Engineering (L1083)	Recitation Section (large)	1	1
Feedback Control in Medical Technology (L0664)	Lecture	2	3
Module Responsible	Dr. Martin Gomse		
Admission Requirements	None		
Recommended Previous Knowledge	None		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence <i>Knowledge</i> - Students are able to express their extended knowledge and discuss the connection of different special fields or application areas of mechatronics - Students are qualified to connect different special fields with each other <i>Skills</i> - Students can apply specialized solution strategies and new scientific methods in selected areas - Students are able to transfer learned skills to new and unknown problems and can develop own solution approaches Personal Competence <i>Social Competence</i> <i>Autonomy</i>	None Students are able to develop their knowledge and skills by autonomous election of courses.		
Workload in Hours	Depends on choice of courses		
Credit points	12		
Assignment for the Following Curricula	Mechatronics: Core Qualification: Elective Compulsory		

Course L1592: Applied Automation	
Typ	Project-/problem-based Learning
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Examination Form	Mündliche Prüfung
Examination duration and scale	30 Minuten
Lecturer	Prof. Thorsten Schüppstuhl
Language	DE
Cycle	WiSe
Content	-Project Based Learning -Robot Operating System -Robot structure and description -Motion description -Calibration -Accuracy
Literature	John J. Craig Introduction to Robotics - Mechanics and Control ISBN: 0131236296 Pearson Education, Inc., 2005 Stefan Hesse Grundlagen der Handhabungstechnik ISBN: 3446418725 München Hanser, 2010 K. Thulasiraman and M. N. S. Swamy Graphs: Theory and Algorithms ISBN: 9781118033104 %CITAVIPICKER£9781118033104£Titel anhand dieser ISBN in Citavi-Projekt übernehmen£% John Wiley & Sons, Inc., 1992

Course L2739: Advanced Training Course SE-ZERT	
Typ	Project-/problem-based Learning
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Examination Form	Klausur
Examination duration and scale	120 min
Lecturer	Prof. Ralf God
Language	DE
Cycle	SoSe
Content	
Literature	INCOSE Systems Engineering Handbuch - Ein Leitfaden für Systemlebenszyklus-Prozesse und -Aktivitäten, GfSE (Hrsg. der deutschen Übersetzung), ISBN 978-3-9818805-0-2. ISO/IEC 15288 System- und Software-Engineering - System-Lebenszyklus-Prozesse (Systems and Software Engineering - System Life Cycle Processes).

Course L1512: Development Management for Mechatronics	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Examination Form	Mündliche Prüfung
Examination duration and scale	30 Minuten
Lecturer	Dr. Johannes Nicolas Gebhardt
Language	DE
Cycle	SoSe
Content	- Processes and methods of product development - from idea to market launch - identification of market and technology potentials - development of a common product architecture - Synchronized product development across all engineering disciplines - product validation incl. customer view - Steering and optimization of product development - Design of processes for product development - IT systems for product development - Establishment of management standards - Typical types of organization
Literature	- Bender: Embedded Systems - qualitätsorientierte Entwicklung - Ehrlenspiel: Integrierte Produktentwicklung: Denkabläufe, Methodeneinsatz, Zusammenarbeit - Gausemeier/Ebbesmeyer/Kallmeyer: Produktinnovation - Strategische Planung und Entwicklung der Produkte von morgen - Haberfellner/de Weck/Fricke/Vössner: Systems Engineering: Grundlagen und Anwendung - Lindemann: Methodische Entwicklung technischer Produkte: Methoden flexibel und situationsgerecht anwenden - Pahl/Beitz: Konstruktionslehre: Grundlagen erfolgreicher Produktentwicklung. Methoden und Anwendung - VDI-Richtlinie 2206: Entwicklungsmethodik für mechatronische Systeme

Course L0310: Fatigue & Damage Tolerance	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Examination Form	Mündliche Prüfung
Examination duration and scale	45 min
Lecturer	Dr. Martin Flamm
Language	EN
Cycle	WiSe
Content	Design principles, fatigue strength, crack initiation and crack growth, damage calculation, counting methods, methods to improve fatigue strength, environmental influences
Literature	Jaap Schijve, Fatigue of Structures and Materials. Kluwer Academic Publisher, Dordrecht, 2001 E. Haibach. Betriebsfestigkeit Verfahren und Daten zur Bauteilberechnung. VDI-Verlag, Düsseldorf, 1989

Course L2012: Industry 4.0 for Engineers	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Examination Form	Klausur
Examination duration and scale	120 min
Lecturer	Prof. Thorsten Schüppstuhl
Language	DE
Cycle	SoSe
Content	
Literature	

Course L0087: Microcontroller Circuits: Implementation in Hardware and Software	
Typ	Seminar
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Examination Form	Schriftliche Ausarbeitung
Examination duration and scale	10 min. Vortrag + anschließende Diskussion
Lecturer	Prof. Siegfried Rump
Language	DE
Cycle	WiSe/SoSe
Content	
Literature	ATmega16A 8-bit Microcontroller with 16K Bytes In-System Programmable Flash - DATASHEET, Atmel Corporation 2014 Atmel AVR 8-bit Instruction Set Instruction Set Manual, Atmel Corporation 2016

Course L0724: Microsystems Technology	
Typ	Lecture
Hrs/wk	2
CP	4
Workload in Hours	Independent Study Time 92, Study Time in Lecture 28
Examination Form	Mündliche Prüfung
Examination duration and scale	30 min
Lecturer	Prof. Hoc Khiem Trieu
Language	EN
Cycle	WiSe
Content	• Introduction (historical view, scientific and economic relevance, scaling laws) • Semiconductor Technology Basics, Lithography (wafer fabrication, photolithography, improving resolution, next-generation lithography, nano-imprinting, molecular imprinting) • Deposition Techniques (thermal oxidation, epitaxy, electroplating, PVD techniques: evaporation and sputtering; CVD techniques: APCVD, LPCVD, PECVD and LECVD; screen printing) • Etching and Bulk Micromachining (definitions, wet chemical etching, isotropic etch with HNA, electrochemical etching, anisotropic etching with KOH/TMAH: theory, corner undercutting, measures for compensation and etch-stop techniques; plasma processes, dry etching: back sputtering, plasma etching, RIE, Bosch process, cryo process, XeF2 etching) • Surface Micromachining and alternative Techniques (sacrificial etching, film stress, stiction: theory and counter measures; Origami microstructures, Epi-Poly, porous silicon, SOI, SCREAM process, LIGA, SU8, rapid prototyping) • Thermal and Radiation Sensors (temperature measurement, self-generating sensors: Seebeck effect and thermopile; modulating sensors: thermo resistor, Pt-100, spreading resistance sensor, pn junction, NTC and PTC; thermal anemometer, mass flow sensor, photometry, radiometry, IR sensor: thermopile and bolometer) • Mechanical Sensors (strain based and stress based principle, capacitive readout, piezoresistivity, pressure sensor: piezoresistive, capacitive and fabrication process; accelerometer: piezoresistive, piezoelectric and capacitive; angular rate sensor: operating principle and fabrication process) • Magnetic Sensors (galvanomagnetic sensors: spinning current Hall sensor and magneto-transistor; magnetoresistive sensors: magneto resistance, AMR and GMR, fluxgate magnetometer) • Chemical and Bio Sensors (thermal gas sensors: pellistor and thermal conductivity sensor; metal oxide semiconductor gas sensor, organic semiconductor gas sensor, Lambda probe, MOSFET gas sensor, pH-FET, SAW sensor, principle of biosensor, Clark electrode, enzyme electrode, DNA chip) • Micro Actuators, Microfluidics and TAS (drives: thermal, electrostatic, piezo electric and electromagnetic; light modulators, DMD, adaptive optics, microscanner, microvalves: passive and active, micropumps, valveless micropump, electrokinetic micropumps, micromixer, filter, inkjet printhead, microdispenser, microfluidic switching elements, microreactor, lab-on-a-chip, microanalytics) • MEMS in medical Engineering (wireless energy and data transmission, smart pill, implantable drug delivery system, stimulators: microelectrodes, cochlear and retinal implant; implantable pressure sensors, intelligent osteosynthesis, implant for spinal cord regeneration) • Design, Simulation, Test (development and design flows, bottom-up approach, top-down approach, testability, modelling: multiphysics, FEM and equivalent circuit simulation; reliability test, physics-of-failure, Arrhenius equation, bath-tub relationship) • System Integration (monolithic and hybrid integration, assembly and packaging, dicing, electrical contact: wire bonding, TAB and flip chip bonding; packages, chip-on-board, wafer-level-package, 3D integration, wafer bonding: anodic bonding and silicon fusion bonding; micro electroplating, 3D-MID)
Literature	M. Madou: Fundamentals of Microfabrication, CRC Press, 2002 N. Schwesinger: Lehrbuch Mikrosystemtechnik, Oldenbourg Verlag, 2009 T. M. Adams, R. A. Layton: Introductory MEMS, Springer, 2010 G. Gerlach; W. Dötzel: Introduction to microsystem technology, Wiley, 2008

Course L1551: Model-Based Systems Engineering (MBSE) with SysML/UML	
Typ	Project-/problem-based Learning
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Examination Form	Schriftliche Ausarbeitung
Examination duration and scale	ca. 10 Seiten
Lecturer	Prof. Ralf God
Language	DE
Cycle	SoSe
Content	Objectives of the problem-oriented course are the acquisition of knowledge on system design using the formal languages SysML/UML, learning about tools for modeling and finally the implementation of a project with methods and tools of Model-Based Systems Engineering (MBSE) on a realistic hardware platform (e.g. Arduino®, Raspberry Pi®): • What is a model? • What is Systems Engineering? • Survey of MBSE methodologies • The modelling languages SysML /UML • Tools for MBSE • Best practices for MBSE • Requirements specification, functional architecture, specification of a solution • From model to software code • Validation and verification: XiL methods • Accompanying MBSE project
Literature	- Skript zur Vorlesung - Weilkiens, T.: Systems Engineering mit SysML/UML: Modellierung, Analyse, Design. 2. Auflage, dpunkt.Verlag, 2008 - Holt, J., Perry, S.A., Brownsword, M.: Model-Based Requirements Engineering. Institution Engineering & Tech, 2011

Course L2863: Sustainable Industrial Production	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Examination Form	Klausur
Examination duration and scale	60 min
Lecturer	Dr. Simon Markus Kothe
Language	DE
Cycle	SoSe
Content	Industrial production deals with the manufacture of physical products to satisfy human needs using various manufacturing processes that change the form and physical properties of raw materials. Manufacturing is a central driver of economic development and has a major impact on the well-being of humanity. However, the scale of current manufacturing activities results in enormous global energy and material demands that are harmful to both the environment and people. Historically, industrial activities were mostly oriented towards economic constraints, while social and environmental consequences were only hardly considered. As a result, today's global consumption rates of many resources and associated emissions often exceed the natural regeneration rate of our planet. In this respect, current industrial production can mostly be described as unsustainable. This is emphasized each year by the Earth Overshoot Day, which marks the day when humanity's ecological footprint exceeds the Earth's annual regenerative capacity. This lecture aims to provide the motivation, analytical methods as well as approaches for sustainable industrial production and to clarify the influence of the production phase in relation to the raw material, use and recycling phases in the entire life cycle of products. For this, the following topics will be highlighted: - Motivation for sustainable production, the 17 Sustainable Development Goals (SDGs) of the UN and their relevance for tomorrow's manufacturing; - raw material vs. production phase vs. use phase vs. recycling/end-of-life phase: importance of the production phase for the environmental impact of manufactured products; - Typical energy- and resource-intensive processes in industrial production and innovative approaches to increase energy and resource efficiency; - Methodology for optimizing the energy and resource efficiency of industrial manufacturing chains with the three steps of modeling (1), evaluating (2) and improving (3); - Resource efficiency of industrial manufacturing value chains and its assessment using life cycle analysis (LCA); - Exercise: LCA analysis of a manufacturing process (thermoplastic joining of an aircraft fuselage segment) as part of a product life cycle assessment.
Literature	Literatur: - Stefan Alexander (2020): Resource efficiency in manufacturing value chains. Cham: Springer International Publishing. - Hauschild, Michael Z.; Rosenbaum, Ralph K.; Olsen, Stig Irving (Hg.) (2018): Life Cycle Assessment. Theory and Practice. Cham: Springer International Publishing. - Kishita, Yusuke; Matsumoto, Mitsutaka; Inoue, Masato; Fukushige, Shinichi (2021): EcoDesign and sustainability. Singapore: Springer. - Schebek, Liselotte; Herrmann, Christoph; Cerdas, Felipe (2019): Progress in Life Cycle Assessment. Cham: Springer International Publishing. - Thiede, Sebastian; Hermann, Christoph (2019): Eco-factories of the future. Cham: Springer Nature Switzerland AG. - Vorlesungsskript.

Course L1077: Process Measurement Engineering	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Examination Form	Mündliche Prüfung
Examination duration and scale	45 Minuten
Lecturer	Prof. Roland Harig
Language	DE/EN
Cycle	SoSe
Content	• Introduction to Process Measurement Engineering • Fundamentals ◦ Characteristics, Statistics, Calibration ◦ Fourier Transform, Convolution, Cross Correlation, Sampling Theorem • Sensors ◦ Distance, Pressure, Strain ◦ Temperature, Infrared • Amplification • Analog to Digital Conversion • Data Transmission and Data Systems • Example: Advanced Process Measurement Systems based on Optical Spectroscopy
Literature	- Färber: „Prozeßrechentechnik“, Springer-Verlag 1994 - Kiencke, Kronmüller: „Meßtechnik“, Springer Verlag Berlin Heidelberg, 1995 - A. Ambardar: „Analog and Digital Signal Processing“ (1), PWS Publishing Company, 1995, NTC 339 - A. Papoulis: „Signal Analysis“ (1), McGraw-Hill, 1987, NTC 312 (LB) - M. Schwartz: „Information Transmission, Modulation and Noise“ (3,4), McGraw-Hill, 1980, 2402095 - S. Haykin: „Communication Systems“ (1,3), Wiley&Sons, 1983, 2419072 - H. Sheingold: „Analog-Digital Conversion Handbook“ (5), Prentice-Hall, 1986, 2440072 - J. Fraden: „AIP Handbook of Modern Sensors“ (5,6), American Institute of Physics, 1993, MTB 346

Course L1083: Process Measurement Engineering	
Typ	Recitation Section (large)
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Examination Form	Mündliche Prüfung
Examination duration and scale	
Lecturer	Prof. Roland Harig
Language	DE/EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Course L0664: Feedback Control in Medical Technology	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Examination Form	Mündliche Prüfung
Examination duration and scale	20 min
Lecturer	Johannes Kreuzer, Christian Neuhaus
Language	DE
Cycle	SoSe
Content	Always viewed from the engineer's point of view, the lecture is structured as follows: • Introduction to the topic • Fundamentals of physiological modelling • Introduction to Breathing and Ventilation • Physiology and Pathology in Cardiology • Introduction to the Regulation of Blood Glucose • kidney function and renal replacement therapy • Representation of the control technology on the concrete ventilator • Excursion to a medical technology company Techniques of modeling, simulation and controller development are discussed. In the models, simple equivalent block diagrams for physiological processes are derived and explained how sensors, controllers and actuators are operated. MATLAB and SIMULINK are used as development tools.
Literature	• Leonhardt, S., & Walter, M. (2016). Medizintechnische Systeme. Berlin, Heidelberg: Springer Vieweg. • Werner, J. (2005). Kooperative und autonome Systeme der Medizintechnik. München: Oldenbourg. • Oczenski, W. (2017). Atmen : Atemhilfen ; Atemphysiologie und Beatmungstechnik: Georg Thieme Verlag KG.

Module M1224: Selected Topics of Mechatronics (Alternative B: 6 LP)			
Courses			
Title	Typ	Hrs/wk	CP
Applied Automation (L1592)	Project-/problem-based Learning	3	3
Advanced Training Course SE-ZERT (L2739)	Project-/problem-based Learning	2	3
Development Management for Mechatronics (L1512)	Lecture	2	3
Fatigue & Damage Tolerance (L0310)	Lecture	2	3
Industry 4.0 for Engineers (L2012)	Lecture	2	3
Microcontroller Circuits: Implementation in Hardware and Software (L0087)	Seminar	2	2
Microsystems Technology (L0724)	Lecture	2	4
Model-Based Systems Engineering (MBSE) with SysML/UML (L1551)	Project-/problem-based Learning	3	3
Sustainable Industrial Production (L2863)	Lecture	2	3
Process Measurement Engineering (L1077)	Lecture	2	3
Process Measurement Engineering (L1083)	Recitation Section (large)	1	1
Feedback Control in Medical Technology (L0664)	Lecture	2	3
Module Responsible	Dr. Martin Gomse		
Admission Requirements	None		
Recommended Previous Knowledge	None		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence	- Students are able to express their extended knowledge and discuss the connection of different special fields or application areas of mechatronics - Students are qualified to connect different special fields with each other - Students can apply specialized solution strategies and new scientific methods in selected areas - Students are able to transfer learned skills to new and unknown problems and can develop own solution approaches		
Knowledge			
Skills			
Personal Competence	Students are able to develop their knowledge and skills by autonomous election of courses.		
Social Competence			
Autonomy			
Workload in Hours	Depends on choice of courses		
Credit points	6		
Assignment for the Following Curricula	Mechatronics: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory		

Course L1592: Applied Automation	
Typ	Project-/problem-based Learning
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Examination Form	Mündliche Prüfung
Examination duration and scale	30 Minuten
Lecturer	Prof. Thorsten Schüppstuhl
Language	DE
Cycle	WiSe
Content	-Project Based Learning -Robot Operating System -Robot structure and description -Motion description -Calibration -Accuracy
Literature	John J. Craig Introduction to Robotics - Mechanics and Control ISBN: 0131236296 Pearson Education, Inc., 2005 Stefan Hesse Grundlagen der Handhabungstechnik ISBN: 3446418725 München Hanser, 2010 K. Thulasiraman and M. N. S. Swamy Graphs: Theory and Algorithms ISBN: 9781118033104 %CITAVIPICKER£9781118033104£Titel anhand dieser ISBN in Citavi-Projekt übernehmen£% John Wiley & Sons, Inc., 1992

Course L2739: Advanced Training Course SE-ZERT	
Typ	Project-/problem-based Learning
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Examination Form	Klausur
Examination duration and scale	120 min
Lecturer	Prof. Ralf God
Language	DE
Cycle	SoSe
Content	
Literature	INCOSE Systems Engineering Handbuch - Ein Leitfaden für Systemlebenszyklus-Prozesse und -Aktivitäten, GfSE (Hrsg. der deutschen Übersetzung), ISBN 978-3-9818805-0-2. ISO/IEC 15288 System- und Software-Engineering - System-Lebenszyklus-Prozesse (Systems and Software Engineering - System Life Cycle Processes).

Course L1512: Development Management for Mechatronics	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Examination Form	Mündliche Prüfung
Examination duration and scale	30 Minuten
Lecturer	Dr. Johannes Nicolas Gebhardt
Language	DE
Cycle	SoSe
Content	- Processes and methods of product development - from idea to market launch - identification of market and technology potentials - development of a common product architecture - Synchronized product development across all engineering disciplines - product validation incl. customer view - Steering and optimization of product development - Design of processes for product development - IT systems for product development - Establishment of management standards - Typical types of organization
Literature	- Bender: Embedded Systems - qualitätsorientierte Entwicklung - Ehrlenspiel: Integrierte Produktentwicklung: Denkabläufe, Methodeneinsatz, Zusammenarbeit - Gausemeier/Ebbesmeyer/Kallmeyer: Produktinnovation - Strategische Planung und Entwicklung der Produkte von morgen - Haberfellner/de Weck/Fricke/Vössner: Systems Engineering: Grundlagen und Anwendung - Lindemann: Methodische Entwicklung technischer Produkte: Methoden flexibel und situationsgerecht anwenden - Pahl/Beitz: Konstruktionslehre: Grundlagen erfolgreicher Produktentwicklung. Methoden und Anwendung - VDI-Richtlinie 2206: Entwicklungsmethodik für mechatronische Systeme

Course L0310: Fatigue & Damage Tolerance	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Examination Form	Mündliche Prüfung
Examination duration and scale	45 min
Lecturer	Dr. Martin Flamm
Language	EN
Cycle	WiSe
Content	Design principles, fatigue strength, crack initiation and crack growth, damage calculation, counting methods, methods to improve fatigue strength, environmental influences
Literature	Jaap Schijve, Fatigue of Structures and Materials. Kluwer Academic Publisher, Dordrecht, 2001 E. Haibach. Betriebsfestigkeit Verfahren und Daten zur Bauteilberechnung. VDI-Verlag, Düsseldorf, 1989

Course L2012: Industry 4.0 for Engineers	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Examination Form	Klausur
Examination duration and scale	120 min
Lecturer	Prof. Thorsten Schüppstuhl
Language	DE
Cycle	SoSe
Content	
Literature	

Course L0087: Microcontroller Circuits: Implementation in Hardware and Software	
Typ	Seminar
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Examination Form	Schriftliche Ausarbeitung
Examination duration and scale	10 min. Vortrag + anschließende Diskussion
Lecturer	Prof. Siegfried Rump
Language	DE
Cycle	WiSe/SoSe
Content	
Literature	ATmega16A 8-bit Microcontroller with 16K Bytes In-System Programmable Flash - DATASHEET, Atmel Corporation 2014 Atmel AVR 8-bit Instruction Set Instruction Set Manual, Atmel Corporation 2016

Course L0724: Microsystems Technology	
Typ	Lecture
Hrs/wk	2
CP	4
Workload in Hours	Independent Study Time 92, Study Time in Lecture 28
Examination Form	Mündliche Prüfung
Examination duration and scale	30 min
Lecturer	Prof. Hoc Khiem Trieu
Language	EN
Cycle	WiSe
Content	• Introduction (historical view, scientific and economic relevance, scaling laws) • Semiconductor Technology Basics, Lithography (wafer fabrication, photolithography, improving resolution, next-generation lithography, nano-imprinting, molecular imprinting) • Deposition Techniques (thermal oxidation, epitaxy, electroplating, PVD techniques: evaporation and sputtering; CVD techniques: APCVD, LPCVD, PECVD and LECVD; screen printing) • Etching and Bulk Micromachining (definitions, wet chemical etching, isotropic etch with HNA, electrochemical etching, anisotropic etching with KOH/TMAH: theory, corner undercutting, measures for compensation and etch-stop techniques; plasma processes, dry etching: back sputtering, plasma etching, RIE, Bosch process, cryo process, XeF2 etching) • Surface Micromachining and alternative Techniques (sacrificial etching, film stress, stiction: theory and counter measures; Origami microstructures, Epi-Poly, porous silicon, SOI, SCREAM process, LIGA, SU8, rapid prototyping) • Thermal and Radiation Sensors (temperature measurement, self-generating sensors: Seebeck effect and thermopile; modulating sensors: thermo resistor, Pt-100, spreading resistance sensor, pn junction, NTC and PTC; thermal anemometer, mass flow sensor, photometry, radiometry, IR sensor: thermopile and bolometer) • Mechanical Sensors (strain based and stress based principle, capacitive readout, piezoresistivity, pressure sensor: piezoresistive, capacitive and fabrication process; accelerometer: piezoresistive, piezoelectric and capacitive; angular rate sensor: operating principle and fabrication process) • Magnetic Sensors (galvanomagnetic sensors: spinning current Hall sensor and magneto-transistor; magnetoresistive sensors: magneto resistance, AMR and GMR, fluxgate magnetometer) • Chemical and Bio Sensors (thermal gas sensors: pellistor and thermal conductivity sensor; metal oxide semiconductor gas sensor, organic semiconductor gas sensor, Lambda probe, MOSFET gas sensor, pH-FET, SAW sensor, principle of biosensor, Clark electrode, enzyme electrode, DNA chip) • Micro Actuators, Microfluidics and TAS (drives: thermal, electrostatic, piezo electric and electromagnetic; light modulators, DMD, adaptive optics, microscanner, microvalves: passive and active, micropumps, valveless micropump, electrokinetic micropumps, micromixer, filter, inkjet printhead, microdispenser, microfluidic switching elements, microreactor, lab-on-a-chip, microanalytics) • MEMS in medical Engineering (wireless energy and data transmission, smart pill, implantable drug delivery system, stimulators: microelectrodes, cochlear and retinal implant; implantable pressure sensors, intelligent osteosynthesis, implant for spinal cord regeneration) • Design, Simulation, Test (development and design flows, bottom-up approach, top-down approach, testability, modelling: multiphysics, FEM and equivalent circuit simulation; reliability test, physics-of-failure, Arrhenius equation, bath-tub relationship) • System Integration (monolithic and hybrid integration, assembly and packaging, dicing, electrical contact: wire bonding, TAB and flip chip bonding; packages, chip-on-board, wafer-level-package, 3D integration, wafer bonding: anodic bonding and silicon fusion bonding; micro electroplating, 3D-MID)
Literature	M. Madou: Fundamentals of Microfabrication, CRC Press, 2002 N. Schwesinger: Lehrbuch Mikrosystemtechnik, Oldenbourg Verlag, 2009 T. M. Adams, R. A. Layton: Introductory MEMS, Springer, 2010 G. Gerlach; W. Dötzel: Introduction to microsystem technology, Wiley, 2008

Course L1551: Model-Based Systems Engineering (MBSE) with SysML/UML	
Typ	Project-/problem-based Learning
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Examination Form	Schriftliche Ausarbeitung
Examination duration and scale	ca. 10 Seiten
Lecturer	Prof. Ralf God
Language	DE
Cycle	SoSe
Content	Objectives of the problem-oriented course are the acquisition of knowledge on system design using the formal languages SysML/UML, learning about tools for modeling and finally the implementation of a project with methods and tools of Model-Based Systems Engineering (MBSE) on a realistic hardware platform (e.g. Arduino®, Raspberry Pi®): • What is a model? • What is Systems Engineering? • Survey of MBSE methodologies • The modelling languages SysML /UML • Tools for MBSE • Best practices for MBSE • Requirements specification, functional architecture, specification of a solution • From model to software code • Validation and verification: XiL methods • Accompanying MBSE project
Literature	- Skript zur Vorlesung - Weilkiens, T.: Systems Engineering mit SysML/UML: Modellierung, Analyse, Design. 2. Auflage, dpunkt.Verlag, 2008 - Holt, J., Perry, S.A., Brownsword, M.: Model-Based Requirements Engineering. Institution Engineering & Tech, 2011

Course L2863: Sustainable Industrial Production	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Examination Form	Klausur
Examination duration and scale	60 min
Lecturer	Dr. Simon Markus Kothe
Language	DE
Cycle	SoSe
Content	Industrial production deals with the manufacture of physical products to satisfy human needs using various manufacturing processes that change the form and physical properties of raw materials. Manufacturing is a central driver of economic development and has a major impact on the well-being of humanity. However, the scale of current manufacturing activities results in enormous global energy and material demands that are harmful to both the environment and people. Historically, industrial activities were mostly oriented towards economic constraints, while social and environmental consequences were only hardly considered. As a result, today's global consumption rates of many resources and associated emissions often exceed the natural regeneration rate of our planet. In this respect, current industrial production can mostly be described as unsustainable. This is emphasized each year by the Earth Overshoot Day, which marks the day when humanity's ecological footprint exceeds the Earth's annual regenerative capacity. This lecture aims to provide the motivation, analytical methods as well as approaches for sustainable industrial production and to clarify the influence of the production phase in relation to the raw material, use and recycling phases in the entire life cycle of products. For this, the following topics will be highlighted: - Motivation for sustainable production, the 17 Sustainable Development Goals (SDGs) of the UN and their relevance for tomorrow's manufacturing; - raw material vs. production phase vs. use phase vs. recycling/end-of-life phase: importance of the production phase for the environmental impact of manufactured products; - Typical energy- and resource-intensive processes in industrial production and innovative approaches to increase energy and resource efficiency; - Methodology for optimizing the energy and resource efficiency of industrial manufacturing chains with the three steps of modeling (1), evaluating (2) and improving (3); - Resource efficiency of industrial manufacturing value chains and its assessment using life cycle analysis (LCA); - Exercise: LCA analysis of a manufacturing process (thermoplastic joining of an aircraft fuselage segment) as part of a product life cycle assessment.
Literature	Literatur: - Stefan Alexander (2020): Resource efficiency in manufacturing value chains. Cham: Springer International Publishing. - Hauschild, Michael Z.; Rosenbaum, Ralph K.; Olsen, Stig Irving (Hg.) (2018): Life Cycle Assessment. Theory and Practice. Cham: Springer International Publishing. - Kishita, Yusuke; Matsumoto, Mitsutaka; Inoue, Masato; Fukushige, Shinichi (2021): EcoDesign and sustainability. Singapore: Springer. - Schebek, Liselotte; Herrmann, Christoph; Cerdas, Felipe (2019): Progress in Life Cycle Assessment. Cham: Springer International Publishing. - Thiede, Sebastian; Hermann, Christoph (2019): Eco-factories of the future. Cham: Springer Nature Switzerland AG. - Vorlesungsskript.

Course L1077: Process Measurement Engineering	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Examination Form	Mündliche Prüfung
Examination duration and scale	45 Minuten
Lecturer	Prof. Roland Harig
Language	DE/EN
Cycle	SoSe
Content	• Introduction to Process Measurement Engineering • Fundamentals ◦ Characteristics, Statistics, Calibration ◦ Fourier Transform, Convolution, Cross Correlation, Sampling Theorem • Sensors ◦ Distance, Pressure, Strain ◦ Temperature, Infrared • Amplification • Analog to Digital Conversion • Data Transmission and Data Systems • Example: Advanced Process Measurement Systems based on Optical Spectroscopy
Literature	- Färber: „Prozeßrechentechnik“, Springer-Verlag 1994 - Kiencke, Kronmüller: „Meßtechnik“, Springer Verlag Berlin Heidelberg, 1995 - A. Ambardar: „Analog and Digital Signal Processing“ (1), PWS Publishing Company, 1995, NTC 339 - A. Papoulis: „Signal Analysis“ (1), McGraw-Hill, 1987, NTC 312 (LB) - M. Schwartz: „Information Transmission, Modulation and Noise“ (3,4), McGraw-Hill, 1980, 2402095 - S. Haykin: „Communication Systems“ (1,3), Wiley&Sons, 1983, 2419072 - H. Sheingold: „Analog-Digital Conversion Handbook“ (5), Prentice-Hall, 1986, 2440072 - J. Fraden: „AIP Handbook of Modern Sensors“ (5,6), American Institute of Physics, 1993, MTB 346

Course L1083: Process Measurement Engineering	
Typ	Recitation Section (large)
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Examination Form	Mündliche Prüfung
Examination duration and scale	
Lecturer	Prof. Roland Harig
Language	DE/EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Course L0664: Feedback Control in Medical Technology	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Examination Form	Mündliche Prüfung
Examination duration and scale	20 min
Lecturer	Johannes Kreuzer, Christian Neuhaus
Language	DE
Cycle	SoSe
Content	Always viewed from the engineer's point of view, the lecture is structured as follows: • Introduction to the topic • Fundamentals of physiological modelling • Introduction to Breathing and Ventilation • Physiology and Pathology in Cardiology • Introduction to the Regulation of Blood Glucose • kidney function and renal replacement therapy • Representation of the control technology on the concrete ventilator • Excursion to a medical technology company Techniques of modeling, simulation and controller development are discussed. In the models, simple equivalent block diagrams for physiological processes are derived and explained how sensors, controllers and actuators are operated. MATLAB and SIMULINK are used as development tools.
Literature	• Leonhardt, S., & Walter, M. (2016). Medizintechnische Systeme. Berlin, Heidelberg: Springer Vieweg. • Werner, J. (2005). Kooperative und autonome Systeme der Medizintechnik. München: Oldenbourg. • Oczenski, W. (2017). Atmen : Atemhilfen ; Atemphysiologie und Beatmungstechnik: Georg Thieme Verlag KG.

Module M1281: Advanced Topics in Vibration				
Courses				
Title	Advanced Topics in Vibration (L1743)		Typ	Hrs/wk
			Project-/problem-based Learning	4
				6
Module Responsible	Prof. Norbert Hoffmann			
Admission Requirements	None			
Recommended Previous Knowledge	Vibration Theory			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	- Students are able to reflect existing terms and concepts of Advanced Vibrations. - Students are able to identify the need to develop and research new terms and concepts in vibrations. - Students are able to apply existing methods and procedures of Advanced Vibrations. - Students are able to develop novel methods and procedures for advanced vibration problems. - Students can reach working results also in groups. - Students can present working results also in groups. - Students are able to approach given research tasks individually - Students are able to identify and follow up novel research tasks by themselves.			
<i>Knowledge</i>				
<i>Skills</i>				
Personal Competence				
<i>Social Competence</i>				
<i>Autonomy</i>				
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	None			
Examination	Written exam			
Examination duration and scale	2 Hours			
Assignment for the Following Curricula	Mechatronics: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Product Development and Production: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Simulation Technology: Elective Compulsory			

Course L1743: Advanced Topics in Vibration	
Typ	Project-/problem-based Learning
Hrs/wk	4
CP	6
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56
Lecturer	Prof. Norbert Hoffmann
Language	DE/EN
Cycle	SoSe
Content	Advanced and Research Topics in Vibrations - Rotor Dynamics - Modal Analysis - Model Order Reduction - Stability of Periodic Solutions - Random Vibrations
Literature	Aktuelle Veröffentlichungen / Recent research publications Bücher/Books: Gasch, Nordmann, Pfützner: Rotordynamik Gasch, Knothe, Liebich: Strukturdynamik

Module M1269: Lab Cyber-Physical Systems				
Courses				
Title	Typ		Hrs/wk	CP
Lab Cyber-Physical Systems (L1740)	Project-/problem-based Learning		4	6
Module Responsible	Prof. Heiko Falk			
Admission Requirements	None			
Recommended Previous Knowledge	Module "Embedded Systems"			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> Cyber-Physical Systems (CPS) are tightly integrated with their surrounding environment, via sensors, A/D and D/A converters, and actors. Due to their particular application areas, highly specialized sensors, processors and actors are common. Accordingly, there is a large variety of different specification approaches for CPS - in contrast to classical software engineering approaches. Based on practical experiments using robot kits and computers, the basics of specification and modelling of CPS are taught. The lab introduces into the area (basic notions, characteristic properties) and their specification techniques (models of computation, hierarchical automata, data flow models, petri nets, imperative approaches). Since CPS frequently perform control tasks, the lab's experiments will base on simple control applications. The experiments will use state-of-the-art industrial specification tools (MATLAB/Simulink, LabVIEW, NX) in order to model cyber-physical models that interact with the environment via sensors and actors. <i>Skills</i> After successful attendance of the lab, students are able to develop simple CPS. They understand the interdependencies between a CPS and its surrounding processes which stem from the fact that a CPS interacts with the environment via sensors, A/D converters, digital processors, D/A converters and actors. The lab enables students to compare modelling approaches, to evaluate their advantages and limitations, and to decide which technique to use for a concrete task. They will be able to apply these techniques to practical problems. They obtain first experiences in hardware-related software development, in industry-relevant specification tools and in the area of simple control applications. Personal Competence <i>Social Competence</i> Students are able to solve similar problems alone or in a group and to present the results accordingly. <i>Autonomy</i> Students are able to acquire new knowledge from specific literature and to associate this knowledge with other classes.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	None			
Examination	Written elaboration			
Examination duration and scale	Execution and documentation of all lab experiments			
Assignment for the Following Curricula	General Engineering Science (German program, 7 semester): Specialisation Computer Science: Elective Compulsory Computer Science: Specialisation II. Mathematics and Engineering Science: Elective Compulsory Computer Science in Engineering: Specialisation II. Mathematics & Engineering Science: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory			

Course L1740: Lab Cyber-Physical Systems	
Typ	Project-/problem-based Learning
Hrs/wk	4
CP	6
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56
Lecturer	Prof. Heiko Falk
Language	EN
Cycle	SoSe
Content	- Experiment 1: Programming in NX - Experiment 2: Programming the Robot in Matlab/Simulink - Experiment 3: Programming the Robot in LabVIEW
Literature	- Peter Marwedel. Embedded System Design - Embedded System Foundations of Cyber-Physical Systems. 2nd Edition, Springer, 2012. - Begleitende Foliensätze

Module M0630: Robotics and Navigation in Medicine				
Courses				
Title		Typ	Hrs/wk	CP
Robotics and Navigation in Medicine (L0335)		Lecture	2	3
Robotics and Navigation in Medicine (L0338)		Project Seminar	2	2
Robotics and Navigation in Medicine (L0336)		Recitation Section (small)	1	1
Module Responsible	Prof. Alexander Schlaefer			
Admission Requirements	None			
Recommended Previous Knowledge	- principles of math (algebra, analysis/calculus) - principles of programming, e.g., in Java or C++ - solid R or Matlab skills			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence				
Knowledge	The students can explain kinematics and tracking systems in clinical contexts and illustrate systems and their components in detail. Systems can be evaluated with respect to collision detection and safety and regulations. Students can assess typical systems regarding design and limitations.			
Skills	The students are able to design and evaluate navigation systems and robotic systems for medical applications.			
Personal Competence				
Social Competence	The students are able to grasp practical tasks in groups, develop solution strategies independently, define work processes and work on them collaboratively. The students are able to collaboratively organize their work processes and software solutions using virtual communication and software management tools. The students can critically reflect on the results of other groups, make constructive suggestions for improvement, and also incorporate them into their own work.			
Autonomy	The students can assess their level of knowledge and independently control their learning processes on this basis as well as document their work results. They can critically evaluate the results achieved and present them in an appropriate argumentative manner to the other groups.			
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	Yes	10 %	Written elaboration	
	Yes	10 %	Presentation	
Examination	Written exam			
Examination duration and scale	90 minutes			
Assignment for the Following Curricula	Computer Science: Specialisation II: Intelligence Engineering: Elective Compulsory Data Science: Specialisation III. Applications: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Medical Technology: Elective Compulsory Electrical Engineering: Specialisation Medical Technology: Elective Compulsory Computer Science in Engineering: Specialisation II. Engineering Science: Elective Compulsory International Management and Engineering: Specialisation II. Electrical Engineering: Elective Compulsory International Management and Engineering: Specialisation II. Process Engineering and Biotechnology: Elective Compulsory International Management and Engineering: Specialisation II. Medical Engineering: Elective Compulsory Mechanical Engineering - Product Development and Production: Specialisation Product Development: Elective Compulsory Mechanical Engineering - Product Development and Production: Specialisation Production: Elective Compulsory Mechanical Engineering - Product Development and Production: Specialisation Materials: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Biomedical Engineering: Specialisation Artificial Organs and Regenerative Medicine: Elective Compulsory Biomedical Engineering: Specialisation Implants and Endoprostheses: Elective Compulsory Biomedical Engineering: Specialisation Medical Technology and Control Theory: Elective Compulsory Biomedical Engineering: Specialisation Management and Business Administration: Elective Compulsory Product Development, Materials and Production: Specialisation Product Development: Elective Compulsory Product Development, Materials and Production: Specialisation Production: Elective Compulsory Product Development, Materials and Production: Specialisation Materials: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Bio- and Medical Technology: Elective Compulsory			

Course L0335: Robotics and Navigation in Medicine	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Alexander Schlaefer
Language	EN
Cycle	SoSe
Content	- kinematics - calibration - tracking systems - navigation and image guidance - motion compensation The seminar extends and complements the contents of the lecture with respect to recent research results.
Literature	Spong et al.: Robot Modeling and Control, 2005 Troccaz: Medical Robotics, 2012 Further literature will be given in the lecture.

Course L0338: Robotics and Navigation in Medicine	
Typ	Project Seminar
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Alexander Schlaefer
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Course L0336: Robotics and Navigation in Medicine	
Typ	Recitation Section (small)
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Prof. Alexander Schlaefer
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1724: Smart Monitoring				
Courses				
Title	Typ		Hrs/wk	CP
Smart Monitoring (L2762)	Integrated Lecture		2	2
Smart Monitoring (L2763)	Recitation Section (small)		2	4
Module Responsible	Prof. Kay Smarsly			
Admission Requirements	None			
Recommended Previous Knowledge	Basic knowledge or interest in object-oriented modeling, programming, and sensor technologies are helpful. Interest in modern research and teaching areas, such as Internet of Things, Industry 4.0 and cyber-physical systems, as well as the will to deepen skills of scientific working, are required. Basic knowledge in scientific writing and good English skills.			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence				
<i>Knowledge</i>	The students will become familiar with the principles and practices of smart monitoring. The students will be able to design decentralized smart systems to be applied for continuous (remote) monitoring of systems in the built and in the natural environment. In addition, the students will learn to design and to implement intelligent sensor systems using state-of-the-art data analysis techniques, modern software design concepts, and embedded computing methodologies. Besides lectures, project work is also part of this module, which will be conducted throughout the semester and will contribute to the grade. In small groups, the students will design smart monitoring systems that integrate a number of "intelligent" sensors to be implemented by the students. Specific focus will be put on the application of machine learning techniques. The smart monitoring systems will be mounted on real-world (built or natural) systems, such as bridges or slopes, or on scaled lab structures for validation purposes. The outcome of every group will be documented in a paper. All students of this module will "automatically" participate with their smart monitoring system in the annual "Smart Monitoring" competition. The written papers and oral examinations form the final grades. The module will be taught in English. Limited enrollment.			
<i>Skills</i>	The students will gain insights into operating state-of-the-art smart sensor systems, used for monitoring a wide range of physical processes relevant to engineering, such as environmental, structural, or comfort monitoring. The students will be capable of devising monitoring strategies of physical processes as part of group projects, tailored to their knowledge backgrounds, and to implement the strategies in smart wireless sensor nodes, using embedded computing and programming. Finally, the students will be able to document the findings of their projects in short reports.			
Personal Competence				
<i>Social Competence</i>	The students will be able to work in groups, share parts of the work for their projects, and develop communication skills, towards achieving the common project goals.			
<i>Autonomy</i>	The students will be able to gain a solid basis on approaching and solving problems in engineering, as well as on documenting results, through their involvement in their monitoring group projects.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	None			
Examination	Written elaboration			
Examination duration and scale	10 pages of work with 15-minute oral presentation			
Assignment for the Following Curricula	Civil Engineering: Specialisation Water and Traffic: Elective Compulsory Civil Engineering: Specialisation Geotechnical Engineering: Elective Compulsory Civil Engineering: Specialisation Coastal Engineering: Elective Compulsory Civil Engineering: Specialisation Structural Engineering: Elective Compulsory Computer Science: Specialisation II: Intelligence Engineering: Elective Compulsory Environmental Engineering: Specialisation Energy and Resources: Elective Compulsory Environmental Engineering: Specialisation Environment and Climate: Elective Compulsory Environmental Engineering: Specialisation Water Quality and Water Engineering: Elective Compulsory Mechatronics: Technical Complementary Course: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory Water and Environmental Engineering: Specialisation Cities: Elective Compulsory Water and Environmental Engineering: Specialisation Environment: Elective Compulsory Water and Environmental Engineering: Specialisation Water: Elective Compulsory			

Course L2762: Smart Monitoring	
Typ	Integrated Lecture
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Kay Smarsly
Language	EN
Cycle	SoSe
Content	In this course, principles of smart monitoring will be taught, focusing on modern concepts of data acquisition, data storage, and data analysis. Also, fundamentals of intelligent sensors and embedded computing will be illuminated. Autonomous software and decentralized data processing are further crucial parts of the course, including concepts of the Internet of Things, Industry 4.0 and cyber-physical systems. Furthermore, measuring principles, data acquisition systems, data management and data analysis algorithms will be discussed. Besides the theoretical background, numerous practical examples will be shown to demonstrate how smart monitoring may advantageously be used for assessing the condition of systems in the built or natural environment.
Literature	The course contents couples different fields, such as signal processing, sensing technologies, data analytics, environmental engineering, civil engineering, artificial intelligence, database systems, and many more. The basics will be taught in this course. However, specific literature that covers all these topics does not exist. Instead, literature will be referenced in the lectures, all of which are papers that are freely available online.

Course L2763: Smart Monitoring	
Typ	Recitation Section (small)
Hrs/wk	2
CP	4
Workload in Hours	Independent Study Time 92, Study Time in Lecture 28
Lecturer	Prof. Kay Smarsly
Language	EN
Cycle	SoSe
Content	The contents of the exercises are based on the lecture contents. In addition to the exercises, project work will be conducted throughout the semester, which will consume the majority of the workload. As part of the project work, students will design smart monitoring systems that will be tested in the laboratory or in the field. As mentioned in the module description, the students will participate in the "Smart Monitoring" competition, hosted annually by the Institute of Digital and Autonomous Construction. Students are encouraged to contribute their own ideas. The tools required to implement the smart monitoring systems will be taught in the group exercises as well as through external sources, such as video tutorials and literature.
Literature	The course contents couples different fields, such as signal processing, sensing technologies, data analytics, environmental engineering, civil engineering, artificial intelligence, database systems, and many more. The basics will be taught in this course. However, specific literature that covers all these topics does not exist. Instead, literature will be referenced in the lectures, all of which are papers that are freely available online.

Module M1203: Applied Dynamics: Numerical and experimental methods				
Courses				
Title	Typ		Hrs/wk	CP
Lab Applied Dynamics (L1631)	Practical Course		1	1
Applied Dynamics (L1630)	Lecture		4	4
Applied Dynamics (L3103)	Recitation Section (small)		1	1
Module Responsible	Prof. Robert Seifried			
Admission Requirements	None			
Recommended Previous Knowledge	Mathematics I, II, III, Mechanics I, II, III, IV Numerical Treatment of Ordinary Differential Equations			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> Students can represent the most important methods of dynamics after successful completion of the module Technical dynamics and have a good understanding of the main concepts in the technical dynamics. <i>Skills</i> Students are able + to think holistically + to independently, securely and critically analyze and optimize basic problems of the dynamics of rigid and flexible multibody systems + to describe dynamics problems mathematically + to investigate dynamics problems both experimentally and numerically Personal Competence <i>Social Competence</i> Students are able to + solve problems in heterogeneous groups and to document the corresponding results. <i>Autonomy</i> Students are able to + assess their knowledge by means of exercises and experiments. + acquaint themselves with the necessary knowledge to solve research oriented tasks.			
Workload in Hours	Independent Study Time 96, Study Time in Lecture 84			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	Yes	None	Subject	theoretical andVersuche Fachlabor
	No	20 %	practical work	
			Excercises	Aufgaben in Matlab
Examination	Written exam			
Examination duration and scale	90 min			
Assignment for the Following Curricula	International Management and Engineering: Specialisation II. Product Development and Production: Elective Compulsory International Management and Engineering: Specialisation II. Mechatronics: Elective Compulsory Mechanical Engineering - Product Development and Production: Core Qualification: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Product Development, Materials and Production: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Core Qualification: Compulsory			

Course L1631: Lab Applied Dynamics	
Typ	Practical Course
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Dr. Marc-André Pick
Language	DE
Cycle	SoSe
Content	Practical exercises are performed in groups. The examples are taken from different areas of applied dynamics, such as numerical simulation, experimental validation and experimental vibration analysis.
Literature	Schiehlen, W.; Eberhard, P.: Technische Dynamik, 4. Auflage, Vieweg+Teubner: Wiesbaden, 2014.

Course L1630: Applied Dynamics	
Typ	Lecture
Hrs/wk	4
CP	4
Workload in Hours	Independent Study Time 64, Study Time in Lecture 56
Lecturer	Prof. Robert Seifried, Dr. Marc-André Pick
Language	DE
Cycle	SoSe
Content	1. Modelling of Multibody Systems 2. Basics from kinematics and kinetics 3. Constraints 4. Multibody systems in minimal coordinates 5. State space, linearization and modal analysis 6. Multibody systems with kinematic constraints 7. Multibody systems as DAE 8. Non-holonomic multibody systems 9. Experimental Methods in Dynamics
Literature	Schiehlen, W.; Eberhard, P.: Technische Dynamik, 4. Auflage, Vieweg+Teubner: Wiesbaden, 2014. Woernle, C.: Mehrkörpersysteme, Springer: Heidelberg, 2011. Seifried, R.: Dynamics of Underactuated Multibody Systems, Springer, 2014.

Course L3103: Applied Dynamics	
Typ	Recitation Section (small)
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Prof. Robert Seifried
Language	DE
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1212: Technical Complementary Course for IMPMEC (according to Subject Specific Regulations)				
Courses				
Title	Typ		Hrs/wk	CP
Module Responsible	NN			
Admission Requirements	None			
Recommended Previous Knowledge	See selected module according to FSPO			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence <i>Knowledge</i>	see selected module according to FSPO			
<i>Skills</i>	see selected module according to FSPO			
Personal Competence <i>Social Competence</i>	see selected module according to FSPO			
<i>Autonomy</i>	see selected module according to FSPO			
Workload in Hours	Depends on choice of courses			
Credit points	6			
Assignment for the Following Curricula	Mechatronics: Core Qualification: Elective Compulsory			

Module M1616: Flight Control Laws: Design and Application				
Courses				
Title			Typ	Hrs/wk
Flight Control Law Design and Application (L2448)			Lecture	2
Flight Control Law Design and Application (L2449)			Project/problem-based Learning	2
Module Responsible	Prof. Frank Thielecke			
Admission Requirements	None			
Recommended Previous Knowledge	Basic knowledge in: - Mathematics (linear algebra and ordinary differential equations) - Control Systems (transfer functions and state space representation) - Mechanics (rigid-body kinetics) - Flight mechanics			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	Knowledge Students are able to: - Describe and understand flight dynamics models for control tasks - Assess handling qualities and understand the need for augmentation through control systems - Identify fundamental performance limitations of control laws Skills Students are able to: - Design model-based control laws for stability augmentation - Design model-based flight control laws - Assess robustness and performance of control laws Personal Competence Social Competence Students are able to: - Design control laws in groups as well as discuss the requirements and results Autonomy Students are able to: - Reflect on the contents of lectures and extend their knowledge through literature research - Solve control design tasks with software tools			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	Yes	None	Attestation	Die in der Vorlesung vermittelten Kenntnisse werden in einem semesterbegleitenden Projekt direkt auf das Modell eines Passagierflugzeugs angewendet.
Examination	Written exam			
Examination duration and scale	60 min			
Assignment for the Following Curricula	Aircraft Systems Engineering: Core Qualification: Elective Compulsory Aeronautics: Core Qualification: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Aircraft Systems Engineering: Elective Compulsory			

Course L2448: Flight Control Law Design and Application	
Typ	Lecture
Hrs/wk	2
CP	4
Workload in Hours	Independent Study Time 92, Study Time in Lecture 28
Lecturer	Dr. Julian Theis
Language	EN
Cycle	SoSe
Content	- Flight dynamics (equations of motion, trim and linearization, linear models of longitudinal and lateral-directional motion, eigenforms) - Stability augmentation (modal dynamics, damper design with root-loci, pole placement and eigenstructure assignment) - Primary flight control laws and autopilots - Design of flight control laws (loopshaping design, robustness criteria and analysis, cascaded control loops, gain-scheduling) - Verification of flight control laws in simulation
Literature	J. Theis: Lecture Notes Flight Control Law Design D. Schmidt: Modern Flight Dynamics B. Stevens, F. Lewis: Aircraft Control and Simulation D. McGruer, D. Graham, I. Ashkenas: Aircraft Dynamics and Automatic Control - SAE Aerospace Standard 94900 - Flight Control Systems - The MathWorks: Control Systems Design Toolbox User Guide

Course L2449: Flight Control Law Design and Application	
Typ	Project-/problem-based Learning
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Dr. Julian Theis
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M0805: Technical Acoustics I (Acoustic Waves, Noise Protection, Psycho Acoustics)				
Courses				
Title			Typ	Hrs/wk
Technical Acoustics I (Acoustic Waves, Noise Protection, Psycho Acoustics) (L0516)			Lecture	2
Technical Acoustics I (Acoustic Waves, Noise Protection, Psycho Acoustics) (L0518)			Recitation Section (large)	2
Module Responsible	Dr. Alessandro Bassetti			
Admission Requirements	None			
Recommended Previous Knowledge	Mechanics I (Statics, Mechanics of Materials) and Mechanics II (Hydrostatics, Kinematics, Dynamics) Mathematics I, II, III (in particular differential equations)			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence				
<i>Knowledge</i>	The students possess an in-depth knowledge in acoustics regarding acoustic waves, noise protection, and psycho acoustics and are able to give an overview of the corresponding theoretical and methodical basis.			
<i>Skills</i>	The students are capable to handle engineering problems in acoustics by theory-based application of the demanding methodologies and measurement procedures treated within the module.			
Personal Competence				
<i>Social Competence</i>	Students can work in small groups on specific problems to arrive at joint solutions.			
<i>Autonomy</i>	The students are able to independently solve challenging acoustical problems in the areas treated within the module. Possible conflicting issues and limitations can be identified and the results are critically scrutinized.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	None			
Examination	Written exam			
Examination duration and scale	90 min			
Assignment for the Following Curricula	Computational Engineering: Core Qualification: Elective Compulsory Aircraft Systems Engineering: Core Qualification: Elective Compulsory International Management and Engineering: Specialisation II. Aviation Systems: Elective Compulsory Aeronautics: Core Qualification: Elective Compulsory Mechanical Engineering - Product Development and Production: Core Qualification: Elective Compulsory Mechanical Engineering and Management: Core Qualification: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Product Development, Materials and Production: Core Qualification: Elective Compulsory Technomathematics: Specialisation III. Engineering Science: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Product Development and Production: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Simulation Technology: Elective Compulsory			

Course L0516: Technical Acoustics I (Acoustic Waves, Noise Protection, Psycho Acoustics)	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Dr. Alessandro Bassetti
Language	EN
Cycle	SoSe
Content	- Introduction and Motivation - Acoustic quantities - Acoustic waves - Sound sources, sound radiation - Sound energy and intensity - Sound propagation - Signal processing - Psycho acoustics - Noise - Measurements in acoustics
Literature	Cremer, L.; Heckl, M. (1996): Körperschall. Springer Verlag, Berlin Veit, I. (1988): Technische Akustik. Vogel-Buchverlag, Würzburg Veit, I. (1988): Flüssigkeitsschall. Vogel-Buchverlag, Würzburg

Course L0518: Technical Acoustics I (Acoustic Waves, Noise Protection, Psycho Acoustics)	
Typ	Recitation Section (large)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Dr. Alessandro Bassetti
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1400: Design of Dependable Systems				
Courses				
Title	Typ		Hrs/wk	CP
Designing Dependable Systems (L2000)	Lecture		2	3
Designing Dependable Systems (L2001)	Recitation Section (small)		2	3
Module Responsible	Prof. Görschwin Fey			
Admission Requirements	None			
Recommended Previous Knowledge	Basic knowledge about data structures and algorithms			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	In the following "dependable" summarizes the concepts Reliability, Availability, Maintainability, Safety and Security. Knowledge about approaches for designing dependable systems, e.g., • Structural solutions like modular redundancy • Algorithmic solutions like handling byzantine faults or checkpointing Knowledge about methods for the analysis of dependable systems <i>Skills</i> Ability to implement dependable systems using the above approaches. Ability to analyze the dependability of systems using the above methods for analysis. Personal Competence <i>Social Competence</i> Students • discuss relevant topics in class and • present their solutions orally. <i>Autonomy</i> Using accompanying material students independently learn in-depth relations between concepts explained in the lecture and additional solution strategies.			
<i>Knowledge</i>				
<i>Skills</i>				
Personal Competence				
<i>Social Competence</i>				
<i>Autonomy</i>				
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	Yes	None	Subject theoretical and practical work	Die Lösung einer Aufgabe ist Zulassungsvoraussetzung für die Prüfung. Die Aufgabe wird in Vorlesung und Übung definiert.
Examination	Oral exam			
Examination duration and scale	30 min			
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Computer Science in Engineering: Specialisation I. Computer Science: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Microelectronics and Microsystems: Specialisation Embedded Systems: Elective Compulsory			

Course L2000: Designing Dependable Systems	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Görschwin Fey
Language	EN
Cycle	SoSe
Content	Description The term dependability comprises various aspects of a system. These are typically: • Reliability • Availability • Maintainability • Safety • Security This makes dependability a core aspect that has to be considered early in system design, no matter whether software, embedded systems or full scale cyber-physical systems are considered. Contents The module introduces the basic concepts for the design and the analysis of dependable systems. Design examples for getting practical hands-on-experience in dependable design techniques. The module focuses towards embedded systems. The following topics are covered: • Modelling • Fault Tolerance • Design Concepts • Analysis Techniques
Literature	The lecture is based on a selection of research papers that is regularly updated. The specific literature references are provided with the lecture slides.

Course L2001: Designing Dependable Systems	
Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Görschwin Fey
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1894: Automation Technology and Systems								
Courses								
Title		Typ	Hrs/wk	CP				
Automation Technology and Systems (L2329)		Lecture	4	4				
Automation Technology and Systems (L2331)		Project-/problem-based Learning	1	1				
Automation Technology and Systems (L2330)		Recitation Section (small)	1	1				
Module Responsible	Prof. Thorsten Schüppstuhl							
Admission Requirements	None							
Recommended Previous Knowledge	without major course assessment							
Educational Objectives	After taking part successfully, students have reached the following learning results							
Professional Competence	Knowledge Students - know the characteristic components of an automation systems and have good understanding of their interaction - know methods for a systematical analysis of automation tasks and are able to use them - have special competences in industrial robot based automation systems Skills Students are able to... - analyze complex Automation tasks - develop application based concepts and solutions - design subsystems and integrate into one system - investigate and evaluate safety of machinery - create simple programs for robots and programmable logic controllers - design of circuit for pneumatic applications Personal Competence Social Competence Students are able to ... - find solutions for automation and handling tasks in groups - develop solutions in a production environment with qualified personnel at technical level and represent decisions. Autonomy Students are able to ... - analyze automation tasks independently - generate programs for robots and programmable logic devices autonomously - develop solutions for practice oriented tasks of automation independently - design safety concepts for automation applications - assess consequences of their professional actions and responsibilities							
Workload in Hours					Independent Study Time 96, Study Time in Lecture 84			
Credit points					6			
Course achievement					Compulsory	Bonus	Form	Description
					No	20 %	Subject theoretical and practical work	Die Studienleistung umfasst die Ergebnisse der PBL basierten Anteile des Moduls sowie der Präsentation in der Gruppe.
Examination					Written exam			
Examination duration and scale					180 min			
Assignment for the Following Curricula					International Management and Engineering: Specialisation II. Product Development and Production: Elective Compulsory Mechanical Engineering - Product Development and Production: Specialisation Product Development: Elective Compulsory Mechanical Engineering - Product Development and Production: Specialisation Production: Compulsory Mechanical Engineering - Product Development and Production: Specialisation Materials: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Product Development, Materials and Production: Specialisation Product Development: Elective Compulsory Product Development, Materials and Production: Specialisation Production: Compulsory Product Development, Materials and Production: Specialisation Materials: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Product Development and Production: Elective Compulsory			

Course L2329: Automation Technology and Systems	
Typ	Lecture
Hrs/wk	4
CP	4
Workload in Hours	Independent Study Time 64, Study Time in Lecture 56
Lecturer	Prof. Thorsten Schüppstuhl
Language	DE
Cycle	SoSe
Content	
Literature	

Course L2331: Automation Technology and Systems	
Typ	Project-/problem-based Learning
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Prof. Thorsten Schüppstuhl
Language	DE
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Course L2330: Automation Technology and Systems	
Typ	Recitation Section (small)
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Prof. Thorsten Schüppstuhl
Language	DE
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1772: Smart Sensors				
Courses				
Title	Typ		Hrs/wk	CP
Smart Sensors (L2904)	Lecture		2	2
Smart Sensors Lab (L2905)	Project-/problem-based Learning		3	4
Module Responsible	Prof. Ulf Kulau			
Admission Requirements	None			
Recommended Previous Knowledge	• Embedded Systems (recommended) • Computer engineering (recommended) • Programming skills in C (recommended) • FPGA experience (desirable)			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> The term smart sensors is actually easy to explain. While data processing in classic sensor systems usually takes place in the backend, the aim of intelligent sensors is to integrate signal processing ever closer to the actual sensor. However, sensors are resource-limited systems and therefore pose many challenges, which are covered in this course. There is a strong focus on the use of ULP FPGAs as flexible processing elements for smart sensors and how these can be implemented using the Verilog hardware description language. The following contents are taught: • Introduction and motivation • Sensors, uncertainties and their compensation • Analog and digital sensor interfaces • Processing elements for smart sensors • General introduction with a focus on ULP FPGAs • Introduction to HDL Verilog • Signal processing (filters and fusion) • Decision making on smart sensors • Reliability, fault tolerance and fault detection In addition to the lecture, there is an exercise in which a simple sensor system is to be implemented using a ULP FPGA. The knowledge learned in the lecture and especially the excursus on the introduction to HDL Verilog will help. <i>Skills</i> After completing this module, students will have a deeper understanding of how smart sensor systems are constructed, which individual components play a role and how they can design their own components: • They have a deeper understanding of the physical and electrotechnical principles of intelligent (smart) sensors • They know the functionalities of the processing elements that can be used for smart sensors, in particular FPGA • They have received a basic introduction to HDL Verilog and are able to carry out small implementations of a smart sensor • They will be able to plan, model, implement and evaluate sensor-related signal processing • They have acquired basic knowledge for obtaining reliable smart sensors Personal Competence <i>Social Competence</i> As part of the module, concepts learned in small groups are to be implemented on an LP FPGA platform. Students learn to work in a team and develop solutions together. Specific tasks are worked on within the group, with cross-group collaboration (exchange) also taking place. The aim is to implement their own small smart sensor system based on the content of the lecture. The results of each group are presented to each other at the end of the semester. <i>Autonomy</i> After completing this module, students will be able to independently develop, optimize and evaluate solutions for Smart Sensor-Systems based on the knowledge they have acquired and further specialist literature.			
Workload in Hours				
Credit points				
Course achievement				
Examination				
Examination duration and scale				
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Information and Communication Systems: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Wireless and Sensor Technologies: Elective Compulsory Electrical Engineering: Specialisation Information and Communication Systems: Elective Compulsory Electrical Engineering: Specialisation Wireless and Sensor Technologies: Elective Compulsory Computer Science in Engineering: Specialisation I. Computer Science: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Microelectronics and Microsystems: Specialisation Embedded Systems: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory			

Course L2904: Smart Sensors	
Typ	Lecture
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Ulf Kulau
Language	EN
Cycle	SoSe
Content	The following contents are taught: • Introduction and motivation • Sensors, uncertainties and their compensation • Analog and digital sensor interfaces • Processing elements for smart sensors • General introduction with a focus on ULP FPGAs • Introduction to HDL Verilog • Signal processing (filters and fusion) • Decision making on smart sensors • Reliability, fault tolerance and fault detection
Literature	• Czichos, Horst. Mechatronik: Grundlagen und Anwendungen technischer Systeme. Springer-Verlag, 2015. • Czichos, Horst, and Werner Daum. "Messtechnik und Sensorik." Dubbel. Springer, Berlin, Heidelberg, 2007. W1-W37. • Iso, I., and BIPM OIML. "Guide to the Expression of Uncertainty in Measurement." Geneva, Switzerland 122 (1995): 16-17 • Hoffmann, Jörg, and Fachbuchverlag Leipzig im Carl-Hanser-Verlag. Taschenbuch der Messtechnik. München, 2010. • Niebuhr, Johannes, and Gerhard Lindner. "Physikalische Meßtechnik mit Sensoren." tm-Technisches Messen 63 (1996):4. • Kiencke, Uwe, and Ralf Eger. Messtechnik. Springer Berlin Heidelberg, 2008. (https://katalog.tub.tuhh.de/Record/1646558138) • Fraden, Jacob. "Handbook of modern sensors: physics, designs, and applications." (2007). • Meijer, Gerard, Kofi Makinwa, and Michiel Pertijs. Smart sensor systems: Emerging technologies and applications. John Wiley & Sons, 2014. • Vansteenkiste, Elias. New FPGA design tools and architectures. Diss. Ghent University, 2016. • Wüst, Klaus. Mikroprozessortechnik: Grundlagen, Architekturen, Schaltungstechnik und Betrieb von Mikroprozessoren und Mikrocontrollern. Springer-Verlag, 2010. • Huddleston, Creed. Intelligent sensor design using the microchip dsPIC. Elsevier, 2006. • Wolf, Marilyn. Computers as components: principles of embedded computing system design. Elsevier, 2012. • Smith, Douglas J. HDL Chip Design: A practical guide for designing, synthesizing and simulating ASICs and FPGAs • Thomas, Donald, and Philip Moorby. The Verilog® hardware description language. Springer Science & Business Media, 2008. • LaMeres, Brock J. Quick Start Guide to Verilog. Springer International Publishing, 2019. (https://katalog.tub.tuhh.de/Record/1067372644) • J. Hennessy, D. Patterson: „Computer Architecture: A Quantitative Approach“, Morgan Kaufmann, 4. Aufl., 2006, • B. Parhami : „Computer Arithmetic: Algorithms and Hardware Designs“, Oxford Univ Press, 2009 • Pei, Yan, et al. "An elementary introduction to Kalman filtering." Communications of the ACM 62.11 (2019): 122-133 • Johnson, Barry W. "An introduction to the design and analysis of fault-tolerant systems." Fault-tolerant computer system design 1 (1996)

Course L2905: Smart Sensors Lab	
Typ	Project-/problem-based Learning
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Lecturer	Prof. Ulf Kulau
Language	EN
Cycle	SoSe
Content	The following contents are taught in the PBL, drawing on previous knowledge from the lecture. The basic aim of this Smart Sensors Lab is to implement a simple smart sensor system using a ULP FPGA. • Getting to know a ULP FPGA platform (simple introductory tasks) • Implementation of a host interface (UART) • Implementation of a sensor interface SPI • Sensor-near data processing (creative part)
Literature	• Thomas, Donald, and Philip Moorby. The Verilog® hardware description language. Springer Science & Business Media, 2008. • LaMeres, Brock J. Quick Start Guide to Verilog. Springer International Publishing, 2019. (https://katalog.tub.tuhh.de/Record/1067372644) • Online-Simulators: ◦ https://www.jdoodle.com/execute-verilog-online/ ◦ https://www.edaplayground.com/home ◦ https://www.tutorialspoint.com/compile_verilog_online.php

Module M1810: Autonomous Cyber-Physical Systems				
Courses				
Title	Typ		Hrs/wk	CP
Autonomous Cyber-Physical Systems (L3000)	Lecture		2	3
Autonomous Cyber-Physical Systems (L3001)	Project/problem-based Learning		3	3
Module Responsible	Prof. Bernd-Christian Renner			
Admission Requirements	None			
Recommended Previous Knowledge	• Very good knowledge and practical experience in programming in the C/C++ language (e.g., module: Procedural Programming for Computer Scientists) • Basic knowledge in software engineering • Basic knowledge in wired and wireless communication protocols • Principal understanding of simple electronic circuits			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence <i>Knowledge</i>	This master's course centers on the advanced Internet of Things (IoT) systems, spotlighting energy sustainability and Real-Time Operating Systems (RTOS) within cyber-physical frameworks. Key focus areas include the exploration of IoT system architecture, especially leveraging RTOS for real-time responsiveness, and energy-efficient technologies for sustainable operations. Upon completion, students will be capable of: • Outlining the structure and operational principles of IoT systems and their significance in modern cyber-physical environments. • Differentiating between RTOS and conventional operating systems in the context of IoT applications. • Evaluating energy harvesting (specifically solar technologies) and storage systems for their applicability and efficiency in IoT devices • Knowing different wireless technologies (e.g., Bluetooth LE, LoRaWAN, Zigbee) and networking protocols (TCP/IP, MQTT) for IoT connectivity and functionality, and pinpointing their dis-/advantages.			
<i>Skills</i>	Students will acquire practical skills to: • Describe, plan, develop and optimize IoT applications using RTOS, focusing on multitasking, efficient memory, and power management. • Select and deploy suitable wireless and networking technologies considering device requirements and operational contexts. • Integrate energy harvesting and storage technologies into IoT solutions, enhancing their sustainability and autonomy. • Conduct comprehensive planning and evaluation of IoT projects, incorporating scientific methodologies for real-world applications.			
Personal Competence <i>Social Competence</i>	• Students are able to work goal-oriented in small mixed groups. • They learn and broaden their teamwork abilities. • They learn to define and split tasks within the team. • After completing the module, the students are able to work on similar tasks alone or in a group and to present the results in a suitable way.			
<i>Autonomy</i>	After completing the module, the students are able to independently work on sub-areas of the subject using specialist literature, to summarize and present the knowledge they have acquired and to link it to the content of other courses.			
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	No	10 %	Attestation	
Examination	Written exam			
Examination duration and scale	90 min			
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Wireless and Sensor Technologies: Elective Compulsory Electrical Engineering: Specialisation Wireless and Sensor Technologies: Elective Compulsory Computer Science in Engineering: Specialisation II. Engineering Science: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems, Focus Software and Signal Processing: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Mechatronics: Technical Complementary Course: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory			

Course L3000: Autonomous Cyber-Physical Systems	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Bernd-Christian Renner
Language	EN
Cycle	SoSe
Content	
Literature	

Course L3001: Autonomous Cyber-Physical Systems	
Typ	Project-/problem-based Learning
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Lecturer	Prof. Bernd-Christian Renner
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M0803: Embedded Systems				
Courses				
Title	Typ		Hrs/wk	CP
Embedded Systems (L0805)	Lecture		3	3
Embedded Systems (L2938)	Project-/problem-based Learning		1	1
Embedded Systems (L0806)	Recitation Section (small)		1	2
Module Responsible	Prof. Heiko Falk			
Admission Requirements	None			
Recommended Previous Knowledge	Computer Engineering			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> Embedded systems can be defined as information processing systems embedded into enclosing products. This course teaches the foundations of such systems. In particular, it deals with an introduction into these systems (notions, common characteristics) and their specification languages (models of computation, hierarchical automata, specification of distributed systems, task graphs, specification of real-time applications, translations between different models). Another part covers the hardware of embedded systems: Sensors, A/D and D/A converters, real-time capable communication hardware, embedded processors, memories, energy dissipation, reconfigurable logic and actuators. The course also features an introduction into real-time operating systems, middleware and real-time scheduling. Finally, the implementation of embedded systems using hardware/software co-design (hardware/software partitioning, high-level transformations of specifications, energy-efficient realizations, compilers for embedded processors) is covered. <i>Skills</i> After having attended the course, students shall be able to realize simple embedded systems. The students shall realize which relevant parts of technological competences to use in order to obtain a functional embedded systems. In particular, they shall be able to compare different models of computations and feasible techniques for system-level design. They shall be able to judge in which areas of embedded system design specific risks exist. Personal Competence <i>Social Competence</i> Students are able to solve similar problems alone or in a group and to present the results accordingly. <i>Autonomy</i> Students are able to acquire new knowledge from specific literature and to associate this knowledge with other classes.			
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	Yes	10 %	Subject	theoretical and practical work
Examination	Written exam			
Examination duration and scale	90 minutes, contents of course and labs			
Assignment for the Following Curricula	General Engineering Science (German program, 7 semester): Specialisation Computer Science: Compulsory Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Electrical Engineering: Core Qualification: Elective Compulsory Electrical Engineering and Information Technology: Core Qualification: Elective Compulsory Engineering Science: Specialisation Electrical Engineering: Elective Compulsory Engineering Science: Specialisation Information and Communication Systems: Compulsory Engineering Science: Specialisation Mechatronics: Elective Compulsory Aircraft Systems Engineering: Core Qualification: Elective Compulsory Computer Science in Engineering: Core Qualification: Compulsory Aeronautics: Core Qualification: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Mechatronics: Specialisation Naval Engineering: Compulsory Mechatronics: Specialisation Electrical Systems: Compulsory Mechatronics: Specialisation Dynamic Systems and AI: Compulsory Mechatronics: Specialisation Robot- and Machine-Systems: Compulsory Mechatronics: Specialisation Medical Engineering: Compulsory Microelectronics and Microsystems: Specialisation Embedded Systems: Elective Compulsory			

Course L0805: Embedded Systems	
Typ	Lecture
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Lecturer	Prof. Heiko Falk
Language	EN
Cycle	SoSe
Content	• Introduction • Specifications and Modeling • Embedded/Cyber-Physical Systems Hardware • System Software • Evaluation and Validation • Mapping of Applications to Execution Platforms • Optimization
Literature	• Peter Marwedel. Embedded System Design - Embedded Systems Foundations of Cyber-Physical Systems. 2nd Edition, Springer, 2012., Springer, 2012.

Course L2938: Embedded Systems	
Typ	Project-/problem-based Learning
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Prof. Heiko Falk
Language	EN
Cycle	SoSe
Content	• Introduction • Specifications and Modeling • Embedded/Cyber-Physical Systems Hardware • System Software • Evaluation and Validation • Mapping of Applications to Execution Platforms • Optimization
Literature	• Peter Marwedel. Embedded System Design - Embedded Systems Foundations of Cyber-Physical Systems. 2nd Edition, Springer, 2012., Springer, 2012.

Course L0806: Embedded Systems	
Typ	Recitation Section (small)
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Prof. Heiko Falk
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M0924: Software for Embedded Systems				
Courses				
Title	Typ		Hrs/wk	CP
Software for Embedded Systems (L1069)	Lecture		2	3
Software for Embedded Systems (L1070)	Project/problem-based Learning		3	3
Module Responsible	Prof. Bernd-Christian Renner			
Admission Requirements	None			
Recommended Previous Knowledge	• Very good knowledge and practical experience in programming in the C language and its compilation process • Basic knowledge in software engineering • Basic understanding of assembly language • Basic knowledge of electrical engineering			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	Knowledge • Students know the contents of the module and can reproduce and link them. • They are able to describe the usage and advantages of event-based programming using interrupts. • Students can explain the basic principles and procedures for creating software for embedded systems. • They know the components and functions of a concrete microcontroller. • The participants can explain requirements of real time systems. • They know at least three scheduling algorithms for real time operating systems including their pros and cons. • They know different memories for embedded systems and their use cases, including advantages and disadvantages. • They understand race conditions, their causes and consequences. • They are familiar with at least two bootloader concepts and can evaluate their suitability in different contexts. • They know mechanisms for optimizing the power consumption of microcontrollers with sleep modes. Skills • Students design and write hardware-oriented software modules in a high-level language (C or similar) for an embedded system based on a specific microcontroller. • They can configure and interact with peripherals (timer, ADC, EEPROM), including bit-level register operations, interrupt-based processing, and program flow. • They implement and use a (preemptive) scheduler for an embedded system. • They learn to write independent, reusable software components. They build a complex software system (library and applications) throughout the semester. • They learn how to model event-based software for embedded systems with UML state diagrams and finite state machines and implement a complex system that they modeled. • They can identify race conditions in unknown software and develop strategies to work around them. Personal Competence Social Competence • Students are able to work goal-oriented in small mixed groups. • They learn and broaden their teamwork abilities. • They learn to define and split tasks within the team. • They practice to present their software, the selection of algorithms and their design choices. Autonomy Students are able • to solve assignments related to this lecture independently with instructional direction. • to design, implement, and test software components for an embedded system independently based on a textual description. • to read and understand data sheets and manuals of electronic components (such as micro-controllers and sensors). • to independently transfer the concepts known from the lecture to the exercises and to select algorithms based on given criteria.			
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	No	10 %	Attestation	
Examination	Written exam			
Examination duration and scale	90 min			
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Information and Communication Systems: Elective Compulsory Electrical Engineering: Specialisation Information and Communication Systems: Elective Compulsory Computer Science in Engineering: Specialisation I. Computer Science: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Software: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Microelectronics and Microsystems: Specialisation Embedded Systems: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory			

Course L1069: Software for Embedded Systems	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Bernd-Christian Renner
Language	EN
Cycle	SoSe
Content	• General-Purpose Processors • Programming the Atmel AVR • Interrupts and Race Conditions • C for Embedded Systems • Standard Single Purpose Processors: Peripherals • Finite-State Machines • Memory • Operating Systems for Embedded Systems • Real-Time Embedded Systems • Boot loader and Power Management
Literature	1. Embedded System Design, F. Vahid and T. Givargis, John Wiley 2. Programming Embedded Systems: With C and Gnu Development Tools, M. Barr and A. Massa, O'Reilly 3. C und C++ für Embedded Systems, F. Bollow, M. Homann, K. Köhn, MITP 4. The Art of Designing Embedded Systems, J. Ganssle, Newnes 5. Mikrocomputertechnik mit Controllern der Atmel AVR-RISC-Familie, G. Schmitt, Oldenbourg 6. Making Embedded Systems: Design Patterns for Great Software, E. White, O'Reilly

Course L1070: Software for Embedded Systems	
Typ	Project-/problem-based Learning
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Lecturer	Prof. Bernd-Christian Renner
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M0627: Machine Learning and Data Mining			
Courses			
Title	Typ	Hrs/wk	CP
Machine Learning and Data Mining (L0340)	Lecture	2	4
Machine Learning and Data Mining (L0510)	Recitation Section (small)	2	2
Module Responsible	Rainer Marrone		
Admission Requirements	None		
Recommended Previous Knowledge	• Calculus • Stochastics		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence <i>Knowledge</i>	Students can explain the difference between instance-based and model-based learning approaches, and they can enumerate basic machine learning technique for each of the two basic approaches, either on the basis of static data, or on the basis of incrementally incoming data . For dealing with uncertainty, students can describe suitable representation formalisms, and they explain how axioms, features, parameters, or structures used in these formalisms can be learned automatically with different algorithms. Students are also able to sketch different clustering techniques. They depict how the performance of learned classifiers can be improved by ensemble learning, and they can summarize how this influences computational learning theory. Algorithms for reinforcement learning can also be explained by students.		
<i>Skills</i>	Student derive decision trees and, in turn, propositional rule sets from simple and static data tables and are able to name and explain basic optimization techniques. They present and apply the basic idea of first-order inductive learning. Students apply the BME, MAP, ML, and EM algorithms for learning parameters of Bayesian networks and compare the different algorithms. They also know how to carry out Gaussian mixture learning. They can contrast kNN classifiers, neural networks, and support vector machines, and name their basic application areas and algorithmic properties. Students can describe basic clustering techniques and explain the basic components of those techniques. Students compare related machine learning techniques, e.g., k-means clustering and nearest neighbor classification. They can distinguish various ensemble learning techniques and compare the different goals of those techniques.		
Personal Competence <i>Social Competence</i> <i>Autonomy</i>			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56		
Credit points	6		
Course achievement	None		
Examination	Written exam		
Examination duration and scale	90 minutes		
Assignment for the Following Curricula	Computer Science: Specialisation II: Intelligence Engineering: Elective Compulsory International Management and Engineering: Specialisation II. Information Technology: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory		

Course L0340: Machine Learning and Data Mining	
Typ	Lecture
Hrs/wk	2
CP	4
Workload in Hours	Independent Study Time 92, Study Time in Lecture 28
Lecturer	Rainer Marrone
Language	EN
Cycle	SoSe
Content	• Decision trees • First-order inductive learning • Incremental learning: Version spaces • Uncertainty • Bayesian networks • Learning parameters of Bayesian networks BME, MAP, ML, EM algorithm • Learning structures of Bayesian networks • Gaussian Mixture Models • kNN classifier, neural network classifier, support vector machine (SVM) classifier • Clustering Distance measures, k-means clustering, nearest neighbor clustering • Kernel Density Estimation • Ensemble Learning • Reinforcement Learning • Computational Learning Theory
Literature	1. Artificial Intelligence: A Modern Approach (Third Edition), Stuart Russel, Peter Norvig, Prentice Hall, 2010, Chapters 13, 14, 18-21 2. Machine Learning: A Probabilistic Perspective, Kevin Murphy, MIT Press 2012

Course L0510: Machine Learning and Data Mining	
Typ	Recitation Section (small)
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Rainer Marrone
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1143: Applied Design Methodology in Mechatronics				
Courses				
Title		Typ	Hrs/wk	CP
Applied Design Methodology in Mechatronics (L1523)		Lecture	2	2
Applied Design Methodology in Mechatronics (L1524)		Project-/problem-based Learning	3	4
Module Responsible	Prof. Thorsten Kern			
Admission Requirements	None			
Recommended Previous Knowledge	Basics of mechanical design, electrical design or computer-sciences			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence				
<i>Knowledge</i>	Science-based working on interdisciplinary product design considering targeted application of specific product design techniques			
<i>Skills</i>	Creative handling of processes used for scientific preparation and formulation of complex product design problems / Application of various product design techniques following theoretical aspects.			
Personal Competence				
<i>Social Competence</i>	Students will solve and execute technical-scientific tasks from an industrial context in small design-teams with application of common, creative methodologies.			
<i>Autonomy</i>	Students are enabled to optimize the design and development process according to the target and topic of the design			
	Students are educated to operate in a development team			
	Students learn about the right application of creative methods in engineering.			
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70			
Credit points	6			
Course achievement	None			
Examination	Subject theoretical and practical work			
Examination duration and scale	30 min Presentation for a group design-work			
Assignment for the Following Curricula	International Management and Engineering: Specialisation II. Product Development and Production: Elective Compulsory International Management and Engineering: Specialisation II. Mechatronics: Elective Compulsory Mechanical Engineering and Management: Specialisation Product Engineering: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Biomedical Engineering: Specialisation Implants and Endoprotheses: Elective Compulsory Biomedical Engineering: Specialisation Medical Technology and Control Theory: Elective Compulsory Biomedical Engineering: Specialisation Management and Business Administration: Elective Compulsory Biomedical Engineering: Specialisation Artificial Organs and Regenerative Medicine: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Product Development and Production: Elective Compulsory			

Course L1523: Applied Design Methodology in Mechatronics	
Typ	Lecture
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Thorsten Kern
Language	EN
Cycle	SoSe
Content	• Systematic analysis and planning of the design process for products combining a multitude of disciplines • Structure of the engineering process with focus on engineering steps (task-definition, functional decomposition, physical principles, elements for solution, combination to systems and products, execution of design, component-tests, system-tests, product-testing and qualification/validation) • Creative methods (Basics, methods like lead-user-method, 6-3-5, BrainStorming, Intergalactic Thinking, ... - Applications in examples all around mechatronics topics) • Several design-supporting methods and tools (functional structures, GALFMOS, AEIOU-method, GAMPFT, simulation and its application, TRIZ, design for SixSigma, continuous integration and testing, ...) • Evaluation and final selection of solution (technical and business-considerations, preference-matrix, pair-comparison), dealing with uncertainties, decision-making • Value-analysis • Derivation of architectures and architectural management • Project-tracking and -guidance (project-lead, guiding of employees, organization of multidisciplinary R&D departments, idea-identification, responsibilities and communication) • Project-execution methods (Scrum, Kanban, ...) • Presentation-skills • Questions of aesthetic product design and design for subjective requirements (industrial design, color, haptic/optic/acoustic interfaces) • Evaluation of selected methods at practical examples in small teams
Literature	• Definition folgt... • Pahl, G.; Beitz, W.; Feldhusen, J.; Grote, K.-H.: Konstruktionslehre: Grundlage erfolgreicher Produktentwicklung, Methoden und Anwendung, 7. Auflage, Springer Verlag, Berlin 2007 • VDI-Richtlinien: 2206; 2221ff

Course L1524: Applied Design Methodology in Mechatronics	
Typ	Project-/problem-based Learning
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Lecturer	Prof. Thorsten Kern
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1156: Systems Engineering				
Courses				
Title	Typ		Hrs/wk	CP
Systems Engineering (L1547)	Lecture		4	4
Systems Engineering (L1548)	Recitation Section (large)		1	2
Module Responsible	Prof. Ralf God			
Admission Requirements	None			
Recommended Previous Knowledge	Basic knowledge in: • Mathematics • Mechanics • Thermodynamics • Electrical Engineering • Control Systems Previous knowledge in: • Aircraft Cabin Systems			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence <i>Knowledge</i> <i>Skills</i> Personal Competence <i>Social Competence</i> <i>Autonomy</i>	Students are able to: • understand systems engineering process models, methods and tools for the development of complex Systems • describe innovation processes and the need for technology Management • explain the aircraft development process and the process of type certification for aircraft • explain the system development process, including requirements for systems reliability • identify environmental conditions and test procedures for airborne Equipment • value the methodology of requirements-based engineering (RBE) and model-based requirements engineering (MBRE) Students are able to: • plan the process for the development of complex Systems • organize the development phases and development Tasks • assign required business activities and technical Tasks • apply systems engineering methods and tools Students are able to: • understand and accept their tasks within a development team • be comfortable with their role their tasks within the overall process • understand and serve their suppliers and customers in large projects • assume responsibility for people and technology in the development of safety-critical systems Students are able to: • interact and communicate in a development team with division of tasks. • independently research and identify certification specifications • formulate requirements on their own • create test plans on their own and accompany certification processes			
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70			
Credit points	6			
Course achievement	None			
Examination	Written exam			
Examination duration and scale	120 Minutes			
Assignment for the Following Curricula	Aircraft Systems Engineering: Core Qualification: Compulsory International Management and Engineering: Specialisation II. Product Development and Production: Elective Compulsory International Management and Engineering: Specialisation II. Aviation Systems: Elective Compulsory Aeronautics: Core Qualification: Compulsory Mechanical Engineering - Product Development and Production: Specialisation Product Development: Compulsory Mechanical Engineering - Product Development and Production: Specialisation Production: Elective Compulsory Mechanical Engineering - Product Development and Production: Specialisation Materials: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Product Development, Materials and Production: Specialisation Product Development: Compulsory Product Development, Materials and Production: Specialisation Production: Elective Compulsory Product Development, Materials and Production: Specialisation Materials: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Aircraft Systems Engineering: Elective Compulsory			

Course L1547: Systems Engineering	
Typ	Lecture
Hrs/wk	4
CP	4
Workload in Hours	Independent Study Time 64, Study Time in Lecture 56
Lecturer	Prof. Ralf God
Language	DE
Cycle	SoSe
Content	The objective of the lecture with the corresponding exercise is to accomplish the prerequisites for the development and integration of complex systems using the example of commercial aircraft and cabin systems. Competences in the systems engineering process, tools and methods is to be achieved. Regulations, guidelines and certification issues will be known. Key aspects of the course are processes for innovation and technology management, system design, system integration and certification as well as tools and methods for systems engineering: • Innovation processes • IP-protection • Technology management • Systems engineering • Aircraft program • Certification issues • Systems development • Safety objectives and fault tolerance • Environmental and operating conditions • Tools for systems engineering • Requirements-based engineering (RBE) • Model-based requirements engineering (MBRE)
Literature	- Skript zur Vorlesung - diverse Normen und Richtlinien (EASA, FAA, RTCA, SAE) - Hauschildt, J., Salomo, S.: Innovationsmanagement. Vahlen, 5. Auflage, 2010 - NASA Systems Engineering Handbook, National Aeronautics and Space Administration, 2007 - Hinsch, M.: Industrielles Luftfahrtmanagement: Technik und Organisation luftfahrttechnischer Betriebe. Springer, 2010 - De Florio, P.: Airworthiness: An Introduction to Aircraft Certification. Elsevier Ltd., 2010 - Pohl, K.: Requirements Engineering. Grundlagen, Prinzipien, Techniken. 2. korrigierte Auflage, dpunkt.Verlag, 2008

Course L1548: Systems Engineering	
Typ	Recitation Section (large)
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Prof. Ralf God
Language	DE
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1340: Introduction to Waveguides, Antennas, and Electromagnetic Compatibility				
Courses				
Title	Typ		Hrs/wk	CP
Introduction to Waveguides, Antennas, and Electromagnetic Compatibility (L1669)	Lecture		3	4
Introduction to Waveguides, Antennas, and Electromagnetic Compatibility (L1877)	Recitation Section (small)		2	2
Module Responsible	Prof. Christian Schuster			
Admission Requirements	None			
Recommended Previous Knowledge	Basic principles of physics and electrical engineering			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	Students can explain the basic principles, relationships, and methods for the design of waveguides and antennas as well as of Electromagnetic Compatibility. Specific topics are: - Fundamental properties and phenomena of electrical circuits - Steady-state sinusoidal analysis of electrical circuits - Fundamental properties and phenomena of electromagnetic fields and waves - Steady-state sinusoidal description of electromagnetic fields and waves - Useful microwave network parameters - Transmission lines and basic results from transmission line theory - Plane wave propagation, superposition, reflection and refraction - General theory of waveguides - Most important types of waveguides and their properties - Radiation and basic antenna parameters - Most important types of antennas and their properties - Numerical techniques and CAD tools for waveguide and antenna design - Fundamentals of Electromagnetic Compatibility - Coupling mechanisms and countermeasures - Shielding, grounding, filtering - Standards and regulations - EMC measurement techniques Students know how to apply various methods and models for characterization and choice of waveguides and antennas. They are able to assess and qualify their basic electromagnetic properties. They can apply results and strategies from the field of Electromagnetic Compatibility to the development of electrical components and systems.			
<i>Knowledge</i>				
<i>Skills</i>				
Personal Competence				
<i>Social Competence</i>				
<i>Autonomy</i>	Students are capable to gather information from subject related, professional publications and relate that information to the context of the lecture. They are able to make a connection between their knowledge obtained in this lecture with the content of other lectures (e.g. theory of electromagnetic fields, fundamentals of electrical engineering / physics). They can discuss technical problems and physical effects in English.			
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70			
Credit points	6			
Course achievement	None			
Examination	Oral exam			
Examination duration and scale	45 min			
Assignment for the Following Curricula	General Engineering Science (German program, 7 semester): Specialisation Electrical Engineering: Elective Compulsory Electrical Engineering: Core Qualification: Elective Compulsory Electrical Engineering and Information Technology: Core Qualification: Elective Compulsory Engineering Science: Specialisation Electrical Engineering: Elective Compulsory Aircraft Systems Engineering: Core Qualification: Elective Compulsory Aeronautics: Core Qualification: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory			

Course L1669: Introduction to Waveguides, Antennas, and Electromagnetic Compatibility	
Typ	Lecture
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Lecturer	Prof. Christian Schuster
Language	DE/EN
Cycle	SoSe
Content	This course is intended as an introduction to the topics of wave propagation, guiding, sending, and receiving as well as Electromagnetic Compatibility (EMC). It will be useful for engineers that face the technical challenge of transmitting high frequency / high bandwidth data in e.g. medical, automotive, or avionic applications. Both circuit and field concepts of wave propagation and Electromagnetic Compatibility will be introduced and discussed. Topics: - Fundamental properties and phenomena of electrical circuits - Steady-state sinusoidal analysis of electrical circuits - Fundamental properties and phenomena of electromagnetic fields and waves - Steady-state sinusoidal description of electromagnetic fields and waves - Useful microwave network parameters - Transmission lines and basic results from transmission line theory - Plane wave propagation, superposition, reflection and refraction - General theory of waveguides - Most important types of waveguides and their properties - Radiation and basic antenna parameters - Most important types of antennas and their properties - Numerical techniques and CAD tools for waveguide and antenna design - Fundamentals of Electromagnetic Compatibility - Coupling mechanisms and countermeasures - Shielding, grounding, filtering - Standards and regulations - EMC measurement techniques
Literature	- Zinke, Brunswig, "Hochfrequenztechnik 1", Springer (1999) - J. Detlefsen, U. Siart, "Grundlagen der Hochfrequenztechnik", Oldenbourg (2012) - D. M. Pozar, "Microwave Engineering", Wiley (2011) - Y. Huang, K. Boyle, "Antenna: From Theory to Practice", Wiley (2008) - H. Ott, "Electromagnetic Compatibility Engineering", Wiley (2009) - A. Schwab, W. Kürner, "Elektromagnetische Verträglichkeit", Springer (2007)

Course L1877: Introduction to Waveguides, Antennas, and Electromagnetic Compatibility	
Typ	Recitation Section (small)
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Christian Schuster
Language	DE/EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M0752: Nonlinear Dynamics				
Courses				
Title	Typ		Hrs/wk	CP
Nonlinear Dynamics (L0702)	Integrated Lecture		4	6
Module Responsible	Prof. Norbert Hoffmann			
Admission Requirements	None			
Recommended Previous Knowledge	• Calculus • Linear Algebra • Engineering Mechanics			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence <i>Knowledge</i>	• Students are able to reflect existing terms and concepts in Nonlinear Dynamics and to develop and research new terms and concepts. • Students are able to denote and expand methods of modeling and analysis for nonlinear dynamical systems.			
<i>Skills</i>	• Students are able to apply existing methods and procedures of Nonlinear Dynamics. • Students are able to develop novel methods and procedures for nonlinear dynamical systems.			
Personal Competence <i>Social Competence</i>	• Students can analyze problems of nonlinear dynamics also in groups. • Students can achieve solution procedures for problems of nonlinear dynamical systems also in groups.			
<i>Autonomy</i>	• Students are able to approach given research tasks on the basis of given methods individually. • Students are able to identify and follow up novel research tasks by themselves.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	None			
Examination	Written exam			
Examination duration and scale	2 Hours			
Assignment for the Following Curricula	Aircraft Systems Engineering: Core Qualification: Elective Compulsory International Management and Engineering: Specialisation II. Mechatronics: Elective Compulsory Mechanical Engineering - Product Development and Production: Core Qualification: Elective Compulsory Mechanical Engineering and Management: Specialisation Mechatronics: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Biomedical Engineering: Specialisation Artificial Organs and Regenerative Medicine: Elective Compulsory Biomedical Engineering: Specialisation Implants and Endoprostheses: Elective Compulsory Biomedical Engineering: Specialisation Medical Technology and Control Theory: Elective Compulsory Biomedical Engineering: Specialisation Management and Business Administration: Elective Compulsory Product Development, Materials and Production: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Core Qualification: Elective Compulsory			

Course L0702: Nonlinear Dynamics	
Typ	Integrated Lecture
Hrs/wk	4
CP	6
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56
Lecturer	Prof. Norbert Hoffmann
Language	DE/EN
Cycle	SoSe
Content	Fundamentals of Nonlinear Dynamics • One dimensional problems ◦ Linear Stability ◦ Local Bifurcations ◦ Synchronisation • Two dimensional problems ◦ Limit Cycles ◦ Global Bifurcations • Chaos ◦ Lorenz Equations ◦ Fractals and Strange Attractors ◦ Predictability and Horizons
Literature	Steven Strogatz: Nonlinear Dynamics and Chaos.

Module M1248: Compilers for Embedded Systems				
Courses				
Title	Typ		Hrs/wk	CP
Compilers for Embedded Systems (L1692)	Lecture		3	4
Compilers for Embedded Systems (L1693)	Project-/problem-based Learning		1	2
Module Responsible	Prof. Heiko Falk			
Admission Requirements	None			
Recommended Previous Knowledge	Module "Embedded Systems" C/C++ Programming skills			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	The relevance of embedded systems increases from year to year. Within such systems, the amount of software to be executed on embedded processors grows continuously due to its lower costs and higher flexibility. Because of the particular application areas of embedded systems, highly optimized and application-specific processors are deployed. Such highly specialized processors impose high demands on compilers which have to generate code of highest quality. After the successful attendance of this course, the students are able - to illustrate the structure and organization of such compilers, - to distinguish and explain intermediate representations of various abstraction levels, and - to assess optimizations and their underlying problems in all compiler phases. The high demands on compilers for embedded systems make effective code optimizations mandatory. The students learn in particular, - which kinds of optimizations are applicable at the source code level, - how the translation from source code to assembly code is performed, - which kinds of optimizations are applicable at the assembly code level, - how register allocation is performed, and - how memory hierarchies can be exploited effectively. Since compilers for embedded systems often have to optimize for multiple objectives (e.g., average- or worst-case execution time, energy dissipation, code size), the students learn to evaluate the influence of optimizations on these different criteria. <i>Skills</i> After successful completion of the course, students shall be able to translate high-level program code into machine code. They will be enabled to assess which kind of code optimization should be applied most effectively at which abstraction level (e.g., source or assembly code) within a compiler. While attending the labs, the students will learn to implement a fully functional compiler including optimizations.			
<i>Knowledge</i>				
Personal Competence				
<i>Social Competence</i>				
<i>Autonomy</i>				
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	None			
Examination	Oral exam			
Examination duration and scale	30 min			
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Information and Communication Systems: Elective Compulsory Electrical Engineering: Specialisation Information and Communication Systems: Elective Compulsory Aircraft Systems Engineering: Core Qualification: Elective Compulsory Aeronautics: Core Qualification: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory			

Course L1692: Compilers for Embedded Systems	
Typ	Lecture
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Lecturer	Prof. Heiko Falk
Language	EN
Cycle	SoSe
Content	• Introduction and Motivation • Compilers for Embedded Systems - Requirements and Dependencies • Internal Structure of Compilers • Pre-Pass Optimizations • HIR Optimizations and Transformations • Code Generation • LIR Optimizations and Transformations • Register Allocation • WCET-Aware Compilation • Outlook
Literature	• Peter Marwedel. Embedded System Design - Embedded Systems Foundations of Cyber-Physical Systems. 2nd Edition, Springer, 2012. • Steven S. Muchnick. Advanced Compiler Design and Implementation. Morgan Kaufmann, 1997. • Andrew W. Appel. Modern compiler implementation in C. Oxford University Press, 1998.

Course L1693: Compilers for Embedded Systems	
Typ	Project-/problem-based Learning
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Prof. Heiko Falk
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1968: Approximation and Stability				
Courses				
Title		Typ	Hrs/wk	CP
Approximation and Stability (L0487)		Lecture	3	4
Approximation and Stability (L0488)		Recitation Section (small)	1	2
Module Responsible	Prof. Marko Lindner			
Admission Requirements	None			
Recommended Previous Knowledge	- Linear Algebra: systems of linear equations, least squares problems, eigenvalues, singular values - Analysis: sequences, series, differentiation, integration			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> Students are able to - sketch and interrelate basic concepts of functional analysis (Hilbert space, operators), - name and understand concrete approximation methods, - name and explain basic stability theorems, - discuss spectral quantities, conditions numbers and methods of regularisation <i>Skills</i> Students are able to - apply basic results from functional analysis, - apply approximation methods, - apply stability theorems, - compute spectral quantities, - apply regularisation methods. Personal Competence <i>Social Competence</i> Students are able to solve specific problems in groups and to present their results appropriately (e.g. as a seminar presentation). <i>Autonomy</i> - Students are capable of checking their understanding of complex concepts on their own. They can specify open questions precisely and know where to get help in solving them. - Students have developed sufficient persistence to be able to work for longer periods in a goal-oriented manner on hard problems.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	None			
Examination	Oral exam			
Examination duration and scale	20 min			
Assignment for the Following Curricula	Electrical Engineering and Information Technology: Specialisation Control and Power Systems Engineering: Elective Compulsory Electrical Engineering: Specialisation Control and Power Systems Engineering: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Technomathematics: Specialisation I. Mathematics: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory			

Course L0487: Approximation and Stability	
Typ	Lecture
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Lecturer	Prof. Marko Lindner
Language	DE/EN
Cycle	SoSe
Content	This course is about solving the following basic problems of Linear Algebra, • systems of linear equations, • least squares problems, • eigenvalue problems but now in function spaces (i.e. vector spaces of infinite dimension) by a stable approximation of the problem in a space of finite dimension. Contents: • crash course on Hilbert spaces: metric, norm, scalar product, completeness • crash course on operators: boundedness, norm, compactness, projections • uniform vs. strong convergence, approximation methods • applicability and stability of approximation methods, Polski's theorem • Galerkin methods, collocation, spline interpolation, truncation • convolution and Toeplitz operators • crash course on C^*-algebras • convergence of condition numbers • convergence of spectral quantities: spectrum, eigen values, singular values, pseudospectra • regularisation methods (truncated SVD, Tichonov)
Literature	• R. Hagen, S. Roch, B. Silbermann: C^*-Algebras in Numerical Analysis • H. W. Alt: Lineare Funktionalanalysis • M. Lindner: Infinite matrices and their finite sections

Course L0488: Approximation and Stability	
Typ	Recitation Section (small)
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Prof. Marko Lindner
Language	DE/EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M2029: Process Imaging				
Courses				
Title	Typ		Hrs/wk	CP
Process Imaging (L2723)	Lecture		3	3
Applied Process Imaging (L2724)	Project-/problem-based Learning		3	3
Module Responsible	Prof. Alexander Penn			
Admission Requirements	None			
Recommended Previous Knowledge	No special prerequisites needed. An interest in imaging techniques and image processing is helpful but not mandatory.			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	The module focuses primarily on discussing established imaging techniques including (a) optical and infrared imaging, (b) magnetic resonance imaging, (c) X-ray imaging and tomography. Moreover, it presents and discusses a range of more recent imaging modalities. The students will learn: 1. what these imaging techniques can measure (such as sample density or concentration, material transport, chemical composition, temperature), 2. how the measurement techniques work (physical measurement principles, hardware requirements, image reconstruction), and 3. how to determine the most suited imaging methods for a given problem. After the successful completion of the course, the students shall: 1. understand the physical principles and practical aspects of the most common imaging methods, 2. be able to assess the pros and cons of these methods with regard to cost, complexity, expected contrasts, spatial and temporal resolution, and based on this assessment 3. be able to identify the most suited imaging modality for any specific engineering challenge in the field of chemical and bioprocess engineering.			
<i>Knowledge</i>				
<i>Skills</i>				
Personal Competence				
<i>Social Competence</i>	In the problem-based interactive course, students work in small teams and set up two process imaging systems and use these systems to measure relevant process parameters in different chemical and bioprocess engineering applications. The teamwork will foster interpersonal communication skills.			
<i>Autonomy</i>	Students are guided to work in self-motivation due to the challenge-based character of this module. A final presentation improves presentation skills.			
Workload in Hours	Independent Study Time 96, Study Time in Lecture 84			
Credit points	6			
Course achievement	None			
Examination	Subject theoretical and practical work			
Examination duration and scale	70% written examination, 30% active participation and final presentation of the problem-based learning units with a 5-10 page report			
Assignment for the Following Curricula	Bioprocess Engineering: Specialisation A - General Bioprocess Engineering: Elective Compulsory Bioprocess Engineering: Specialisation B - Industrial Bioprocess Engineering: Elective Compulsory Bioprocess Engineering: Specialisation C - Bioeconomic Process Engineering, Focus Energy and Bioprocess Technology: Elective Compulsory Chemical and Bioprocess Engineering: Specialisation Chemical and Bioprocess Engineering: Elective Compulsory Chemical and Bioprocess Engineering: Specialisation Chemical and Bioprocess Engineering: Elective Compulsory Chemical and Bioprocess Engineering: Core Qualification: Elective Compulsory Computer Science: Specialisation II: Intelligence Engineering: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Signal Processing: Elective Compulsory International Management and Engineering: Specialisation II. Process Engineering and Biotechnology: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory Process Engineering: Specialisation Process Engineering: Elective Compulsory Process Engineering: Specialisation Chemical Process Engineering: Elective Compulsory Process Engineering: Specialisation Environmental Process Engineering: Elective Compulsory			

Course L2723: Process Imaging	
Typ	Lecture
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Lecturer	Prof. Alexander Penn
Language	EN
Cycle	SoSe
Content	The lecture focuses primarily on presenting and discussing established imaging techniques relevant to the field of engineering including (a) optical and infrared imaging, (b) magnetic resonance imaging, (c) X-ray imaging and tomography. Moreover, it presents and discusses a range of more recent imaging modalities. The students will learn: 1. what these imaging techniques can measure (such as sample density or concentration, material transport, chemical composition, temperature), 2. how the measurement techniques work (physical measurement principles, hardware requirements, image reconstruction), and 3. how to determine the most suited imaging methods for a given problem.
Literature	Wang, M. (2015). Industrial Tomography. Cambridge, UK: Woodhead Publishing. Available as e-book in the library of TUHH: https://katalog.tub.tuhh.de/Record/823579395

Course L2724: Applied Process Imaging	
Typ	Project-/problem-based Learning
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Lecturer	Prof. Alexander Penn, Dr. Stefan Benders
Language	EN
Cycle	SoSe
Content	Content: The module focuses primarily on discussing established imaging techniques including (a) optical and infrared imaging, (b) magnetic resonance imaging, (c) X-ray imaging and tomography, and (d) ultrasound imaging and also covers a range of more recent imaging modalities. The students will learn: 1. what these imaging techniques can measure (such as sample density or concentration, material transport, chemical composition, temperature), 2. how the measurements work (physical measurement principles, hardware requirements, image reconstruction), and 3. how to determine the most suited imaging methods for a given problem. Learning goals: After the successful completion of the course, the students shall: 1. understand the physical principles and practical aspects of the most common imaging methods, 2. be able to assess the pros and cons of these methods with regard to cost, complexity, expected contrasts, spatial and temporal resolution, and based on this assessment 3. be able to identify the most suited imaging modality for any specific engineering challenge in the field of chemical and bioprocess engineering.
Literature	Wang, M. (2015). Industrial Tomography. Cambridge, UK: Woodhead Publishing. Available as e-book in the library of TUHH: https://katalog.tub.tuhh.de/Record/823579395

Module M2035: Mobile Robotics				
Courses				
Title	Typ		Hrs/wk	CP
Applied Automation (L1592)	Project-/problem-based Learning		3	3
Mobile Robotics (L3286)	Lecture		2	2
Mobile Robotics (L3287)	Recitation Section (small)		1	1
Module Responsible	Prof. Thorsten Schüppstuhl			
Admission Requirements	None			
Recommended Previous Knowledge	Robotics, computer science, measurement technology, control engineering			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> Students can name examples and applications of mobile robotics. Students can name procedures and methods of typical elements of mobile robotics for both hardware and software components (e.g. localization with LiDaRs, mapping, path planning). Students can name the common tools for collaborative working on a task in this field. <i>Skills</i> Students can evaluate and compare procedures and methods of typical elements of mobile robotics for both hardware and software components in relation to a task. Students can select a suitable procedure or method and apply it to a task. Students are able to work in groups on research and application-related tasks, develop solution strategies independently, define work processes and work on them collaboratively with the aid and use of common tools. Personal Competence <i>Social Competence</i> Students are able to work goal-oriented in small mixed groups represent the results of their work. <i>Autonomy</i> Students are able to recognize and improve knowledge deficits independently. With instructor assistance, students are able to evaluate their own knowledge level and define a further course of action.			
Workload in Hours	Independent Study Time 96, Study Time in Lecture 84			
Credit points	6			
Course achievement	None			
Examination	Subject theoretical and practical work			
Examination duration and scale	30 min			
Assignment for the Following Curricula	Mechanical Engineering - Product Development and Production: Specialisation Product Development: Elective Compulsory Mechanical Engineering - Product Development and Production: Specialisation Materials: Elective Compulsory Mechanical Engineering - Product Development and Production: Specialisation Production: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Product Development, Materials and Production: Specialisation Product Development: Elective Compulsory Product Development, Materials and Production: Specialisation Production: Elective Compulsory Product Development, Materials and Production: Specialisation Materials: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory			

Course L1592: Applied Automation	
Typ	Project-/problem-based Learning
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Lecturer	Prof. Thorsten Schüppstuhl
Language	DE
Cycle	WiSe
Content	-Project Based Learning -Robot Operating System -Robot structure and description -Motion description -Calibration -Accuracy
Literature	John J. Craig Introduction to Robotics - Mechanics and Control ISBN: 0131236296 Pearson Education, Inc., 2005 Stefan Hesse Grundlagen der Handhabungstechnik ISBN: 3446418725 München Hanser, 2010 K. Thulasiraman and M. N. S. Swamy Graphs: Theory and Algorithms ISBN: 9781118033104 %CITAVIPICKER£9781118033104£Titel anhand dieser ISBN in Citavi-Projekt übernehmen£% John Wiley & Sons, Inc., 1992

Course L3286: Mobile Robotics	
Typ	Lecture
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Thorsten Schüppstuhl
Language	DE
Cycle	SoSe
Content	Examples and applications Hardware: Sensors, actuators and control hardware Concepts: Environmental perception, robot control architectures, autonomous behavior Methods: Localization, mapping, path & motion planning, navigation & obstacle avoidance Actions: Autonomous movement execution and manipulation
Literature	Siciliano, Bruno, and Oussama Khatib. "Robotics and the Handbook." Springer Handbook of Robotics. Cham: Springer International Publishing, 2016. Hertzberg, Joachim, Kai Lingemann, and Andreas Nüchter. Mobile Roboter: Eine Einführung aus Sicht der Informatik. Springer-Verlag, 2012. Thrun, Sebastian, Wolfram Burgard, and Dieter Fox. Probabilistic Robotics. MIT Press, 2005.

Course L3287: Mobile Robotics	
Typ	Recitation Section (small)
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Prof. Thorsten Schüppstuhl
Language	DE
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M2036: Advanced Topics in Production Engineering				
Courses				
Title		Typ	Hrs/wk	CP
Advanced Topics in Production Engineering (L3288)		Project-/problem-based Learning	4	6
Module Responsible	Dr. Martin Gomse			
Admission Requirements	None			
Recommended Previous Knowledge	Depending on the type of project: e.g. Robotics, computer science, measurement technology, control engineering, machine learning			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence <i>Knowledge</i> <i>Skills</i> <				

Course L3288: Advanced Topics in Production Engineering	
Typ	Project-/problem-based Learning
Hrs/wk	4
CP	6
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56
Lecturer	Dr. Martin Gomse
Language	DE/EN
Cycle	WiSe/SoSe
Content	Advanced and Research Topics in Production Engineering
Literature	Aktuelle Veröffentlichungen / Recent publications

Module M0714: Numerical Methods for Ordinary Differential Equations					
Courses					
Title		Typ	Hrs/wk	CP	
Numerical Treatment of Ordinary Differential Equations (L0576)		Lecture	2	3	
Numerical Treatment of Ordinary Differential Equations (L0582)		Recitation Section (small)	2	3	
Module Responsible	Prof. Daniel Ruprecht				
Admission Requirements	None				
Recommended Previous Knowledge	- Mathematik I, II, III for Engineers (German or English) or Analysis & Linear Algebra I + II plus Analysis III for Technomathematiker. - Basic knowledge of MATLAB, Python or a similar programming language.				
Educational Objectives	After taking part successfully, students have reached the following learning results				
Professional Competence					
Knowledge					Students are able to - name numerical methods for the solution of ordinary differential equations and explain their core ideas, - formulate convergence statements for the taught numerical methods (including the necessary assumptions about the solved problem), - explain aspects regarding the practical realisation of a method, - select the appropriate numerical method for specific problems, implement the numerical algorithms efficiently and interpret the numerical results.
Skills					Students are able to - implement, apply and compare numerical methods for the solution of ordinary differential equations, - explain the convergence behaviour of numerical methods, taking into consideration the solved problem and selected algorithm, - develop a suitable solution approach for a given problem, if necessary by combining multiple algorithms, realise this approach and critically evaluate results.
Personal Competence					
Social Competence	Students are able to work together in heterogeneous teams (i.e., teams from different study programs and with different background knowledge), explain theoretical foundations and support each other with practical aspects regarding the implementation of algorithms.				
Autonomy	Students are capable - to assess whether the provided theoretical and practical exercises are better solved individually or in a team and - to assess their individual progress and, if necessary, to ask questions and seek help.				
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56				
Credit points	6				
Course achievement	None				
Examination	Written exam				
Examination duration and scale	90 min				
Assignment for the Following Curricula	Bioprocess Engineering: Specialisation A - General Bioprocess Engineering: Elective Compulsory Chemical and Bioprocess Engineering: Technical Complementary Course: Elective Compulsory Chemical and Bioprocess Engineering: Technical Complementary Course: Elective Compulsory Computational Engineering: Core Qualification: Elective Compulsory Computer Science: Specialisation III. Mathematics: Elective Compulsory Data Science: Specialisation I. Data Science & Mathematics: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Control and Power Systems Engineering: Elective Compulsory Electrical Engineering: Specialisation Control and Power Systems Engineering: Elective Compulsory Energy Systems: Core Qualification: Elective Compulsory Aircraft Systems Engineering: Core Qualification: Elective Compulsory Interdisciplinary Mathematics: Specialisation II. Numerical - Modelling Training: Compulsory Mechatronics: Core Qualification: Elective Compulsory Technomathematics: Specialisation I. Mathematics: Elective Compulsory Theoretical Mechanical Engineering: Core Qualification: Compulsory Process Engineering: Specialisation Chemical Process Engineering: Elective Compulsory Process Engineering: Specialisation Process Engineering: Elective Compulsory				

Course L0576: Numerical Treatment of Ordinary Differential Equations	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Daniel Ruprecht
Language	DE/EN
Cycle	SoSe
Content	Numerical methods for Initial Value Problems • single step methods • multistep methods • stiff problems • differential algebraic equations (DAE) of index 1 Numerical methods for Boundary Value Problems • multiple shooting method • difference methods
Literature	• E. Hairer, S. Nørsett, G. Wanner: Solving Ordinary Differential Equations I: Nonstiff Problems. • E. Hairer, G. Wanner: Solving Ordinary Differential Equations II: Stiff and Differential-Algebraic Problems. • D. Griffiths, D. Higham: Numerical Methods for Ordinary Differential Equations.

Course L0582: Numerical Treatment of Ordinary Differential Equations	
Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Daniel Ruprecht
Language	DE/EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M2061: From Data to Models (D2M)				
Courses				
Title	Typ		Hrs/wk	CP
From Data to Models (D2M) (L3328)	Lecture		2	4
From Data to Models (D2M) (L3327)	Recitation Section (large)		1	2
Module Responsible	Prof. Timm Faulwasser			
Admission Requirements	None			
Recommended Previous Knowledge	Basics of control engineering (suggested), differential equations (suggested), basic understanding of modelling techniques from atleast one engineering science (EE, CE, ME, ...)			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> The students learn how to use data to formulate models of dynamic systems. They know how to use linear and nonlienar methods appropriately. They know how to estimate parameters for models based on first principles. <i>Skills</i> Identifcation of state-space models for linear systems via classic system identification and subapce methods. The students are able to use nonlinear kernel regression for dynamic systems as well as Gaussian process regresson. The students also know how to modeldynamics with neural networks. <i>Personal Competence</i> <i>Social Competence</i> The students are able to collaborate across disciplinary and culturalboundaries. <i>Autonomy</i> The students are able to formulate questions for complex topics.They are able to implement the discussed methods numerically.			
Workload in Hours	Independent Study Time 138, Study Time in Lecture 42			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	No	20 %	Midterm	
Examination	Written exam			
Examination duration and scale	90 min			
Assignment for the Following Curricula	Electrical Engineering and Information Technology: Specialisation Control and Power Systems Engineering: Elective Compulsory Electrical Engineering: Specialisation Control and Power Systems Engineering: Elective Compulsory Energy Systems: Core Qualification: Elective Compulsory Mechanical Engineering - Product Development and Production: Specialisation Product Development: Elective Compulsory Mechanical Engineering - Product Development and Production: Specialisation Production: Elective Compulsory Mechanical Engineering - Product Development and Production: Specialisation Materials: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Product Development, Materials and Production: Specialisation Product Development: Elective Compulsory Theoretical Mechanical Engineering: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Core Qualification: Elective Compulsory			

Course L3328: From Data to Models (D2M)	
Typ	Lecture
Hrs/wk	2
CP	4
Workload in Hours	Independent Study Time 92, Study Time in Lecture 28
Lecturer	Prof. Timm Faulwasser
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Course L3327: From Data to Models (D2M)	
Typ	Recitation Section (large)
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Prof. Timm Faulwasser
Language	EN
Cycle	SoSe
Content	See modul description
Literature	

Module M2067: Control Lab A				
Courses				
Title		Typ	Hrs/wk	CP
Control Lab A (L3336)		Practical Course	4	4
Module Responsible	Prof. Timm Faulwasser			
Admission Requirements	None			
Recommended Previous Knowledge	Basics of Control Engineering, Control System Theory and Design			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	Knowledge In the control lab, students can test out the knowledge acquired in the control engineering lectures on real test systems in the laboratory of the Institute of Control Systems. Different experiments are offered every semester. Various linear and nonlinear methods can be tested. Under flatness-based feedforward, LQR, MPC and passivity-based design methods Skills The students know how to implement different designs on laboratory hardware. They are familiar with state esimtaiotn concepts and tune their controllers according to given speifications. Personal Competence Social Competence The students can collaborate across boundaries of disciplinesand in international teams. Autonomy The student can formulate questions and problems with respect to complex issues. They can program selected techniques on their own in Matlab & Python.			
Workload in Hours	Independent Study Time 64, Study Time in Lecture 56			
Credit points	4			
Course achievement	None			
Examination	Subject theoretical and practical work			
Examination duration and scale	60 minutes			
Assignment for the Following Curricula	Electrical Engineering and Information Technology: Specialisation Control and Power Systems Engineering: Elective Compulsory Electrical Engineering: Specialisation Control and Power Systems Engineering: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory			
Course L3336: Control Lab A				
Typ	Practical Course			
Hrs/wk	4			
CP	4			
Workload in Hours	Independent Study Time 64, Study Time in Lecture 56			
Lecturer	Prof. Timm Faulwasser			
Language	EN			
Cycle	WiSe/SoSe			
Content	Four experiments in control theory.			
Literature	Experiment Guides			

Module M2068: Control Lab B				
Courses				
Title	Typ		Hrs/wk	CP
Control Lab B (L3337)	Practical Course		3	3
Module Responsible	Prof. Timm Faulwasser			
Admission Requirements	None			
Recommended Previous Knowledge	Basics of Control Engineering, Control System Theory and Design			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> In the control lab, students can test out the knowledge acquired in the control engineering lectures on real test systems in the laboratory of the Institute of Control Systems. Different experiments are offered every semester. Various linear and nonlinear methods can be tested. Under flatness-based feedforward, LQR, MPC and passivity-based design methods <i>Skills</i> The students know how to implement different designs on laboratory hardware. They are familiar with state estimation concepts and tune their controllers according to given specifications. <i>Personal Competence</i> <i>Social Competence</i> The students can collaborate across boundaries of disciplines and in international teams. <i>Autonomy</i> The student can formulate questions and problems with respect to complex issues. They can program selected techniques on their own in Matlab & Python.			
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42			
Credit points	3			
Course achievement	None			
Examination	Subject theoretical and practical work			
Examination duration and scale	60 minutes			
Assignment for the Following Curricula	Electrical Engineering and Information Technology: Specialisation Control and Power Systems Engineering: Elective Compulsory Electrical Engineering: Specialisation Control and Power Systems Engineering: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Core Qualification: Elective Compulsory			

Course L3337: Control Lab B	
Typ	Practical Course
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Lecturer	Prof. Timm Faulwasser
Language	EN
Cycle	WiSe/SoSe
Content	Three experiments in control theory.
Literature	Experiment Guides

Module M2069: Control Lab C			
Courses			
Title Control Lab C (L3338)	Typ Practical Course	Hrs/wk 2	CP 2
Module Responsible	Prof. Timm Faulwasser		
Admission Requirements	None		
Recommended Previous Knowledge	Basics of Control Engineering, Control System Theory and Design		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence	<i>Knowledge</i> In the control lab, students can test out the knowledge acquired in the control engineering lectures on real test systems in the laboratory of the Institute of Control Systems. Different experiments are offered every semester. Various linear and nonlinear methods can be tested. Under flatness-based feedforward, LQR, MPC and passivity-based design methods <i>Skills</i> The students know how to implement different designs on laboratory hardware. They are familiar with state estimation concepts and tune their controllers according to given specifications. <i>Personal Competence</i> <i>Social Competence</i> The students can collaborate across boundaries of disciplines and in international teams. <i>Autonomy</i> The student can formulate questions and problems with respect to complex issues. They can program selected techniques on their own in Matlab & Python.		
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28		
Credit points	2		
Course achievement	None		
Examination	Subject theoretical and practical work		
Examination duration and scale	60 minutes		
Assignment for the Following Curricula	Electrical Engineering and Information Technology: Specialisation Control and Power Systems Engineering: Elective Compulsory Electrical Engineering: Specialisation Control and Power Systems Engineering: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory		

Course L3338: Control Lab C	
Typ	Practical Course
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Timm Faulwasser
Language	EN
Cycle	WiSe/SoSe
Content	
Literature	

Module M2081: Network Control Systems				
Courses				
Title	Typ		Hrs/wk	CP
Network Control Systems (L3447)	Lecture		2	4
Network Control Systems (L3448)	Recitation Section (large)		1	2
Module Responsible	Prof. Timm Faulwasser			
Admission Requirements	None			
Recommended Previous Knowledge	Linear algebra, Basics of control engineering, differential equations			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	The students learn basic techniques of network control systems. We investigate analysis techniques for consensus problems and synchronization problems. Examples from engineering and science will be analyzed. The students can analyze consensus problems in discrete and continuous time. They can formulate and solve synchronization problems for homogenous and heterogenous sets of LT agents. The students know the basic of analysing nonlinear network control problems The students can collaborate across boundaries of disciplines and in international teams. The student can formulate questions and problems with respect to complex issues. They can program selected techniques on their own in Python.			
<i>Knowledge</i>				
<i>Skills</i>				
Personal Competence				
<i>Social Competence</i>				
<i>Autonomy</i>				
Workload in Hours	Independent Study Time 138, Study Time in Lecture 42			
Credit points	6			
Course achievement	None			
Examination	Oral exam			
Examination duration and scale	30 min			
Assignment for the Following Curricula	Mechatronics: Core Qualification: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory			

Course L3447: Network Control Systems	
Typ	Lecture
Hrs/wk	2
CP	4
Workload in Hours	Independent Study Time 92, Study Time in Lecture 28
Lecturer	Prof. Timm Faulwasser
Language	EN
Cycle	SoSe
Content	see Module description
Literature	

Course L3448: Network Control Systems	
Typ	Recitation Section (large)
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Prof. Timm Faulwasser
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M2082: Nonlinear Control Systems				
Courses				
Title	Typ		Hrs/wk	CP
Nonlinear Control Systems (L3451)	Lecture		2	4
Nonlinear Control Systems (L3452)	Recitation Section (large)		1	2
Module Responsible	Prof. Timm Faulwasser			
Admission Requirements	None			
Recommended Previous Knowledge	Basics of Control Engineering, Control System Theory and Design			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> In the lecture, students are taught methods for analyzing and controlling nonlinear systems. This includes nonlinear normal forms, zero dynamics, differential flatness, stability, controllability, observability, Lyapunov functions, dissipativity and passivity. The basics of port-Hamiltonian systems are also discussed using examples from technical applications. <i>Skills</i> The students can analyze nonlinear control systems for stability, passivity and dissipativity. They can use nonlinear control design techniques such as flatness-based control, back-stepping, passivity-based feedback. They can transform nonlinear systems into structured representations such as the Byrnes-Isidori normal form. They can analyze and model simple port-Hamiltonian systems. Personal Competence <i>Social Competence</i> The students can collaborate across boundaries of disciplines and in international teams. <i>Autonomy</i> The student can formulate questions and problems with respect to complex issues. They can program selected techniques on their own in Matlab & Python.			
Workload in Hours	Independent Study Time 138, Study Time in Lecture 42			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	Yes	20 %	Written elaboration	Durchführung und Dokumentation eines Projekts in Berichtsform
Examination	Oral exam			
Examination duration and scale	30 min			
Assignment for the Following Curricula	Mechanical Engineering - Product Development and Production: Specialisation Product Development: Elective Compulsory Mechanical Engineering - Product Development and Production: Specialisation Production: Elective Compulsory Mechanical Engineering - Product Development and Production: Specialisation Materials: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Product Development, Materials and Production: Specialisation Product Development: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory			

Course L3451: Nonlinear Control Systems	
Typ	Lecture
Hrs/wk	2
CP	4
Workload in Hours	Independent Study Time 92, Study Time in Lecture 28
Lecturer	Prof. Timm Faulwasser
Language	EN
Cycle	SoSe
Content	see Module description
Literature	

Course L3452: Nonlinear Control Systems	
Typ	Recitation Section (large)
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Prof. Timm Faulwasser
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1211: Research Project Mechatronics				
Courses				
Title	Typ		Hrs/wk	CP
Module Responsible	Dozenten des Studiengangs			
Admission Requirements	None			
Recommended Previous Knowledge	Subjects of the program of studies.			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	The students are able to demonstrate their detailed knowledge in the field of mechatronics engineering. They can exemplify the state of technology and application and discuss critically in the context of actual problems and general conditions of science and society. The students can develop solving strategies and approaches for fundamental and practical problems in mechatronics engineering. They may apply theory based procedures and integrate safety-related, ecological, ethical, and economic view points of science and society. Scientific work techniques that are used can be described and critically reviewed. The students are able to independently select methods for the project work and to justify this choice. They can explain how these methods relate to the field of work and how the context of application has to be adjusted. General findings and further developments may essentially be outlined. The students are able to condense the relevance and the structure of the project work, the work steps and the sub-problems for the presentation and discussion in front of a bigger group. They can lead the discussion and give a feedback on the project to their colleagues. The students are capable of independently planning and documenting the work steps and procedures while considering the given deadlines. This includes the ability to accurately procure the newest scientific information. Furthermore, they can obtain feedback from experts with regard to the progress of the work, and to accomplish results on the state of the art in science and technology.			
<i>Knowledge</i>				
<i>Skills</i>				
Personal Competence				
<i>Social Competence</i>				
<i>Autonomy</i>				
Workload in Hours	Independent Study Time 360, Study Time in Lecture 0			
Credit points	12			
Course achievement	None			
Examination	Study work			
Examination duration and scale	It. FSPO			
Assignment for the Following Curricula	Mechatronics: Core Qualification: Compulsory			

Module M0836: Communication Networks								
Courses								
Title		Typ	Hrs/wk	CP				
Selected Topics of Communication Networks (L0899)		Project-/problem-based Learning	2	2				
Communication Networks (L0897)		Lecture	2	2				
Communication Networks Excercise (L0898)		Project-/problem-based Learning	1	2				
Module Responsible	Dr. Koojana Kuladinithi							
Admission Requirements	None							
Recommended Previous Knowledge	- Fundamental stochastics - Basic understanding of computer networks and/or communication technologies is beneficial							
Educational Objectives	After taking part successfully, students have reached the following learning results							
Professional Competence	Knowledge Students are able to describe the principles and structures of communication networks in detail. They can explain the formal description methods of communication networks and their protocols. They are able to explain how current and complex communication networks work and describe the current research in these examples. Skills Students are able to evaluate the performance of communication networks using the learned methods. They are able to work out problems themselves and apply the learned methods. They can apply what they have learned autonomously on further and new communication networks. Personal Competence Social Competence Students are able to define tasks themselves in small teams and solve these problems together using the learned methods. They can present the obtained results. They are able to discuss and critically analyse the solutions. Autonomy Students are able to obtain the necessary expert knowledge for understanding the functionality and performance capabilities of new communication networks independently.							
Workload in Hours					Independent Study Time 110, Study Time in Lecture 70			
Credit points					6			
Course achievement					None			
Examination					Presentation			
Examination duration and scale	1.5 hours colloquium with three students, therefore about 30 min per student. Topics of the colloquium are the posters from the previous poster session and the topics of the module.							
Assignment for the Following Curricula	Electrical Engineering and Information Technology: Specialisation Information and Communication Systems: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Control and Power Systems Engineering: Elective Compulsory Electrical Engineering: Specialisation Information and Communication Systems: Elective Compulsory Electrical Engineering: Specialisation Control and Power Systems Engineering: Elective Compulsory Aircraft Systems Engineering: Core Qualification: Elective Compulsory Computer Science in Engineering: Specialisation I. Computer Science: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems, Focus Networks: Elective Compulsory International Management and Engineering: Specialisation II. Information Technology: Elective Compulsory Aeronautics: Core Qualification: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Microelectronics and Microsystems: Specialisation Communication and Signal Processing: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory							

Course L0899: Selected Topics of Communication Networks	
Typ	Project-/problem-based Learning
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Dr. Koojana Kuladinithi
Language	EN
Cycle	WiSe
Content	Example networks selected by the students will be researched on in a PBL course by the students in groups and will be presented in a poster session at the end of the term.
Literature	see lecture

Course L0897: Communication Networks	
Typ	Lecture
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Dr. Koojana Kuladinithi
Language	EN
Cycle	WiSe
Content	consisting of 3 modules. - Lecture on the analysis of communication networks - Introduction to Communication Networks - Evolution of PSTN and Blocking Probability - TCP/IP Based Networks and Mobile Cellular Networks - Point-to-Point Communication - Internet - Part 1: Network Layer & Part 2: Transport Layer - Local Area Networks - Ethernet - Wireless Local Area Networks - PBL Exercises of communication networks - Performance Metrics - Trial & Error vs. Engineering Approach - Reliable Medium Access Control - Mini poster: Fast and Reliable End-to-End Data Transmission - Selected topics in communication networks This is organised as a Poster presentation of the PBL5 done for the examination. The students are asked to work on a selected topic in communication networks. This topic is selected based on the current related topic.
Literature	The students should have the basic knowledge of communication networks. Lecture 1 explains some terminologies in brief. The following book references are recommended. - Keith W. Ross James F. Kurose. Computer Networking - A Top-Down Approach. 7th ed. Pearson, 2017. A. S. Tanenbaum. Computer Networks. 5th ed. Prentice Hall, 2011. - Schiller: Mobile Communications, 8th edition. - Internet, Wikipedia

Course L0898: Communication Networks Exercise	
Typ	Project-/problem-based Learning
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Dr. Koojana Kuladinithi
Language	EN
Cycle	WiSe
Content	Part of the content of the lecture Communication Networks are reflected in computing tasks in groups, others are motivated and addressed in the form of a PBL exercise.
Literature	announced during lecture

Module M1598: Image Processing				
Courses				
Title	Typ		Hrs/wk	CP
Image Processing (L2443)	Lecture		2	4
Image Processing (L2444)	Recitation Section (small)		2	2
Module Responsible	Prof. Tobias Knopp			
Admission Requirements	None			
Recommended Previous Knowledge	Signal and Systems			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> The students know about • visual perception • multidimensional signal processing • sampling and sampling theorem • filtering • image enhancement • edge detection • multi-resolution procedures: Gauss and Laplace pyramid, wavelets • image compression • image segmentation • morphological image processing <i>Skills</i> The students can • analyze, process, and improve multidimensional image data • implement simple compression algorithms • design custom filters for specific applications Personal Competence <i>Social Competence</i> Students can work on complex problems both independently and in teams. They can exchange ideas with each other and use their individual strengths to solve the problem. <i>Autonomy</i> Students are able to independently investigate a complex problem and assess which competencies are required to solve it.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	None			
Examination	Written exam			
Examination duration and scale	90 min			
Assignment for the Following Curricula	Computer Science: Specialisation II: Intelligence Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Data Science: Specialisation I. Mathematics/Computer Science: Elective Compulsory Data Science: Specialisation IV. Special Focus Area: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Information and Communication Systems: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Medical Technology: Elective Compulsory Electrical Engineering: Specialisation Information and Communication Systems: Elective Compulsory Electrical Engineering: Specialisation Medical Technology: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Signal Processing: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems, Focus Software and Signal Processing: Elective Compulsory International Management and Engineering: Specialisation II. Information Technology: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Microelectronics and Microsystems: Specialisation Communication and Signal Processing: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory			

Course L2443: Image Processing	
Typ	Lecture
Hrs/wk	2
CP	4
Workload in Hours	Independent Study Time 92, Study Time in Lecture 28
Lecturer	Prof. Tobias Knopp
Language	EN
Cycle	WiSe
Content	• Visual perception • Multidimensional signal processing • Sampling and sampling theorem • Filtering • Image enhancement • Edge detection • Multi-resolution procedures: Gauss and Laplace pyramid, wavelets • Image Compression • Segmentation • Morphological image processing
Literature	Bredies/Lorenz, Mathematische Bildverarbeitung, Vieweg, 2011 Pratt, Digital Image Processing, Wiley, 2001 Bernd Jähne: Digitale Bildverarbeitung - Springer, Berlin 2005

Course L2444: Image Processing	
Typ	Recitation Section (small)
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Tobias Knopp
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M0677: Digital Signal Processing and Digital Filters			
Courses			
Title	Typ	Hrs/wk	CP
Digital Signal Processing and Digital Filters (L0446)	Lecture	3	4
Digital Signal Processing and Digital Filters (L0447)	Recitation Section (large)	2	2
Module Responsible	Prof. Gerhard Bauch		
Admission Requirements	None		
Recommended Previous Knowledge	- Mathematics 1-3 - Signals and Systems - Fundamentals of signal and system theory as well as random processes. - Fundamentals of spectral transforms (Fourier series, Fourier transform, Laplace transform)		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence	<i>Knowledge</i> The students know and understand basic algorithms of digital signal processing. They are familiar with the spectral transforms of discrete-time signals and are able to describe and analyse signals and systems in time and image domain. They know basic structures of digital filters and can identify and assess important properties including stability. They are aware of the effects caused by quantization of filter coefficients and signals. They are familiar with the basics of adaptive filters. They can perform traditional and parametric methods of spectrum estimation, also taking a limited observation window into account. The students are familiar with the contents of lecture and tutorials. They can explain and apply them to new problems. <i>Skills</i> The students are able to apply methods of digital signal processing to new problems. They can choose and parameterize suitable filter structures. In particular, they can design adaptive filters according to the minimum mean squared error (MMSE) criterion and develop an efficient implementation, e.g. based on the LMS or RLS algorithm. Furthermore, the students are able to apply methods of spectrum estimation and to take the effects of a limited observation window into account. Personal Competence <i>Social Competence</i> The students can jointly solve specific problems. <i>Autonomy</i> The students are able to acquire relevant information from appropriate literature sources. They can control their level of knowledge during the lecture period by solving tutorial problems, software tools, clicker system.		
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70		
Credit points	6		
Course achievement	None		
Examination	Written exam		
Examination duration and scale	90 min		
Assignment for the Following Curricula	Electrical Engineering and Information Technology: Specialisation Control and Power Systems Engineering: Elective Compulsory Electrical Engineering: Specialisation Control and Power Systems Engineering: Elective Compulsory Computer Science in Engineering: Specialisation II. Engineering Science: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Signal Processing: Elective Compulsory Mechanical Engineering and Management: Specialisation Mechatronics: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Microelectronics and Microsystems: Specialisation Communication and Signal Processing: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory		

Course L0446: Digital Signal Processing and Digital Filters	
Typ	Lecture
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Lecturer	Prof. Gerhard Bauch
Language	EN
Cycle	WiSe
Content	- Transforms of discrete-time signals: - Discrete-time Fourier Transform (DTFT) - Discrete Fourier-Transform (DFT), Fast Fourier Transform (FFT) - Z-Transform - Correspondence of continuous-time and discrete-time signals, sampling, sampling theorem - Fast convolution, Overlap-Add-Method, Overlap-Save-Method - Fundamental structures and basic types of digital filters - Characterization of digital filters using pole-zero plots, important properties of digital filters - Quantization effects - Design of linear-phase filters - Fundamentals of stochastic signal processing and adaptive filters - MMSE criterion - Wiener Filter - LMS- and RLS-algorithm - Traditional and parametric methods of spectrum estimation
Literature	K.-D. Kammeyer, K. Kroschel: Digitale Signalverarbeitung. Vieweg Teubner. V. Oppenheim, R. W. Schaffer, J. R. Buck: Zeitdiskrete Signalverarbeitung. Pearson StudiumA. V. W. Hess: Digitale Filter. Teubner. Oppenheim, R. W. Schaffer: Digital signal processing. Prentice Hall. S. Haykin: Adaptive filter theory. L. B. Jackson: Digital filters and signal processing. Kluwer. T.W. Parks, C.S. Burrus: Digital filter design. Wiley.

Course L0447: Digital Signal Processing and Digital Filters	
Typ	Recitation Section (large)
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Gerhard Bauch
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M1749: Energy Efficiency in Embedded Systems			
Courses			
Title	Typ	Hrs/wk	CP
Energy Efficiency in Embedded Systems (L2870)	Lecture	2	3
Energy Efficiency in Embedded Systems (L2872)	Project/problem-based Learning	2	2
Energy Efficiency in Embedded Systems (L2871)	Recitation Section (large)	1	1
Module Responsible	Prof. Ulf Kulau		
Admission Requirements	None		
Recommended Previous Knowledge	• Computer Engineering (mandatory) • Programming Skills in C (mandatory) • Computer Architecture (recommended) ENG: The following books are very good sources for the necessary background knowledge: 1. Harris, David, and N. Weste: CMOS VLSI Design ed., Pearson Education, 2010 2. Rabaey, Jan: Low Power Design Essentials (Integrated Circuits and Systems), Springer, 2009		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence <i>Knowledge</i>	Motivation: In the field of computer science we have only limited possibilities to influence the efficiency of the hardware directly, respectively we are dependent on the manufacturers (e.g. of microcontrollers). However, in order to exploit the full potential of the hardware we are given at the system level, we need a deeper understanding of the background, processes and mechanisms of power dissipation in embedded systems. Where does the power dissipation come from, what happens at the hardware level, what mechanisms can I use directly/indirectly, what is the tradeoff between flexibility and efficiency,..... are only a few questions, which will be elaborated and discussed in this event. Contents of teaching: • Motivation and power dissipation on semiconductor level • Power dissipation of digital circuits, in particular CMOS • Power Management in Hard- and Software (Sleep Modes, DVS, FS, Undervolting) • Energy efficient system design (applications) • Energy Harvesting and Transiently Powered Computing (TPC)		
<i>Skills</i>	Upon completion of this module, students will have a deeper understanding of hardware and software mechanisms for evaluating and developing energy-efficient embedded systems • They have a deeper understanding of the electrotechnical basics of power dissipation in digital systems • They can analyze the power dissipation of systems at any level and apply appropriate methods to increase efficiency • They can use a variety of standard techniques to achieve "Energy Efficiency by Design" • They can model, evaluate as well as implement energy-autonomous systems		
Personal Competence <i>Social Competence</i>	As part of the module, concepts learned in the lecture will be implemented on a hardware platform within small groups. Students learn to work in a team and to develop solutions together. Specific tasks are worked on within the group, whereby cross-group collaboration (exchange) also takes place. The second part is a challenge-based project in which the groups find the most energy-efficient solutions possible in healthy competition with each other. This strengthens the cohesion in the groups and reinforces mutual motivation, support and creativity.		
<i>Autonomy</i>	After completing this module, students will be able to independently develop, optimize and evaluate solutions for embedded systems based on the knowledge they have acquired and further technical literature.		
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70		
Credit points	6		
Course achievement	None		
Examination	Written exam		
Examination duration and scale	90 min		
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Nanoelectronics and Microsystems Technology: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Wireless and Sensor Technologies: Elective Compulsory Electrical Engineering: Specialisation Wireless and Sensor Technologies: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Signal Processing: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Microelectronics and Microsystems: Specialisation Embedded Systems: Elective Compulsory		

Course L2870: Energy Efficiency in Embedded Systems	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Ulf Kulau
Language	EN
Cycle	WiSe
Content	Motivation: In the field of computer science we have only limited possibilities to influence the efficiency of the hardware directly, respectively we are dependent on the manufacturers (e.g. of microcontrollers). However, in order to exploit the full potential of the hardware we are given at the system level, we need a deeper understanding of the background, processes and mechanisms of power dissipation in embedded systems. Where does the power dissipation come from, what happens at the hardware level, what mechanisms can I use directly/indirectly, what is the tradeoff between flexibility and efficiency,..... are only a few questions, which will be elaborated and discussed in this event. Contents of teaching: • Motivation and power dissipation on semiconductor level • Power dissipation of digital circuits, in particular CMOS • Power Management in Hard- and Software (Sleep Modes, DVS, FS, Undervolting) • Energy efficient system design (applications) • Energy Harvesting and Transiently Powered Computing (TPC)
Literature	DE: Die Vorlesungsinhalte wurden in einem Paper veröffentlicht. All Quellen, welche in diesem Kurs verwendet wurden sind in folgenden Paper aufgelistet: ENG: The lecture contents were published in a paper. All sources used in this course are listed in the following paper: 1. Kulau, Ulf: Course: Energy Efficiency in Embedded Systems-A System-Level Perspective for Computer Scientists, EWME, 2018. DE: Als sehr gute Quellen für notwendiges Hintergrundwissen eignen sich die folgenden Bücher: ENG: The following books are very good sources for the necessary background knowledge: 1. Harris, David, and N. Weste: CMOS VLSI Design ed., Pearson Education, 2010 2. Rabaey, Jan: Low Power Design Essentials (Integrated Circuits and Systems), Springer, 2009

Course L2872: Energy Efficiency in Embedded Systems	
Typ	Project-/problem-based Learning
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Ulf Kulau
Language	EN
Cycle	WiSe
Content	In this project-based exercise, the learned aspects for achieving energy-efficient embedded systems are implemented and consolidated in practical environments in a small project. First, a tool set for the implementation of energy efficiency mechanisms is implemented in common exercises by means of defined tasks. In the second part, a challenge-based exercise is carried out in which a system that is as efficient as possible is to be implemented independently. A system based on an AVR micro-controller is used, which can be operated autonomously by a Solar-Energy Harvester. 1. Task phase: 6 "hands-on" tasks to gain experience and to create a SW library. 2. Project phase: Implementation of an energy autonomous system with the goal of highest possible energy efficiency (Challenge)
Literature	DE: Die Vorlesungsinhalte wurden in einem Paper veröffentlicht. Alle Quellen, welche in diesem Kurs (inklusive Übungen und PBL) verwendet wurden sind in folgenden Paper aufgelistet: ENG: The lecture contents were published in a paper. All sources used in this course (including exercises and PBL) are listed in the following paper: 1. Kulau, Ulf: Course: Energy Efficiency in Embedded Systems-A System-Level Perspective for Computer Scientists, EWME, 2018. DE: Als sehr gute Quellen für notwendiges Hintergrundwissen eignen sich die folgenden Bücher: ENG: The following books are very good sources for the necessary background knowledge: 1. Harris, David, and N. Weste: CMOS VLSI Design ed., Pearson Education, 2010 2. Rabaey, Jan: Low Power Design Essentials (Integrated Circuits and Systems), Springer, 2009

Course L2871: Energy Efficiency in Embedded Systems	
Typ	Recitation Section (large)
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Prof. Ulf Kulau
Language	EN
Cycle	WiSe
Content	In the lecture hall exercise, the theoretical basics taught in the lecture are deepened. This is done through in-depth discussion of relevant aspects, but also through calculation examples, in which a deeper understanding of the topic of energy efficiency in embedded systems is gained. Exercises will be distributed in advance and solutions will be presented in the lecture hall exercise. Contents of the exercise are as follows: • Basics and calculation of power dissipation on semiconductor • Power dissipation of CMOS using the example of an inverter • Influence of the activity factor and external components • DVS and scheduling • Evaluation to show the benefit of undervolting • Aspects of energy harvesting (MPPT) •
Literature	DE: Die Vorlesungsinhalte wurden in einem Paper veröffentlicht. Alle Quellen, welche in diesem Kurs (also auch den Übungen) verwendet wurden sind in folgenden Paper aufgelistet: ENG: The lecture contents were published in a paper. All sources used in this course (including exercises, and PBL) are listed in the following paper: 1. Kulau, Ulf: Course: Energy Efficiency in Embedded Systems-A System-Level Perspective for Computer Scientists, EWME, 2018. DE: Als sehr gute Quellen für notwendiges Hintergrundwissen eignen sich die folgenden Bücher: ENG: The following books are very good sources for the necessary background knowledge: 1. Harris, David, and N. Weste: CMOS VLSI Design ed., Pearson Education, 2010 2. Rabaey, Jan: Low Power Design Essentials (Integrated Circuits and Systems), Springer, 2009

Module M1596: Engineering Haptic Systems					
Courses					
Title		Typ		Hrs/wk	CP
Haptic Technology for Human-Machine-Interfaces (HMI) (L2439)		Lecture		4	3
Haptic Technology for Human-Machine-Interfaces (HMI) (L2859)		Project-/problem-based Learning		2	3
Module Responsible	Prof. Thorsten Kern				
Admission Requirements	None				
Recommended Previous Knowledge	We recommend knowledge in the areas of general engineering sciences, mechatronics and/or control-engineering. However also neighbouring technical areas like mechanical-engineering or even process-engineers can join the course and will be introduced into the content properly.				
Educational Objectives	After taking part successfully, students have reached the following learning results				
Professional Competence	<i>Knowledge</i> This course is an introduction to the design methods and design-requirements to consider when creating haptic systems from scratch. It covers a physiological part, an actuator development part, and goes up to fundamentals of higher system integration with consideration on control theory for more complex projects. Beside design-related topics, it gives a valuable overview on existing haptic applications and research in that field with many examples. This is supported by on-site experiments in the laboratories of M-4. • Motivation and application of haptic systems • Haptic perception • The role of the user in direct system interaction • Development of haptic systems • Identification of requirements • System-structure and control • Kinematic fundamentals • Actuation & Sensors technology for haptic applications • Control and system-design aspects • Fundamental considerations in simulating haptics <i>Skills</i> Executing the course the competency will be developed to apply the general engineering capabilities of the individual course towards the design and application of active haptic systems. The resulting competencies will open an entry into specialized position in avionic-industries, automotive-industry and consumer-device-development. Personal Competence <i>Social Competence</i> As a side-effect this module teaches basics of a general design for human-machine-interfaces, independent from the specific application of "haptics". It teaches methods to execute user-studies, judge on user-feedback and how to deal with soft design-requirements which are common when dealing with subjective perception. <i>Autonomy</i> Independent design-capability of haptic systems, general competency in engineering from a design-perspective				
Workload in Hours					
Credit points					
Course achievement					
Examination					
Examination duration and scale	30 min				
Assignment for the Following Curricula	Mechatronics: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Product Development and Production: Elective Compulsory				

Course L2439: Haptic Technology for Human-Machine-Interfaces (HMI)	
Typ	Lecture
Hrs/wk	4
CP	3
Workload in Hours	Independent Study Time 34, Study Time in Lecture 56
Lecturer	Prof. Thorsten Kern
Language	EN
Cycle	WiSe
Content	This course is an introduction to the design methods and design-requirements to consider when creating haptic systems from scratch. It covers a physiological part, an actuator development part, and goes up to fundamentals of higher system integration with consideration on control theory for more complex projects. Beside design-related topics, it gives a valuable overview on existing haptic applications and research in that field with many examples. • Motivation and application of haptic systems • Haptic perception • The role of the user in direct system interaction • Development of haptic systems • Identification of requirements • System-structure and control • Kinematic fundamentals • Actuation & Sensors technology for haptic applications • Control and system-design aspects • Fundamental considerations in simulating haptics
Literature	

Course L2859: Haptic Technology for Human-Machine-Interfaces (HMI)	
Typ	Project-/problem-based Learning
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Thorsten Kern
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M1552: Advanced Machine Learning			
Courses			
Title	Typ	Hrs/wk	CP
Advanced Machine Learning (L2322)	Lecture	2	3
Advanced Machine Learning (L2323)	Recitation Section (small)	2	3
Module Responsible	Dr. Jens-Peter Zemke		
Admission Requirements	None		
Recommended Previous Knowledge	1. Mathematics I-III 2. Numerical Mathematics 1/ Numerics 3. Programming skills, preferably in Python		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence	<i>Knowledge</i> Students are able to name, state and classify state-of-the-art neural networks and their corresponding mathematical basics. They can assess the difficulties of different neural networks. <i>Skills</i> Students are able to implement, understand, and, tailored to the field of application, apply neural networks. <i>Social Competence</i> Students can • develop and document joint solutions in small teams; • form groups to further develop the ideas and transfer them to other areas of applicability; • form a team to develop, build, and advance a software library. <i>Autonomy</i> Students are able to • correctly assess the time and effort of self-defined work; • assess whether the supporting theoretical and practical exercises are better solved individually or in a team; • define test problems for testing and expanding the methods; • assess their individual progress and, if necessary, to ask questions and seek help.		
<i>Knowledge</i>			
<i>Skills</i>			
<i>Social Competence</i>			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56		
Credit points	6		
Course achievement	None		
Examination	Written exam		
Examination duration and scale	90 min		
Assignment for the Following Curricula	Computational Engineering: Core Qualification: Elective Compulsory Computer Science: Specialisation III. Mathematics: Elective Compulsory Data Science: Core Qualification: Compulsory Computer Science in Engineering: Specialisation III. Mathematics: Elective Compulsory Interdisciplinary Mathematics: Specialisation Computational Methods in Biomedical Imaging: Compulsory Mechatronics: Core Qualification: Elective Compulsory Technomathematics: Specialisation I. Mathematics: Elective Compulsory Technomathematics: Specialisation II. Informatics: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory		

Course L2322: Advanced Machine Learning	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Dr. Jens-Peter Zemke
Language	EN
Cycle	WiSe
Content	1. Basics: analogy; layout of neural nets, universal approximation, NP-completeness 2. Feedforward Neural Networks: backpropagation, variants of Stochastic Gradients 3. Convolutional Neural Networks: idea, layout, FFT and Winograds algorithms, implementation details 4. Adversarial Attacks 5. Recurrent Neural Networks: idea, dynamical systems, training, LSTM 6. Residual Neural Networks 7. Neural ODEs 8. Autoencoder and Generative Adversarial Networks 9. Attention and Transformers
Literature	1. Skript 2. Online-Werke: ◦ http://neuralnetworksanddeeplearning.com/ ◦ https://www.deeplearningbook.org/

Course L2323: Advanced Machine Learning	
Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Dr. Jens-Peter Zemke
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M0629: Intelligent Autonomous Agents and Cognitive Robotics				
Courses				
Title		Typ	Hrs/wk	CP
Intelligent Autonomous Agents and Cognitive Robotics (L0341)		Lecture	2	4
Intelligent Autonomous Agents and Cognitive Robotics (L0512)		Recitation Section (small)	2	2
Module Responsible	Rainer Marrone			
Admission Requirements	None			
Recommended Previous Knowledge	Vectors, matrices, Calculus			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence				
<i>Knowledge</i>	Students can explain the agent abstraction, define intelligence in terms of rational behavior, and give details about agent design (goals, utilities, environments). They can describe the main features of environments. The notion of adversarial agent cooperation can be discussed in terms of decision problems and algorithms for solving these problems. For dealing with uncertainty in real-world scenarios, students can summarize how Bayesian networks can be employed as a knowledge representation and reasoning formalism in static and dynamic settings. In addition, students can define decision making procedures in simple and sequential settings, with and with complete access to the state of the environment. In this context, students can describe techniques for solving (partially observable) Markov decision problems, and they can recall techniques for measuring the value of information. Students can identify techniques for simultaneous localization and mapping, and can explain planning techniques for achieving desired states. Students can explain coordination problems and decision making in a multi-agent setting in term of different types of equilibria, social choice functions, voting protocol, and mechanism design techniques.			
<i>Skills</i>	Students can select an appropriate agent architecture for concrete agent application scenarios. For simplified agent application students can derive decision trees and apply basic optimization techniques. For those applications they can also create Bayesian networks/dynamic Bayesian networks and apply bayesian reasoning for simple queries. Students can also name and apply different sampling techniques for simplified agent scenarios. For simple and complex decision making students can compute the best action or policies for concrete settings. In multi-agent situations students will apply techniques for finding different equilibria states, e.g., Nash equilibria. For multi-agent decision making students will apply different voting protocols and compare and explain the results.			
Personal Competence				
<i>Social Competence</i>	Students are able to discuss their solutions to problems with others. They communicate in English			
<i>Autonomy</i>	Students are able of checking their understanding of complex concepts by solving variants of concrete problems			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	None			
Examination	Written exam			
Examination duration and scale	90 minutes			
Assignment for the Following Curricula	Computer Science: Specialisation II: Intelligence Engineering: Elective Compulsory International Management and Engineering: Specialisation II. Information Technology: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Biomedical Engineering: Specialisation Artificial Organs and Regenerative Medicine: Elective Compulsory Biomedical Engineering: Specialisation Implants and Endoprostheses: Elective Compulsory Biomedical Engineering: Specialisation Medical Technology and Control Theory: Elective Compulsory Biomedical Engineering: Specialisation Management and Business Administration: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory			

Course L0341: Intelligent Autonomous Agents and Cognitive Robotics	
Typ	Lecture
Hrs/wk	2
CP	4
Workload in Hours	Independent Study Time 92, Study Time in Lecture 28
Lecturer	Rainer Marrone
Language	EN
Cycle	WiSe
Content	- Definition of agents, rational behavior, goals, utilities, environment types - Adversarial agent cooperation: Agents with complete access to the state(s) of the environment, games, Minimax algorithm, alpha-beta pruning, elements of chance - Uncertainty: Motivation: agents with no direct access to the state(s) of the environment, probabilities, conditional probabilities, product rule, Bayes rule, full joint probability distribution, marginalization, summing out, answering queries, complexity, independence assumptions, naive Bayes, conditional independence assumptions - Bayesian networks: Syntax and semantics of Bayesian networks, answering queries revised (inference by enumeration), typical-case complexity, pragmatics: reasoning from effect (that can be perceived by an agent) to cause (that cannot be directly perceived). - Probabilistic reasoning over time: Environmental state may change even without the agent performing actions, dynamic Bayesian networks, Markov assumption, transition model, sensor model, inference problems: filtering, prediction, smoothing, most-likely explanation, special cases: hidden Markov models, Kalman filters, Exact inferences and approximations - Decision making under uncertainty: Simple decisions: utility theory, multivariate utility functions, dominance, decision networks, value of information Complex decisions: sequential decision problems, value iteration, policy iteration, MDPs Decision-theoretic agents: POMDPs, reduction to multidimensional continuous MDPs, dynamic decision networks - Simultaneous Localization and Mapping - Planning - Game theory (Golden Balls: Split or Share) Decisions with multiple agents, Nash equilibrium, Bayes-Nash equilibrium - Social Choice Voting protocols, preferences, paradoxes, Arrow's Theorem, - Mechanism Design Fundamentals, dominant strategy implementation, Revelation Principle, Gibbard-Satterthwaite Impossibility Theorem, Direct mechanisms, incentive compatibility, strategy-proofness, Vickrey-Groves-Clarke mechanisms, expected externality mechanisms, participation constraints, individual rationality, budget balancedness, bilateral trade, Myerson-Satterthwaite Theorem
Literature	- Artificial Intelligence: A Modern Approach (Third Edition), Stuart Russell, Peter Norvig, Prentice Hall, 2010, Chapters 2-5, 10-11, 13-17 - Probabilistic Robotics, Thrun, S., Burgard, W., Fox, D. MIT Press 2005 - Multiagent Systems: Algorithmic, Game-Theoretic, and Logical Foundations, Yoav Shoham, Kevin Leyton-Brown, Cambridge University Press, 2009

Course L0512: Intelligent Autonomous Agents and Cognitive Robotics	
Typ	Recitation Section (small)
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Rainer Marrone
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M1024: Methods of Product Development			
Courses			
Title	Typ	Hrs/wk	CP
Methods of Product Development (L1254)	Lecture	3	3
Methods of Product Development (L1255)	Project-/problem-based Learning	2	3
Module Responsible	Prof. Dieter Krause		
Admission Requirements	None		
Recommended Previous Knowledge	Basic knowledge of Integrated product development and applying CAE systems		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence			
<i>Knowledge</i>	After passing the module students are able to: • explain technical terms of design methodology, • describe essential elements of construction management, • describe current problems and the current state of research of integrated product development.		
<i>Skills</i>	After passing the module students are able to: • select and apply proper construction methods for non-standardized solutions of problems as well as adapt new boundary conditions, • solve product development problems with the assistance of a workshop based approach, • choose and execute appropriate moderation techniques.		
Personal Competence			
<i>Social Competence</i>	After passing the module students are able to: • prepare and lead team meetings and moderation processes, • work in teams on complex tasks, • represent problems and solutions and advance ideas.		
<i>Autonomy</i>	After passing the module students are able to: • give a structured feedback and accept a critical feedback, • implement the accepted feedback autonomous.		
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70		
Credit points	6		
Course achievement	None		
Examination	Oral exam		
Examination duration and scale	30 Minuten		
Assignment for the Following Curricula	Aircraft Systems Engineering: Core Qualification: Elective Compulsory International Management and Engineering: Specialisation II. Product Development and Production: Elective Compulsory Aeronautics: Core Qualification: Elective Compulsory Mechanical Engineering - Product Development and Production: Specialisation Product Development: Compulsory Mechanical Engineering - Product Development and Production: Specialisation Production: Elective Compulsory Mechanical Engineering - Product Development and Production: Specialisation Materials: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Product Development, Materials and Production: Specialisation Product Development: Compulsory Product Development, Materials and Production: Specialisation Production: Elective Compulsory Product Development, Materials and Production: Specialisation Materials: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Product Development and Production: Elective Compulsory		

Course L1254: Methods of Product Development	
Typ	Lecture
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Lecturer	Prof. Dieter Krause
Language	DE
Cycle	WiSe
Content	Lecture The lecture extends and enhances the learned content of the module "Integrated Product Development and lightweight design" and is based on the knowledge and skills acquired there. Topics of the course include in particular: • Methods of product development, • Presentation techniques, • Industrial Design, • Design for variety • Modularization methods, • Design catalogs, • Adapted QFD matrix, • Systematic material selection, • Assembly oriented design, Construction management • CE mark, declaration of conformity including risk assessment, • Patents, patent rights, patent monitoring • Project management (cost, time, quality) and escalation principles, • Development management for mechatronics, • Technical Supply Chain Management. Exercise (PBL) In the exercise the content presented in the lecture "Integrated Product Development II" and methods of product development and design management will be enhanced. Students learn an independently moderated and workshop based approach through industry related practice examples to solve complex and currently existing issues in product development. They will learn the ability to apply important methods of product development and design management autonomous and acquire further expertise in the field of integrated product development. Besides personal skills, such as teamwork, guiding discussions and representing work results will be acquired through the workshop based structure of the event under its own planning and management.
Literature	• Andreasen, M.M., Design for Assembly, Berlin, Springer 1985. • Ashby, M. F.: Materials Selection in Mechanical Design, München, Spektrum 2007. • Beckmann, H.: Supply Chain Management, Berlin, Springer 2004. • Hartmann, M., Rieger, M., Funk, R., Rath, U.: Zielgerichtet moderieren. Ein Handbuch für Führungskräfte, Berater und Trainer, Weinheim, Beltz 2007. • Pahl, G., Beitz, W.: Konstruktionslehre, Berlin, Springer 2006. • Roth, K.H.: Konstruieren mit Konstruktionskatalogen, Band 1-3, Berlin, Springer 2000. • Simpson, T.W., Siddique, Z., Jiao, R.J.: Product Platform and Product Family Design. Methods and Applications, New York, Springer 2013.

Course L1255: Methods of Product Development	
Typ	Project-/problem-based Learning
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Dieter Krause
Language	DE
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M0746: Microsystem Engineering				
Courses				
Title	Typ		Hrs/wk	CP
Microsystem Engineering (L0680)	Lecture		2	4
Microsystem Engineering (L0682)	Project-/problem-based Learning		2	2
Module Responsible	Dr. Timo Lipka			
Admission Requirements	None			
Recommended Previous Knowledge	Basic courses in physics, mathematics and electric engineering			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence				
<i>Knowledge</i>	The students know about the most important technologies and materials of MEMS as well as their applications in sensors and actuators.			
<i>Skills</i>	Students are able to analyze and describe the functional behaviour of MEMS components and to evaluate the potential of microsystems.			
Personal Competence				
<i>Social Competence</i>	Students are able to solve specific problems alone or in a group and to present the results accordingly.			
<i>Autonomy</i>	Students are able to acquire particular knowledge using specialized literature and to integrate and associate this knowledge with other fields.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	No	None	Exercises	
Examination	Written exam			
Examination duration and scale	2h			
Assignment for the Following Curricula	Electrical Engineering and Information Technology: Core Qualification: Compulsory Electrical Engineering: Core Qualification: Compulsory International Management and Engineering: Specialisation II. Electrical Engineering: Elective Compulsory International Management and Engineering: Specialisation II. Mechatronics: Elective Compulsory Mechanical Engineering and Management: Specialisation Mechatronics: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Microelectronics and Microsystems: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Bio- and Medical Technology: Elective Compulsory			

Course L0680: Microsystem Engineering	
Typ	Lecture
Hrs/wk	2
CP	4
Workload in Hours	Independent Study Time 92, Study Time in Lecture 28
Lecturer	Dr. Timo Lipka
Language	EN
Cycle	WiSe
Content	Object and goal of MEMS Scaling Rules Wafers and thin film deposition Lithography Structuring and etching Laser processing and bonding Energy conversion and force generation Electromagnetic actuators Electrostatic actuators, comb drives Reluctance motors Piezoelectric actuators, bi-metal actuator Transducer principles Signal detection and signal processing Mechanical and physical sensors based on beams and diaphragms Optical sensors and sensor arrays System integration aspects Yield, test and reliability
Literature	M. Kasper: Mikrosystementwurf, Springer (2000) M. Madou: Fundamentals of Microfabrication, CRC Press (1997)

Course L0682: Microsystem Engineering	
Typ	Project-/problem-based Learning
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Dr. Timo Lipka
Language	EN
Cycle	WiSe
Content	Scaling effects in microsystems Process flow charts Design aspects and layout consideration Examples of MEMS components Electric, thermal and mechanical behaviour Electrostatic actuators, comb drives Piezoelectric actuators, bi-metal actuator Signal detection and signal processing Mechanical and physical sensors based on beams and diaphragms Yield estimation and reliability issues
Literature	Wird in der Veranstaltung bekannt gegeben

Module M1268: Linear and Nonlinear Waves				
Courses				
Title		Typ	Hrs/wk	CP
Linear and Nonlinear Waves (L1737)		Project-/problem-based Learning	4	6
Module Responsible	Prof. Norbert Hoffmann			
Admission Requirements	None			
Recommended Previous Knowledge	Calculus, Algebra, Engineering Mechanics, Vibrations.			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence <i>Knowledge</i> <i>Skills</i> Personal Competence <i>Social Competence</i> <i>Autonomy</i>	• Students are able to reflect existing terms and concepts in Wave Mechanics • Students are able to identify and express the need to develop and research new terms and concepts. • Students are able to apply existing research methods and procedures of wave mechanics. • Students are able to develop novel research methods and procedures in wave mechanics. • Students can reach working results also in groups. • Students can present and communicate working results also in groups. • Students are able to approach given research tasks individually. • Studetns are able to identify and follow up novel research tasks by themselves.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	None			
Examination	Written exam			
Examination duration and scale	2 Hours			
Assignment for the Following Curricula	Mechatronics: Core Qualification: Elective Compulsory Naval Architecture and Ocean Engineering: Core Qualification: Elective Compulsory Naval Architecture and Ocean Engineering: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Maritime Technology: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Simulation Technology: Elective Compulsory			

Course L1737: Linear and Nonlinear Waves	
Typ	Project-/problem-based Learning
Hrs/wk	4
CP	6
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56
Lecturer	Prof. Norbert Hoffmann
Language	DE/EN
Cycle	WiSe
Content	Introduction into the Dynamics of Linear and Nonlinear Waves - Linear Waves - Dispersion - Phase and Group Velocity - Envelopes - Discrete Systems - Nonlinear Waves - Model Equations - Solitons, Breathers, Extreme Waves - Water Waves, Ocean Waves - Airy and Stokes - Natural Sea State - Kinetic Modelling - Other topics
Literature	F.K. Kneubühl: Oscillations and Waves. Springer. G.B. Witham, Linear and Nonlinear Waves. Wiley. C.C. Mei, Theory and Applications of Ocean Surface Waves. World Scientific. L.H. Holthuijsen, Waves in Oceanic and Coastal Waters. Cambridge. And others.

Module M1398: Selected Topics in Multibody Dynamics and Robotics				
Courses				
Title			Typ	Hrs/wk
Formulas and Vehicles - Dynamics and Control of Autonomous Vehicles (L2869)			Integrated Lecture	1
Formulas and Vehicles - Introduction into Mobile Underwater Robotics (L1981)			Project-/problem-based Learning	4
Module Responsible	Prof. Robert Seifried			
Admission Requirements	None			
Recommended Previous Knowledge	Computational Mechanics, Applied Dynamics or Robotics Numerical Treatment of Ordinary Differential Equations Control Systems Theory and Design			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	After successful completion of the module students demonstrate deeper knowledge and understanding in selected application areas of multibody dynamics and robotics Students are able + to think holistically + to independently, securely and critically analyze and optimize basic problems of the dynamics of rigid and flexible multibody systems + to describe dynamics problems mathematically + to implement dynamical problems on hardware			
<i>Knowledge</i>				
<i>Skills</i>				
Personal Competence				
<i>Social Competence</i>				
<i>Autonomy</i>	Students are able to + solve problems in heterogeneous groups and to document the corresponding results and present them Students are able to + assess their knowledge by means of exercises and projects. + acquaint themselves with the necessary knowledge to solve research oriented tasks.			
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70			
Credit points	6			
Course achievement	None			
Examination	Presentation			
Examination duration and scale	TBA			
Assignment for the Following Curricula	Mechatronics: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Core Qualification: Elective Compulsory			

Course L2869: Formulas and Vehicles - Dynamics and Control of Autonomous Vehicles	
Typ	Integrated Lecture
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Prof. Robert Seifried
Language	DE
Cycle	WiSe
Content	Mechatronic Design Modeling of AUVs Localisation Control Design Application in ROS Experiments
Literature	Schiehlen, W.; Eberhard, P.: Technische Dynamik, 4. Auflage, Vieweg+Teubner: Wiesbaden, 2014. Woernle, C.: Mehrkörpersysteme, Springer: Heidelberg, 2011. Seifried, R.: Dynamics of Underactuated Multibody Systems, Springer, 2014.

Course L1981: Formulas and Vehicles - Introduction into Mobile Underwater Robotics	
Typ	Project-/problem-based Learning
Hrs/wk	4
CP	5
Workload in Hours	Independent Study Time 94, Study Time in Lecture 56
Lecturer	Prof. Robert Seifried
Language	DE
Cycle	WiSe
Content	• Interdisciplinary between applied mathematics (systems theory) and engineering (mechanical engineering). • Working on issues relating to autonomous robotics in small interdisciplinary groups • Development of theoretical control procedures and their implementation on test vehicles
Literature	Seifried, R.: Dynamics of underactuated multibody systems, Springer, 2014 Popp, K.; Schiehlen, W.: Ground vehicle dynamics, Springer, 2010

Module M1748: Construction Robotics				
Courses				
Title		Typ	Hrs/wk	CP
Construction Robotics (L2867)		Project-/problem-based Learning	6	6
Module Responsible	Prof. Kay Smarsly			
Admission Requirements	None			
Recommended Previous Knowledge	Basics of project-oriented programming			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	Knowledge	Basics of robotics		
		Applications in civil engineering		
		Kinematics		
	Skills	Use of specific hardware		
		Development of software routines		
		Python programming language		
		Image processing		
	Basics of localization (LIDAR, SLAM)			
	Personal Competence	Social Competence	Teamwork	
			Communication skills	
Autonomy		Independent work		
		Independent decisions		
Workload in Hours	Independent Study Time 96, Study Time in Lecture 84			
Credit points	6			
Course achievement	None			
Examination	Written elaboration			
Examination duration and scale	ca. 10 Seiten			
Assignment for the Following Curricula	Civil Engineering: Specialisation Structural Engineering: Elective Compulsory			
	Civil Engineering: Specialisation Water and Traffic: Elective Compulsory			
	Civil Engineering: Specialisation Coastal Engineering: Elective Compulsory			
	Civil Engineering: Specialisation Geotechnical Engineering: Elective Compulsory			
	Civil Engineering: Specialisation Computational Engineering: Elective Compulsory			
	Computer Science: Specialisation II: Intelligence Engineering: Elective Compulsory			
	Mechatronics: Core Qualification: Elective Compulsory			
	Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory			

Course L2867: Construction Robotics	
Typ	Project-/problem-based Learning
Hrs/wk	6
CP	6
Workload in Hours	Independent Study Time 96, Study Time in Lecture 84
Lecturer	Prof. Kay Smarsly, Jan Stührenberg
Language	EN
Cycle	WiSe
Content	1. Introduction: Robotics in civil engineering 2. Presentation of potential topics 3. Programming of algorithms in Python 4. Application of software systems: LINUX distribution, ROS, CloudCompare, ... 5. Application of hardware systems: Petoï Bittle Dog, Raspberry Pi, Arduino, sensing ... 6. Topics considered for robotics using the Petoï Bittle Dog: 1. Movement 2. Use of sensors (camera, infrared, ...) 3. Data structures/data acquisition 4. Programming 7. Topics technically relevant to building inspection: 1. Geodetic evaluations 2. Image processing 3. Localization
Literature	Bock/Linner: Construction Robotics Verl et al.: Soft Robotics Pasquale: New Laws of robotics

Module M0603: Nonlinear Structural Analysis			
Courses			
Title	Typ	Hrs/wk	CP
Nonlinear Structural Analysis (L0277)	Lecture	3	4
Nonlinear Structural Analysis (L0279)	Recitation Section (small)	1	2
Module Responsible	Prof. Alexander Düster		
Admission Requirements	None		
Recommended Previous Knowledge	Knowledge of partial differential equations is recommended.		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence			
<i>Knowledge</i>	Students are able to + give an overview of the different nonlinear phenomena in structural mechanics. + explain the mechanical background of nonlinear phenomena in structural mechanics. + to specify problems of nonlinear structural analysis, to identify them in a given situation and to explain their mathematical and mechanical background.		
<i>Skills</i>	Students are able to + model nonlinear structural problems. + select for a given nonlinear structural problem a suitable computational procedure. + apply finite element procedures for nonlinear structural analysis. + critically verify and judge results of nonlinear finite elements. + to transfer their knowledge of nonlinear solution procedures to new problems.		
Personal Competence			
<i>Social Competence</i>	Students are able to + solve problems in heterogeneous groups. + present and discuss their results in front of others. + give and accept professional constructive criticism.		
<i>Autonomy</i>	Students are able to + assess their knowledge by means of exercises and E-Learning. + acquaint themselves with the necessary knowledge to solve research oriented tasks. + to transform the acquired knowledge to similar problems.		
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56		
Credit points	6		
Course achievement	None		
Examination	Written exam		
Examination duration and scale	120 min		
Assignment for the Following Curricula	Civil Engineering: Specialisation Structural Engineering: Elective Compulsory Civil Engineering: Specialisation Computational Engineering: Compulsory Computational Engineering: Core Qualification: Elective Compulsory International Management and Engineering: Specialisation II. Civil Engineering: Elective Compulsory Mechanical Engineering - Product Development and Production: Core Qualification: Elective Compulsory Materials Science and Engineering: Specialisation Modeling: Elective Compulsory Materials Science: Specialisation Modeling: Elective Compulsory Mechatronics: Technical Complementary Course: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Product Development, Materials and Production: Core Qualification: Elective Compulsory Naval Architecture and Ocean Engineering: Core Qualification: Elective Compulsory Naval Architecture and Ocean Engineering: Core Qualification: Elective Compulsory Ship and Offshore Technology: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Simulation Technology: Elective Compulsory		

Course L0277: Nonlinear Structural Analysis	
Typ	Lecture
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Lecturer	Prof. Alexander Düster
Language	EN
Cycle	WiSe
Content	1. Introduction 2. Nonlinear phenomena 3. Mathematical preliminaries 4. Basic equations of continuum mechanics 5. Spatial discretization with finite elements 6. Solution of nonlinear systems of equations 7. Solution of elastoplastic problems 8. Stability problems 9. Contact problems
Literature	[1] Alexander Düster, Nonlinear Structural Analysis, Lecture Notes, Technische Universität Hamburg-Harburg, 2014. [2] Peter Wriggers, Nonlinear Finite Element Methods, Springer 2008. [3] Peter Wriggers, Nichtlineare Finite-Elemente-Methoden, Springer 2001. [4] Javier Bonet and Richard D. Wood, Nonlinear Continuum Mechanics for Finite Element Analysis, Cambridge University Press, 2008.

Course L0279: Nonlinear Structural Analysis	
Typ	Recitation Section (small)
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Prof. Alexander Düster
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M0623: Intelligent Systems in Medicine				
Courses				
Title	Typ		Hrs/wk	CP
Intelligent Systems in Medicine (L0331)	Lecture		2	3
Intelligent Systems in Medicine (L0334)	Project Seminar		2	2
Intelligent Systems in Medicine (L0333)	Recitation Section (small)		1	1
Module Responsible	Prof. Alexander Schlaefer			
Admission Requirements	None			
Recommended Previous Knowledge	• principles of math (algebra, analysis/calculus) • principles of stochastics • principles of programming, Java/C++ and R/Matlab • advanced programming skills			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence				
<i>Knowledge</i>	The students are able to analyze and solve clinical treatment planning and decision support problems using methods for search, optimization, and planning. They are able to explain methods for classification and their respective advantages and disadvantages in clinical contexts. The students can compare different methods for representing medical knowledge. They can evaluate methods in the context of clinical data and explain challenges due to the clinical nature of the data and its acquisition and due to privacy and safety requirements.			
<i>Skills</i>	The students can give reasons for selecting and adapting methods for classification, regression, and prediction. They can assess the methods based on actual patient data and evaluate the implemented methods.			
Personal Competence				
<i>Social Competence</i>	The students are able to grasp practical tasks in groups, develop solution strategies independently, define work processes and work on them collaboratively. The students can critically reflect on the results of other groups, make constructive suggestions for improvement and also incorporate them into their own work.			
<i>Autonomy</i>	The students can assess their level of knowledge and document their work results. They can critically evaluate the results achieved and present them in an appropriate argumentative manner to the other groups.			
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	Yes	10 %	Written elaboration	
	Yes	10 %	Presentation	
Examination	Written exam			
Examination duration and scale	90 minutes			
Assignment for the Following Curricula	Computer Science: Specialisation II: Intelligence Engineering: Elective Compulsory Data Science: Specialisation IV. Special Focus Area: Elective Compulsory Data Science: Specialisation III. Applications: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Medical Technology: Elective Compulsory Electrical Engineering: Specialisation Medical Technology: Elective Compulsory Interdisciplinary Mathematics: Specialisation Computational Methods in Biomedical Imaging: Compulsory Mechatronics: Core Qualification: Elective Compulsory Biomedical Engineering: Specialisation Artificial Organs and Regenerative Medicine: Elective Compulsory Biomedical Engineering: Specialisation Implants and Endoprotheses: Elective Compulsory Biomedical Engineering: Specialisation Management and Business Administration: Elective Compulsory Biomedical Engineering: Specialisation Medical Technology and Control Theory: Compulsory Theoretical Mechanical Engineering: Specialisation Bio- and Medical Technology: Elective Compulsory			

Course L0331: Intelligent Systems in Medicine	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Alexander Schläefer
Language	EN
Cycle	WiSe
Content	- methods for search, optimization, planning, classification, regression and prediction in a clinical context - representation of medical knowledge - understanding challenges due to clinical and patient related data and data acquisition The students will work in groups to apply the methods introduced during the lecture using problem based learning.
Literature	Russel & Norvig: Artificial Intelligence: a Modern Approach, 2012 Berner: Clinical Decision Support Systems: Theory and Practice, 2007 Greenes: Clinical Decision Support: The Road Ahead, 2007 Further literature will be given in the lecture

Course L0334: Intelligent Systems in Medicine	
Typ	Project Seminar
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Alexander Schläefer
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Course L0333: Intelligent Systems in Medicine	
Typ	Recitation Section (small)
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Prof. Alexander Schläefer
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M0806: Technical Acoustics II (Room Acoustics, Computational Methods)				
Courses				
Title		Typ	Hrs/wk	CP
Technical Acoustics II (Room Acoustics, Computational Methods) (L0519)		Lecture	2	3
Technical Acoustics II (Room Acoustics, Computational Methods) (L0521)		Recitation Section (large)	2	3
Module Responsible	Dr. Alessandro Bassetti			
Admission Requirements	None			
Recommended Previous Knowledge	Technical Acoustics I (Acoustic Waves, Noise Protection, Psycho Acoustics) Mechanics I (Statics, Mechanics of Materials) and Mechanics II (Hydrostatics, Kinematics, Dynamics) Mathematics I, II, III (in particular differential equations)			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> The students possess an in-depth knowledge in acoustics regarding room acoustics and computational methods and are able to give an overview of the corresponding theoretical and methodical basis. <i>Skills</i> The students are capable to handle engineering problems in acoustics by theory-based application of the demanding computational methods and procedures treated within the module. <i>Personal Competence</i> <i>Social Competence</i> Students can work in small groups on specific problems to arrive at joint solutions. <i>Autonomy</i> The students are able to independently solve challenging acoustical problems in the areas treated within the module. Possible conflicting issues and limitations can be identified and the results are critically scrutinized.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	None			
Examination	Oral exam			
Examination duration and scale	20 min			
Assignment for the Following Curricula	Computational Engineering: Core Qualification: Elective Compulsory Aircraft Systems Engineering: Core Qualification: Elective Compulsory Aeronautics: Core Qualification: Elective Compulsory Mechanical Engineering - Product Development and Production: Core Qualification: Elective Compulsory Mechanical Engineering and Management: Core Qualification: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Product Development, Materials and Production: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Product Development and Production: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Simulation Technology: Elective Compulsory			

Course L0519: Technical Acoustics II (Room Acoustics, Computational Methods)	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Dr. Alessandro Bassetti
Language	EN
Cycle	WiSe
Content	- Room acoustics - Sound absorber - Standard computations - Statistical Energy Approaches - Finite Element Methods - Boundary Element Methods - Geometrical acoustics - Special formulations - Practical applications - Hands-on Sessions: Programming of elements (Matlab)
Literature	Cremer, L.; Heckl, M. (1996): Körperschall. Springer Verlag, Berlin Veit, I. (1988): Technische Akustik. Vogel-Buchverlag, Würzburg Veit, I. (1988): Flüssigkeitsschall. Vogel-Buchverlag, Würzburg Gaul, L.; Fiedler, Ch. (1997): Methode der Randelemente in Statik und Dynamik. Vieweg, Braunschweig, Wiesbaden Bathe, K.-J. (2000): Finite-Elemente-Methoden. Springer Verlag, Berlin

Course L0521: Technical Acoustics II (Room Acoustics, Computational Methods)	
Typ	Recitation Section (large)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Dr. Alessandro Bassetti
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M0832: Advanced Topics in Control			
Courses			
Title	Typ	Hrs/wk	CP
Advanced Topics in Control (L0661)	Lecture	2	3
Advanced Topics in Control (L0662)	Recitation Section (small)	2	3
Module Responsible	Prof. Timm Faulwasser		
Admission Requirements	None		
Recommended Previous Knowledge	Grundlagen der Regelungstechnik or Introduction to Control Systems, Control System Theory and Design		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence <i>Knowledge</i> - Students can explain the advantages and shortcomings of the complex advanced control methods - The students can test the topics of the course in simulation <i>Skills</i> Personal Competence <i>Social Competence</i> Students can work in small groups and arrive at joint results. <i>Autonomy</i> Students can find required information in sources provided (lecture notes, literature, software documentation) and use it to solve given problems.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56		
Credit points	6		
Course achievement	None		
Examination	Oral exam		
Examination duration and scale	30 min		
Assignment for the Following Curricula	Electrical Engineering and Information Technology: Specialisation Control and Power Systems Engineering: Elective Compulsory Electrical Engineering: Specialisation Control and Power Systems Engineering: Elective Compulsory Aircraft Systems Engineering: Core Qualification: Elective Compulsory Aeronautics: Core Qualification: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Biomedical Engineering: Specialisation Implants and Endoprostheses: Elective Compulsory Biomedical Engineering: Specialisation Medical Technology and Control Theory: Elective Compulsory Biomedical Engineering: Specialisation Management and Business Administration: Elective Compulsory Biomedical Engineering: Specialisation Artificial Organs and Regenerative Medicine: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory		

Course L0661: Advanced Topics in Control	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	NN
Language	EN
Cycle	WiSe
Content	- Linear Parameter-Varying (LPV) Gain Scheduling - Linearizing gain scheduling, hidden coupling - Jacobian linearization vs. quasi-LPV models - Stability and induced L2 norm of LPV systems - Synthesis of LPV controllers based on the two-sided projection lemma - Simplifications: controller synthesis for polytopic and LFT models - Experimental identification of LPV models - Controller synthesis based on input/output models - Applications: LPV torque vectoring for electric vehicles, LPV control of a robotic manipulator - Control of Multi-Agent Systems - Communication graphs - Spectral properties of the graph Laplacian - First and second order consensus protocols - Formation control, stability and performance - LPV models for agents subject to nonholonomic constraints - Application: formation control for a team of quadrotor helicopters - Linear and Nonlinear Model Predictive Control based on LMIs
Literature	- Werner, H., Lecture Notes "Advanced Topics in Control" - Selection of relevant research papers made available as pdf documents via StudIP

Course L0662: Advanced Topics in Control	
Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	NN
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M1919: Sustainable operation of technical assets				
Courses				
Title		Typ	Hrs/wk	CP
Fundamentals of Maintenance, Repair and Overhaul (MRO) (L3160)		Lecture	3	4
Fundamentals of Maintenance, Repair and Overhaul (MRO) (L3161)		Recitation Section (large)	1	2
Module Responsible	Prof. Gerko Wende			
Admission Requirements	None			
Recommended Previous Knowledge	We recommend knowledge in the areas of general engineering sciences, aeronautics and aircraft systems engineering. Technical fields like mechanical engineering, mechatronics and production engineering will be introduced into the relevant aeronautical content.			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> The students are able to describe fundamental correlations for the sustainable operation of technical assets and to identify solution approaches for complex optimization problems. <i>Skills</i> The students are enabled to apply the general engineering capabilities of the individual course towards the optimization of the sustainability in operation of technical assets. The resulting competencies will open an entry into positions in the development, production and technical operation of sustainable products in the mobility and engineering industries. Personal Competence <i>Social Competence</i> The students are able to work in mixed groups with a clear focus on the approached solutions by respecting the complex environment of multiple stakeholders. <i>Autonomy</i> The students are enabled to find solutions for optimization problems and to take required decision for the assessment of determining factors independently.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	None			
Examination	Oral exam			
Examination duration and scale	30 min			
Assignment for the Following Curricula	Aeronautics: Core Qualification: Elective Compulsory Mechanical Engineering - Product Development and Production: Specialisation Product Development: Elective Compulsory Mechanical Engineering - Product Development and Production: Specialisation Production: Elective Compulsory Mechanical Engineering - Product Development and Production: Specialisation Materials: Elective Compulsory Mechanical Engineering and Management: Core Qualification: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Product Development, Materials and Production: Specialisation Product Development: Elective Compulsory Product Development, Materials and Production: Specialisation Production: Elective Compulsory Product Development, Materials and Production: Specialisation Materials: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Product Development and Production: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Aircraft Systems Engineering: Elective Compulsory			

Course L3160: Fundamentals of Maintenance, Repair and Overhaul (MRO)	
Typ	Lecture
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Lecturer	Prof. Gerko Wende
Language	DE
Cycle	WiSe
Content	Fundamentals for the sustainable operation of technical assets by means of maintenance, repair and overhaul (MRO): - Life cycle analytics - Material circularity and service products - Rules and regulations - Processes and production methods - Tools and technologies - Data handling and usage - Design for maintenance - Self-healing technical systems
Literature	- The Greenhouse Gas Protocol, World Resources Institute and World Business Council for Sustainable Development, 2004, ISBN 1-56973-568-9 - The Environmental Performance Index 2022, Ranking country performance on sustainability issues, Yale Center for Environmental Law & Policy, Yale University - Business model innovation in the aerospace industry: Strategic options for maintenance, repair and overhaul firms, Oliver Wirths, Dissertation, Universität zu Köln, 2019 - Industrielles Luftfahrtmanagement, Technik und Organisation luftfahrttechnischer Betriebe, Martin Hinsch, Springer, 2019, ISBN 978-3-662-58803-1

Course L3161: Fundamentals of Maintenance, Repair and Overhaul (MRO)	
Typ	Recitation Section (large)
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Prof. Gerko Wende
Language	DE
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M0633: Industrial Process Automation				
Courses				
Title	Typ		Hrs/wk	CP
Industrial Process Automation (L0344)	Lecture		2	3
Industrial Process Automation (L0345)	Recitation Section (small)		2	3
Module Responsible	Prof. Alexander Schlaefer			
Admission Requirements	None			
Recommended Previous Knowledge	mathematics and optimization methods principles of automata principles of algorithms and data structures programming skills			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence <i>Knowledge</i> <i>Skills</i> Personal Competence <i>Social Competence</i> <i>Autonomy</i>	The students can evaluate and assess discrete event systems. They can evaluate properties of processes and explain methods for process analysis. The students can compare methods for process modelling and select an appropriate method for actual problems. They can discuss scheduling methods in the context of actual problems and give a detailed explanation of advantages and disadvantages of different programming methods. The students can relate process automation to methods from robotics and sensor systems as well as to recent topics like 'cyberphysical systems' and 'industry 4.0'. The students are able to develop and model processes and evaluate them accordingly. This involves taking into account optimal scheduling, understanding algorithmic complexity, and implementation using PLCs. The students can independently define work processes within their groups, distribute tasks within the group and develop solutions collaboratively. The students are able to assess their level of knowledge and to document their work results adequately.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	No	10 %	Exercises	
Examination	Written exam			
Examination duration and scale	90 minutes			
Assignment for the Following Curricula	Bioprocess Engineering: Specialisation A - General Bioprocess Engineering: Elective Compulsory Chemical and Bioprocess Engineering: Specialisation General Process Engineering: Elective Compulsory Chemical and Bioprocess Engineering: Specialisation Chemical Process Engineering: Elective Compulsory Computer Science: Specialisation II: Intelligence Engineering: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Control and Power Systems Engineering: Elective Compulsory Electrical Engineering: Specialisation Control and Power Systems Engineering: Elective Compulsory Aircraft Systems Engineering: Core Qualification: Elective Compulsory International Management and Engineering: Specialisation II. Mechatronics: Elective Compulsory International Management and Engineering: Specialisation II. Product Development and Production: Elective Compulsory Mechanical Engineering and Management: Specialisation Mechatronics: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory Process Engineering: Specialisation Chemical Process Engineering: Elective Compulsory Process Engineering: Specialisation Process Engineering: Elective Compulsory			

Course L0344: Industrial Process Automation	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Alexander Schlaefer
Language	EN
Cycle	WiSe
Content	- foundations of problem solving and system modeling, discrete event systems - properties of processes, modeling using automata and Petri-nets - design considerations for processes (mutex, deadlock avoidance, liveness) - optimal scheduling for processes - optimal decisions when planning manufacturing systems, decisions under uncertainty - software design and software architectures for automation, PLCs
Literature	J. Lunze: „Automatisierungstechnik“, Oldenbourg Verlag, 2012 Reisig: Petrinetze: Modellierungstechnik, Analysemethoden, Fallstudien; Vieweg+Teubner 2010 Hrúz, Zhou: Modeling and Control of Discrete-event Dynamic Systems; Springer 2007 Li, Zhou: Deadlock Resolution in Automated Manufacturing Systems, Springer 2009 Pinedo: Planning and Scheduling in Manufacturing and Services, Springer 2009

Course L0345: Industrial Process Automation	
Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Alexander Schlaefer
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M0720: Matrix Algorithms				
Courses				
Title	Typ		Hrs/wk	CP
Matrix Algorithms (L0984)	Lecture		2	3
Matrix Algorithms (L0985)	Recitation Section (small)		2	3
Module Responsible	Dr. Jens-Peter Zemke			
Admission Requirements	None			
Recommended Previous Knowledge	- Mathematics I - III - Numerical Mathematics 1/ Numerics - Basic knowledge of the programming language Python			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> Students are able to - name, state and classify state-of-the-art Krylov subspace methods for the solution of the core problems of the engineering sciences, namely, eigenvalue problems, solution of linear systems, and model reduction; - state approaches for the solution of matrix equations (Sylvester, Lyapunov, Riccati). <i>Skills</i> Students are capable to - implement and assess basic Krylov subspace methods for the solution of eigenvalue problems, linear systems, and model reduction; - assess methods used in modern software with respect to computing time, stability, and domain of applicability; - adapt the approaches learned to new, unknown types of problem. Personal Competence <i>Social Competence</i> Students can - develop and document joint solutions in small teams; - form groups to further develop the ideas and transfer them to other areas of applicability; - form a team to develop, build, and advance a software library. <i>Autonomy</i> Students are able to - correctly assess the time and effort of self-defined work; - assess whether the supporting theoretical and practical exercises are better solved individually or in a team; - define test problems for testing and expanding the methods; - assess their individual progress and, if necessary, to ask questions and seek help.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	None			
Examination	Oral exam			
Examination duration and scale	25 min			
Assignment for the Following Curricula	Computational Engineering: Core Qualification: Elective Compulsory Data Science: Specialisation IV. Special Focus Area: Elective Compulsory Data Science: Specialisation I. Data Science & Mathematics: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Mechatronics: Technical Complementary Course: Elective Compulsory Technomathematics: Specialisation I. Mathematics: Elective Compulsory Technomathematics: Specialisation II. Informatics: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Simulation Technology: Elective Compulsory			

Course L0984: Matrix Algorithms	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Dr. Jens-Peter Zemke
Language	EN
Cycle	WiSe
Content	- Part A: Krylov Subspace Methods: - Basics (derivation, basis, Ritz, OR, MR) - Arnoldi-based methods (Arnoldi, GMRes) - Lanczos-based methods (Lanczos, CG, BiCG, QMR, SymmLQ, PVL) - Sonneveld-based methods (IDR, BiCGStab, TFQMR, IDR(s)) - Part B: Matrix Equations: - Sylvester Equation - Lyapunov Equation - Algebraic Riccati Equation
Literature	Skript (224 Seiten) Ergänzend können die folgenden Lehrbücher herangezogen werden: - Saad, Yousef. Numerical methods for large eigenvalue problems: revised edition. Society for Industrial and Applied Mathematics, 2011. - Saad, Yousef. Iterative methods for sparse linear systems. Society for Industrial and Applied Mathematics, 2003. - Van der Vorst, Henk A. Iterative Krylov methods for large linear systems. No. 13. Cambridge University Press, 2003. - Liesen, Jörg, and Zdenek Strakos. Krylov subspace methods: principles and analysis. Oxford University Press, 2013.

Course L0985: Matrix Algorithms	
Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Dr. Jens-Peter Zemke
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M1966: Mathematical Image Processing			
Courses			
Title	Typ	Hrs/wk	CP
Mathematical Image Processing (L0991)	Lecture	3	4
Mathematical Image Processing (L0992)	Recitation Section (small)	1	2
Module Responsible	Prof. Marko Lindner		
Admission Requirements	None		
Recommended Previous Knowledge	- Analysis: partial derivatives, gradient, directional derivative - Linear Algebra: eigenvalues, least squares solution of a linear system		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence <i>Knowledge</i>	Students are able to - characterize and compare diffusion equations - explain elementary methods of image processing - explain methods of image segmentation and registration - sketch and interrelate basic concepts of functional analysis		
<i>Skills</i>	Students are able to - implement and apply elementary methods of image processing - explain and apply modern methods of image processing		
Personal Competence <i>Social Competence</i>	Students are able to work together in heterogeneously composed teams (i.e., teams from different study programs and background knowledge) and to explain theoretical foundations.		
<i>Autonomy</i>	- Students are capable of checking their understanding of complex concepts on their own. They can specify open questions precisely and know where to get help in solving them. - Students have developed sufficient persistence to be able to work for longer periods in a goal-oriented manner on hard problems.		
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56		
Credit points	6		
Course achievement	None		
Examination	Written exam		
Examination duration and scale	20 min		
Assignment for the Following Curricula	Bioprocess Engineering: Specialisation A - General Bioprocess Engineering: Elective Compulsory Computer Science: Specialisation III. Mathematics: Elective Compulsory Computer Science in Engineering: Specialisation III. Mathematics: Elective Compulsory Interdisciplinary Mathematics: Specialisation Computational Methods in Biomedical Imaging: Compulsory Mechatronics: Core Qualification: Elective Compulsory Technomathematics: Specialisation I. Mathematics: Elective Compulsory Technomathematics: Specialisation II. Informatics: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory Process Engineering: Specialisation Process Engineering: Elective Compulsory		

Course L0991: Mathematical Image Processing	
Typ	Lecture
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Lecturer	Prof. Marko Lindner
Language	DE/EN
Cycle	WiSe
Content	- basic methods of image processing - smoothing filters - the diffusion / heat equation - variational formulations in image processing - edge detection - de-convolution - inpainting - image segmentation - image registration
Literature	Bredies/Lorenz: Mathematische Bildverarbeitung

Course L0992: Mathematical Image Processing	
Typ	Recitation Section (small)
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Prof. Marko Lindner
Language	DE/EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M1986: Applied Statistics				
Courses				
Title	Typ		Hrs/wk	CP
Applied Statistics (L1584)	Lecture		2	3
Applied Statistics (L1586)	Project-/problem-based Learning		2	2
Applied Statistics (L1585)	Recitation Section (small)		1	1
Module Responsible	Prof. Matthias Schulte			
Admission Requirements	None			
Recommended Previous Knowledge	Basic knowledge of statistical methods			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	Students can explain the statistical methods and the conditions of their use. Students are able to use the statistics program to solve statistics problems and to interpret and depict the results Team Work, joined presentation of results To understand and interpret the question and solve			
<i>Knowledge</i>				
<i>Skills</i>				
Personal Competence				
<i>Social Competence</i>				
<i>Autonomy</i>				
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	Yes	10 %	Subject	theoretical and practical work
Examination	Written exam			
Examination duration and scale	90 minutes			
Assignment for the Following Curricula	Mechanical Engineering - Product Development and Production: Core Qualification: Elective Compulsory Mechanical Engineering and Management: Specialisation Management: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Biomedical Engineering: Core Qualification: Compulsory Product Development, Materials and Production: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Core Qualification: Elective Compulsory			

Course L1584: Applied Statistics	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Matthias Schulte
Language	DE/EN
Cycle	WiSe
Content	The goal is to introduce students to the basic statistical methods and their application to simple problems. The topics include: • Chi square test • Simple regression and correlation • Multiple regression and correlation • One way analysis of variance • Two way analysis of variance • Discriminant analysis • Analysis of categorical data • Choosing the appropriate statistical method • Determining critical sample sizes
Literature	Applied Regression Analysis and Multivariable Methods, 3rd Edition, David G. Kleinbaum Emory University, Lawrence L. Kupper University of North Carolina at Chapel Hill, Keith E. Muller University of North Carolina at Chapel Hill, Azhar Nizam Emory University, Published by Duxbury Press, CB © 1998, ISBN/ISSN: 0-534-20910-6

Course L1586: Applied Statistics	
Typ	Project-/problem-based Learning
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Matthias Schulte
Language	DE/EN
Cycle	WiSe
Content	The students receive a problem task, which they have to solve in small groups (n=5). They do have to collect their own data and work with them. The results have to be presented in an executive summary at the end of the course.
Literature	Selbst zu finden

Course L1585: Applied Statistics	
Typ	Recitation Section (small)
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Prof. Matthias Schulte
Language	DE/EN
Cycle	WiSe
Content	The different statistical tests are applied for the solution of realistic problems using actual data sets and the most common used commercial statistical software package (SPSS).
Literature	Student Solutions Manual for Kleinbaum/Kupper/Muller/Nizam's Applied Regression Analysis and Multivariable Methods, 3rd Edition, David G. Kleinbaum Emory University Lawrence L. Kupper University of North Carolina at Chapel Hill, Keith E. Muller University of North Carolina at Chapel Hill, Azhar Nizam Emory University, Published by Duxbury Press, Paperbound © 1998, ISBN/ISSN: 0-534-20913-0

Module M2018: Research Trends in Systems and Control (seminar)				
Courses				
Title		Typ	Hrs/wk	CP
Research Trends in Systems and Control (L1803)		Seminar	2	2
Module Responsible	Prof. Timm Faulwasser			
Admission Requirements	None			
Recommended Previous Knowledge	• Introduction to control systems • Control theory and design • optimal and robust control			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence <i>Knowledge</i> <i>Skills</i>	• Students can explain modern control. • Students learn to apply basic control concepts for different tasks • Students acquire knowledge about selected aspects of modern control, based on specified literature • Students generalize developed results and present them to the participants • Students practice to prepare and give a presentation			
Personal Competence <i>Social Competence</i> <i>Autonomy</i>	• Students are capable of developing solutions and present them • They are able to provide appropriate feedback and handle constructive criticism of their own results • Students evaluate advantages and drawbacks of different forms of presentation for specific tasks and select the best solution • Students familiarize themselves with a scientific field, are able of introduce it and follow presentations of other students, such that a scientific discussion develops			
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28			
Credit points	2			
Course achievement	None			
Examination	Presentation			
Examination duration and scale	90 min			
Assignment for the Following Curricula	Mechatronics: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory			
Course L1803: Research Trends in Systems and Control				
Typ	Seminar			
Hrs/wk	2			
CP	2			
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28			
Lecturer	Prof. Timm Faulwasser			
Language	EN			
Cycle	WiSe/SoSe			
Content	• Seminar on selected topics in modern control			
Literature	• To be specified			

Module M2020: Special topics in Sensor/Actuator Design				
Courses				
Title	Typ		Hrs/wk	CP
Special topics in Sensor/Actuator Design (L3272)	Lecture		3	3
Special topics in Sensor/Actuator Design (L3273)	Practical Course		2	3
Module Responsible	Dr. Mohammad Sadeghi			
Admission Requirements	None			
Recommended Previous Knowledge	Measurement technology Basic Understanding of Structural mechanics, Acoustic, Magnetic and electrostatic field physics General knowledge of Arduino coding			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence <i>Knowledge</i> <i>Skills</i>	Comprehensive understanding of sensor and actuator design principles via Multiphysics simulation methods. Using the theoretical knowledge in a real-world application by different measurement schemes Group projects involve designing and evaluating innovative sensor and actuator solutions for specific applications. How to process the sensor/actuator data and write a scientific report on it Skill in experimental measurements and data analysis for sensor/actuator performance evaluation. Competence in interpreting physical phenomena and translating them into sensor/actuator design parameters. Capability to integrate knowledge from multiple disciplines (mechanical engineering, electrical engineering, physics) for sensor/actuator design. Students will engage in innovative approaches to prepare and formulate complex sensor/actuator design challenges through scientific methods. They will apply diverse parametric design techniques grounded in theory to address the intricacies of sensor and actuator design.			
Personal Competence <i>Social Competence</i> <i>Autonomy</i>	Effective communication skills for presenting practical projects and engaging in interactive discussions. Collaborative skills for working in groups during practical sessions and project presentations. Respectful and open-minded attitude towards diverse perspectives and ideas within a multidisciplinary learning environment. Students will collaborate in small design teams to tackle technical-scientific challenges in the realm of sensor and actuator design. Through the application of innovative methodologies, both in simulation and practical experimentation, they will work on real-world application scenarios. Students' initiative to engage in independent study and research on sensors/actuators design principles and methodologies. Ability to manage time effectively to meet course requirements and deadlines. Capacity to adapt to new technologies and methodologies independently for developing new sensor/actuator. Self-assessment and reflection skills to evaluate personal progress and learning outcomes throughout the course.			
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	Yes	None	Subject	theoretical and Practical Work and Presentation practical work
Examination	Written exam			
Examination duration and scale	90 min			
Assignment for the Following Curricula	Mechatronics: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory			

Course L3272: Special topics in Sensor/Actuator Design	
Typ	Lecture
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Lecturer	Dr. Mohammad Sadeghi
Language	EN
Cycle	WiSe
Content	1. Introduction to course rules Introduction to sensor/actuator characteristics (Sensitivity, linearity, limit of detection,) Introduction to finite element method 2. Introduction to different FEM solver types, (Stationary, time dependent, frequency domain) Introduction to the standard solving procedure for FEM model Solving step-by-step example on dynamic vibration (2D) 3. Parametric evaluation example (2D) Modal analyses of Electro-Mechanical transducer (3D) 4. Multiphysics Simulation Modeling of Piezoelectric actuator 5. Multiphysics Simulation Modeling of Magnetostrictive actuator 6. Multiphysics Simulation Modeling of IMU MEMS Accelerometer 7. Multiphysics Simulation Modeling Electric impedance tomography/ Electric Impedance Sensor 8. Multiphysics Simulation Modeling of Hall Effect Sensor 9. Multiphysics Simulation Modeling a Capacitive Position Sensor 10. Multiphysics Simulation Modeling of Ultrasonic Car Parking Sensor 11. Build an application with FEA software (Parametrize the model) 12. Application of inverse problem in Multiphysics model optimization
Literature	[1] Multiphysics Modeling Using COMSOL? A First Principles Approach, Jones & Bartlett Learning, Roger Pryor, 2011. [2] COMSOL Multiphysics help source, version 5 or later. [3] Application Gallery, https://www.comsol.com/models. [4] Arduino Project Handbook, By Mark Geddes, SBN:9781593276904 , 2016. [5] Sensors and Actuators By D.A. Hall, ISBN:9781000122459, 2020.

Course L3273: Special topics in Sensor/Actuator Design	
Typ	Practical Course
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Dr. Mohammad Sadeghi
Language	EN
Cycle	WiSe
Content	1. Making familiar with useful sensors/actuators measurements electronic devices 2. Making familiar with useful sensors/actuators measurements electronic devices 3. Digital signal processing for sensor and actuator design 4. Digital signal processing for sensor and actuator design 5. Utilizing FEM simulation knowledge to gain a comprehensive understanding of sensor application in hand impedance measurement 6. Utilizing FEM simulation knowledge to gain a comprehensive understanding of sensor application in detecting magnetic fields with Hall effect 7. Utilizing FEM simulation knowledge to gain a comprehensive understanding of sensor application for measuring velocity, force, and orientation 8. Utilizing FEM simulation knowledge to gain a comprehensive understanding of sensors for detecting a barrier in car parking application 9. Group Project - (Max 4 students): Group A: Design, Fabrication, and Evaluation of a Rotary Encoder for Motor Speed Control Group B: Design, Fabrication, and Optimization of FSR Sensor for Pressure Sensing Group C: Design, Fabrication, and Evaluation of Electromagnetic Actuator for Precise Actuation
Literature	[1] Multiphysics Modeling Using COMSOL? A First Principles Approach, Jones & Bartlett Learning, Roger Pryor, 2011. [2] COMSOL Multiphysics help source, version 5 or later. [3] Application Gallery, https://www.comsol.com/models. [4] Arduino Project Handbook, By Mark Geddes, SBN:9781593276904 , 2016. [5] Sensors and Actuators By D.A. Hall, ISBN:9781000122459 , 2020.

Module M1725: Scientific Working in Computational Engineering				
Courses				
Title	Scientific Working in Computational Engineering (L2764)		Typ	Hrs/wk
			Project-/problem-based Learning	6
Module Responsible	Prof. Kay Smarsly			
Admission Requirements	None			
Recommended Previous Knowledge	Basic knowledge in scientific writing. String interest in topics related to computing in civil engineering.			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence				
<i>Knowledge</i>	The students will learn to apply concepts and methods of scientific working in computational engineering. In interaction with the course instructors and in collaboration with each other, the students will also learn to understand the complex process of scientific thinking, being able to accurately plan, implement and analyze scientific projects, such as prospective master theses. A project will be conducted throughout the semester, which will contribute to the grade. Since scientific writing is of particular importance in this course, a scientific paper will be developed based, which is a prerequisite for the final examination. The paper will be written based on the project conducted within this course. Project meetings in small groups, presentations, and critical discussions of scientific publications are further key activities.			
<i>Skills</i>	The students will be capable (i) of solving a scientific problem following a scientific methodology, (ii) of documenting their work effectively in the form of a paper, and (iii) of sharing their work in a presentation.			
Personal Competence				
<i>Social Competence</i>	The students will be able to work in a multidisciplinary team and develop communication skills necessary for problem solving.			
<i>Autonomy</i>	The students will be able to extend their knowledge and apply it to solve scientific problems by working independently in a project.			
Workload in Hours	Independent Study Time 96, Study Time in Lecture 84			
Credit points	6			
Course achievement	None			
Examination	Written elaboration			
Examination duration and scale	10 pages of work with 15-minute oral presentation			
Assignment for the Following Curricula	Civil Engineering: Specialisation Water and Traffic: Elective Compulsory Civil Engineering: Specialisation Geotechnical Engineering: Elective Compulsory Civil Engineering: Specialisation Coastal Engineering: Elective Compulsory Civil Engineering: Specialisation Structural Engineering: Elective Compulsory Civil Engineering: Specialisation Computational Engineering: Elective Compulsory Computer Science: Specialisation II: Intelligence Engineering: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Mechatronics: Technical Complementary Course: Elective Compulsory			

Course L2764: Scientific Working in Computational Engineering	
Typ	Project-/problem-based Learning
Hrs/wk	6
CP	6
Workload in Hours	Independent Study Time 96, Study Time in Lecture 84
Lecturer	Prof. Kay Smarsly
Language	EN
Cycle	WiSe/SoSe
Content	In the course, a scientific problem of practical relevance will first be defined, taking into account the interests of the students participating in the course. The scientific problem will then systematically be solved within the framework of a comprehensive project. The principles of scientific working will be taught based on the scientific problem defined previously. As an integral part of scientific working, fundamentals of scientific writing will be presented and applied to a scientific paper to be written during the course. Topics related to scientific writing include structuring in scientific writing (structuring the abstract, the introduction, the main part, the summary and conclusions, and the acknowledgments and references) and recommendations on effective scientific writing (principles of composition, use of English in scientific writing, useful tips, creating figures, writing in mathematics, referencing, and formal email correspondence). A final paper and a final presentation will be assembled by the students.
Literature	Smarsly, K. & Dragos, K., 2019. Scientific Writing in Engineering. Tredition, Hamburg, Germany.

Module M1971: Flexible Multibody Systems				
Courses				
Title		Typ	Hrs/wk	CP
Flexible Multibody Systems (L1632)		Integrated Lecture	3	6
Module Responsible	Prof. Robert Seifried			
Admission Requirements	None			
Recommended Previous Knowledge				
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence				
Knowledge				
Skills				
Personal Competence				
Social Competence				
Autonomy				
Workload in Hours	Independent Study Time 138, Study Time in Lecture 42			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	No	20 %	Exercises	
Examination	Subject theoretical and practical work			
Examination duration and scale	individuell			
Assignment for the Following Curricula	Aircraft Systems Engineering: Core Qualification: Elective Compulsory International Management and Engineering: Specialisation II. Product Development and Production: Elective Compulsory International Management and Engineering: Specialisation II. Mechatronics: Elective Compulsory Aeronautics: Core Qualification: Elective Compulsory Mechanical Engineering - Product Development and Production: Core Qualification: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Product Development, Materials and Production: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Core Qualification: Elective Compulsory			

Course L1632: Flexible Multibody Systems	
Typ	Integrated Lecture
Hrs/wk	3
CP	6
Workload in Hours	Independent Study Time 138, Study Time in Lecture 42
Lecturer	Prof. Robert Seifried, Dr. Alexander Held
Language	DE
Cycle	WiSe
Content	1. Basics of Multibody Systems 2. Basics of Continuum Mechanics 3. Linear finite element models and model reduction 4. Nonlinear finite element Models: absolute nodal coordinate formulation 5. Kinematics of an elastic body 6. Kinetics of an elastic body 7. System assembly
Literature	Schwertassek, R. und Wallrapp, O.: Dynamik flexibler Mehrkörpersysteme. Braunschweig, Vieweg, 1999. Seifried, R.: Dynamics of Underactuated Multibody Systems, Springer, 2014. Shabana, A.A.: Dynamics of Multibody Systems. Cambridge Univ. Press, Cambridge, 2004, 3. Auflage.

Module M2019: Nonlinear Model Predictive Control - Theory and Application				
Courses				
Title			Typ	Hrs/wk
Nonlinear Model Predictive Control - Theory and Application (L3283)			Lecture	3
Nonlinear Model Predictive Control - Theory and Application (L3284)			Project-/problem-based Learning	2
Module Responsible	Prof. Timm Faulwasser			
Admission Requirements	None			
Recommended Previous Knowledge	Basic of control engineering (stability, simple control designs), state space models in control, differential equations.			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence				
<i>Knowledge</i>	Static and dynamic optimization methods, optimal control and numerical solution methods, design and implementation of model predictive control schemes in sampled-data fashion, dissipativity notions for optimal control.			
<i>Skills</i>	The students are able to formulate and to solve problems of operation and control of technical systems on their own. The students are able to understand and to analyze the interplay of problem formulation and efficiency aspects of numerical solutions and to deduce problem-specific formulations. They know how to apply and to implement optimization methods to practical problems. Furthermore, the students can tackle complex problems of predictive control by means of abstraction, they are able to document their results in written form. The students are able to design predictive controllers for nonlinear systems and to validate them by means of simulation.			
Personal Competence				
<i>Social Competence</i>	Interaction in interdisciplinary teams, meeting of project deadlines.			
<i>Autonomy</i>	Compare to Fachkompetenz (Fertigkeiten)			
Workload in Hours	Independent Study Time 200, Study Time in Lecture 70			
Credit points	9			
Course achievement	Compulsory	Bonus	Form	Description
	No	20 %	Subject	theoretical and practical work
Examination	Oral exam			
Examination duration and scale	40 min			
Assignment for the Following Curricula	Electrical Engineering and Information Technology: Specialisation Control and Power Systems Engineering: Elective Compulsory Electrical Engineering: Specialisation Control and Power Systems Engineering: Elective Compulsory Energy Systems: Core Qualification: Elective Compulsory Aeronautics: Core Qualification: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Core Qualification: Elective Compulsory Process Engineering: Specialisation Process Engineering: Elective Compulsory Process Engineering: Specialisation Environmental Process Engineering: Elective Compulsory Process Engineering: Specialisation Chemical Process Engineering: Elective Compulsory			

Course L3283: Nonlinear Model Predictive Control - Theory and Application	
Typ	Lecture
Hrs/wk	3
CP	6
Workload in Hours	Independent Study Time 138, Study Time in Lecture 42
Lecturer	Prof. Timm Faulwasser
Language	EN
Cycle	WiSe
Content	
Literature	

Course L3284: Nonlinear Model Predictive Control - Theory and Application	
Typ	Project-/problem-based Learning
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Timm Faulwasser
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M1402: Machine Learning in Logistics				
Courses				
Title	Typ		Hrs/wk	CP
Digitalization in Traffic and Logistics (L2004)	Lecture		1	2
Basics of Machine Learning (L2003)	Lecture		1	2
Machine Learning in Logistics (L2005)	Recitation Section (small)		2	2
Module Responsible	Prof. Carlos Jahn			
Admission Requirements	None			
Recommended Previous Knowledge	None			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> Students understand specific methods of machine learning. They are able to select appropriate procedures for given data. They can explain the principals of different learning methods. In addition, they can explain the major conceptual differences of learning methods. <i>Skills</i> Students can inspect, describe, and apply selected machine learning techniques to provided data sets. Additionally they can prepare raw data for machine learning algorithms. They are able to evaluate the usability in concrete company-relevant contexts and they know how to derive the requirements and potentials of an effective application, e.g. in relation to controlling or forecasting for the operational planning of companies or other organizations. Personal Competence <i>Social Competence</i> Students are capable of: • Discussing and organizing extensive research tasks in small groups • Jointly describing, differentiating between and evaluating problems <i>Autonomy</i> Students are able: • To research and select specialized literature • Read existing code, interpret it and modify it for new tasks			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	No	15 %	Presentation	
Examination	Written exam			
Examination duration and scale	90 minutes			
Assignment for the Following Curricula	Data Science: Specialisation III. Applications: Elective Compulsory International Management and Engineering: Specialisation II. Logistics: Elective Compulsory Logistics, Infrastructure and Mobility: Specialisation Infrastructure and Mobility: Elective Compulsory Logistics, Infrastructure and Mobility: Specialisation Production and Logistics: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory			

Course L2004: Digitalization in Traffic and Logistics	
Typ	Lecture
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Prof. Carlos Jahn
Language	EN
Cycle	WiSe
Content	When dealing with large amounts of data (big data), it is no longer possible for humans to spot all relevant data by simply looking at the raw data. In the context of logistics, the handling of temporal data and movement data plays a particularly important role. In this course the visualization, the calculation of statistics, and the application of machine learning algorithms are covered. Students are given various tools for later practical application. The course utilizes the machine learning methods learned in "Basics of Machine Learning". These are used and evaluated in the context of practical application in the field of traffic and logistics. In addition, various pre-processing steps for raw data are presented and it is discussed, under which conditions these measurements are applicable. The lecture contents are: • The project structure for Machine Learning in science and industry • Use cases for Machine Learning in logistics • ML-based Computer Vision in road traffic and logistics • Temporal and movement data in traffic • Automated anomaly detection
Literature	• Géron, Aurélien (2022). Hands-On Machine Learning with Scikit-Learn, Keras, and Tensorflow: Concepts, Tools, and Techniques to Build Intelligent Systems. O'Reilly. • VanderPlas, Jake (2023). Python Data Science Handbook: Essential Tools for Working with Data. O'Reilly. • Chapman, Peter and Clinton, Janet and Kerber, Randy and Khabaza, Tom and Reinartz, Thomas and Russel H. Shearer, C and Wirth, Robert (2000). CRISP-DM 1.0 : Step-by-step data mining guide. • Ayyadevara, V Kishore; Reddy, Yeshwanth (2024). Modern Computer Vision with PyTorch: A practical roadmap from deep learning fundamentals to advanced applications and Generative AI. pact. • Aggarwal, Charu C. (2017). Outlier Analysis. Springer International Publishing Switzerland.

Course L2003: Basics of Machine Learning	
Typ	Lecture
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Prof. Sibylle Schupp
Language	EN
Cycle	WiSe
Content	Students are able to understand specific procedures of machine learning and to use on real life examples. Students are able to use appropriate procedures for given data. Students are able to explain the differences between instance and model based learning approaches and are able to use specific approaches in machine learning on the base of static and incremental growing data. By the use of uncertainty the students can explain how axioms, parameter or structures can be learned. Additional the students learn to develop different cluster techniques. Planned content: • Supervised Learning: ◦ Regressions ◦ Decision trees ◦ Bayesian networks ◦ K-next neighbors ◦ Logistical regressions ◦ Neuronal Networks ◦ Support Vector Machines ◦ Ensemble Learning • Unsupervised Learning: ◦ Hierarchical Clustering, K-Mean
Literature	John D. Kelleher, Fundamentals of Machine Learning for Predictive Data Analytics: Algorithms, Worked Examples, and Case Studies (MIT Press) Tom M. Mitchell, Machine Learning Kevin P. Murphy, Machine Learning: A Probabilistic Perspective

Course L2005: Machine Learning in Logistics	
Typ	Recitation Section (small)
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Carlos Jahn
Language	EN
Cycle	WiSe
Content	In the exercise, the skills which the students acquired in the lectures will be applied to real life examples.
Literature	• Aggarwal, Charu C. (2017). Outlier Analysis. Springer International Publishing Switzerland. • Chapman, Peter and Clinton, Janet and Kerber, Randy and Khabaza, Tom and Reinartz, Thomas and Russel H. Shearer, C and Wirth, Robert (2000). DM 1.0 : Step-by-step data mining guide. • Géron, Aurélien (2018). Praxiseinstieg Machine Learning mit Scikit-Learn und TensorFlow: Konzepte, Tools und Techniken für intelligente Systeme. O'Reilly. • Haneke, Uwe and Trahasch, Stephan and Zimmer, Michael and Felden, Carsten (2019). Data Science - Grundlagen, Architekturen und Anwendungen. dpunkt. • Kelleher, John D. (2015) Fundamentals of Machine Learning for Predictive Data Analytics: Algorithms, Worked Examples, and Case Studies. MIT Press. • Mitchell, Tom M. (2005) Machine Learning. McGraw-Hill. • Murphy, Kevin P. (2012) Machine Learning: A Probabilistic Perspective. MIT Press. • VanderPlas, Jake (2017). Data Science mit Python : das Handbuch für den Einsatz von IPython, Jupyter, NumPy, Pandas, Matplotlib, Scikit-Learn. MIT Press.

Module M2079: GSD - Generational Sheet-Metal Development				
Courses				
Title	Typ		Hrs/wk	CP
GSD - Generational Sheet-Metal Development (L3064)	Project-/problem-based Learning		4	6
Module Responsible	Prof. Nikola Bursac			
Admission Requirements	None			
Recommended Previous Knowledge	No specific prior knowledge is required. Experience in engineering design and basic manufacturing techniques is advantageous.			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> The students gain a comprehensive understanding of product development processes. Through practical examples and the teaching of methodological knowledge, they learn how to directly apply these concepts in development projects, allowing them to gain valuable hands-on experience. They understand the fundamental principles and methods of system generation development and practice these within the scope of their own product development project. A particular focus is placed on design for manufacturability in sheet metal construction, the development process, and problem-solving strategies. Students learn how to apply design principles to develop innovative and practical solutions that are both functional and feasible. Additionally, CAD tools are used to optimize the preparation of their ideas for manufacturing. The course combines theoretical knowledge with creativity and practical experience. This enables students to actively participate in product development and further enhance their skills. <i>Skills</i> Students acquire the ability to: • Learn and apply methods for problem-solving within the development context • Analyze and evaluate functional principles • Identify and incorporate customer requirements • Understand, deepen, and implement system generation development • Develop designs optimized for manufacturability • Use CAD tools for design tasks • Leverage creative ideas in product development Personal Competence <i>Social Competence</i> Students are capable of collaborating in interdisciplinary teams to develop generation-driven solutions for real-world product development challenges. In the process, they enhance their communication skills and strengthen their ability to give and receive constructive feedback. Additionally, they foster self-responsibility and self-organization to efficiently achieve project goals together with their team. <i>Autonomy</i> Students are capable of independently identifying problems in product development, devising appropriate measures to solve them, and applying these solutions directly. They can independently structure development projects, set priorities, and implement solution approaches in a focused and goal-oriented manner.			
Workload in Hours				
Credit points				
Course achievement	Compulsory	Bonus	Form	Description
	Yes	None	Presentation	GSD Konstruktion
Examination	Oral exam			
Examination duration and scale	40 min			
Assignment for the Following Curricula	Mechanical Engineering - Product Development and Production: Specialisation Product Development: Elective Compulsory Mechanical Engineering - Product Development and Production: Specialisation Materials: Elective Compulsory Mechanical Engineering - Product Development and Production: Specialisation Production: Elective Compulsory Mechanical Engineering and Management: Core Qualification: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Product Development, Materials and Production: Specialisation Product Development: Elective Compulsory Product Development, Materials and Production: Specialisation Materials: Elective Compulsory Product Development, Materials and Production: Specialisation Production: Elective Compulsory			

Course L3064: GSD - Generational Sheet-Metal Development	
Typ	Project-/problem-based Learning
Hrs/wk	4
CP	6
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56
Lecturer	Prof. Nikola Bursac
Language	DE
Cycle	WiSe
Content	Experience in mechanical engineering design and the fundamentals of manufacturing engineering After successful completion of the course, students will be able to explain development projects using the theory of product generation engineering and explain design rules for sheet metal development. After successful completion of the course, students will be able to apply the theory of product generation engineering to development tasks and develop sheet-metal products suitable for production. After successful completion of the course, students will be able to develop a product in a team and to compete against other teams. After successful completion of the course, students will be able to independently access knowledge required for sheet metal development.
Literature	[1] Albers, A., Rapp, S., Spadinger, M., Richter, T., Birk, C., Marthaler, M., Heimicke, J., Kurtz, V., Wessels, H. (2019). Das Referenzsystem im Modell der PGE - Produktgenerationsentwicklung: Vorschlag einer generalisierten Beschreibung von Referenzprodukten und ihrer Wechselbeziehungen, ICED 2019 [2] Albers et al. (2017). PGE - Produktgenerationsentwicklung am Beispiel des Zweimassenschwungrads, Forschung im Ingenieurwesen 81(1), 13-31, 2017 [3] Albers, A., Heimicke, J., et. al. (2018). Product Profiles: Modelling customer benefits as a foundation to bring inventions to innovations. Procedia CIRP. (70), 253-258 [4] Stachowiak, H., (1973). Allgemeine Modelltheorie. ISBN 3-211-81106-0 [5] Rophol, G., (2009). Systemtheorie der Technik. ISBN: 978-3-86644-374-7 [6] Herstatt, C., & Verworn, B. (2007). Management der frühen Innovationsphasen: Grundlagen - Methoden - Neue Ansätze. Gabler Verlag.

Module M2161: Analog IC Design				
Courses				
Title	Typ		Hrs/wk	CP
Analog IC Design (L0691)	Lecture		3	4
Analog IC Design (L0998)	Recitation Section (small)		1	2
Module Responsible	Prof. Qiang Li			
Admission Requirements	None			
Recommended Previous Knowledge	Basic knowledge of (solid-state) physics and mathematics. Knowledge in fundamentals of electrical engineering and electrical networks.			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence <i>Knowledge</i>	- Students can explain basic concepts of electron transport in semiconductor devices (energy bands, generation/recombination, carrier concentrations, drift and diffusion current densities, semiconductor device equations). - Students are able to explain functional principles of pn-diodes, MOS capacitors, and MOSFETs using energy band diagrams. - Students can present and discuss current-voltage relationships and small-signal equivalent circuits of these devices. - Students can explain the physics and current-voltage behavior transistors based on charged carrier flow. - Students are able to explain the basic concepts for static and dynamic logic gates for integrated circuits - Students can exemplify approaches for low power consumption on the device and circuit level - Students can describe the potential and limitations of analytical expression for device and circuit analysis. - Students can explain characterization techniques for MOS devices.			
<i>Skills</i>				
Personal Competence <i>Social Competence</i>				
<i>Autonomy</i>				
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	None			
Examination	Written exam			
Examination duration and scale	90 min			
Assignment for the Following Curricula	Electrical Engineering and Information Technology: Specialisation Nanoelectronics and Microsystems Technology: Elective Compulsory International Management and Engineering: Specialisation II. Electrical Engineering: Elective Compulsory Mechanical Engineering and Management: Specialisation Mechatronics: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Microelectronics and Microsystems: Core Qualification: Elective Compulsory			

Course L0691: Analog IC Design	
Typ	Lecture
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Lecturer	Prof. Qiang Li
Language	EN
Cycle	WiSe
Content	• Electron transport in semiconductors • Electronic operating principles of diodes, MOS capacitors, and MOS field-effect transistors • MOS transistor as four terminal device • Performance degradation due to short channel effects • Scaling-down of MOS technology • Digital logic circuits • Basic analog circuits • Operational amplifiers • Bipolar and BiCMOS circuits
Literature	• Yuan Taur, Tak H. Ning: Fundamentals of Modern VLSI Devices, Cambridge University Press 1998 • R. Jacob Baker: CMOS, Circuit Design, Layout and Simulation, IEEE Press, Wiley Interscience, 3rd Edition, 2010 • Neil H.E. Weste and David Money Harris, Integrated Circuit Design, Pearson, 4th International Edition, 2013 • John E. Ayers, Digital Integrated Circuits: Analysis and Design, CRC Press, 2009 • Richard C. Jaeger and Travis N. Blalock: Microelectronic Circuit Design, Mc Graw-Hill, 4rd. Edition, 2010

Course L0998: Analog IC Design	
Typ	Recitation Section (small)
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Prof. Qiang Li
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Supplement Modules

Module M0604: High-Order FEM

Courses

Title	Typ	Hrs/wk	CP
High-Order FEM (L0280)	Lecture	3	4
High-Order FEM (L0281)	Recitation Section (large)	1	2

Module Responsible	Prof. Alexander Düster		
Admission Requirements	None		
Recommended Previous Knowledge	Knowledge of partial differential equations is recommended.		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence	<i>Knowledge</i> Students are able to + give an overview of the different (h, p, hp) finite element procedures. + explain high-order finite element procedures. + specify problems of finite element procedures, to identify them in a given situation and to explain their mathematical and mechanical background. <i>Skills</i> Students are able to + apply high-order finite elements to problems of structural mechanics. + select for a given problem of structural mechanics a suitable finite element procedure. + critically judge results of high-order finite elements. + transfer their knowledge of high-order finite elements to new problems. Personal Competence <i>Social Competence</i> Students are able to + solve problems in heterogeneous groups. + present and discuss their results in front of others. + give and accept professional constructive criticism. <i>Autonomy</i> Students are able to + assess their knowledge by means of exercises and E-Learning. + acquaint themselves with the necessary knowledge to solve research oriented tasks. + to transform the acquired knowledge to similar problems.		
Workload in Hours			
Credit points			
Course achievement	Compulsory	Bonus	Form
	No	10 %	Presentation
Examination	Written exam		
Examination duration and scale	120 min		
Assignment for the Following Curricula	Civil Engineering: Specialisation Computational Engineering: Elective Compulsory Computational Engineering: Core Qualification: Elective Compulsory International Management and Engineering: Specialisation II. Product Development and Production: Elective Compulsory Mechanical Engineering - Product Development and Production: Core Qualification: Elective Compulsory Materials Science and Engineering: Specialisation Modeling: Elective Compulsory Mechanical Engineering and Management: Specialisation Product Engineering: Elective Compulsory Mechatronics: Technical Complementary Course: Elective Compulsory Product Development, Materials and Production: Core Qualification: Elective Compulsory Naval Architecture and Ocean Engineering: Core Qualification: Elective Compulsory Naval Architecture and Ocean Engineering: Core Qualification: Elective Compulsory Technomathematics: Specialisation III. Engineering Science: Elective Compulsory Theoretical Mechanical Engineering: Core Qualification: Elective Compulsory		

Course L0280: High-Order FEM	
Typ	Lecture
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Lecturer	Prof. Alexander Düster
Language	EN
Cycle	SoSe
Content	1. Introduction 2. Motivation 3. Hierarchic shape functions 4. Mapping functions 5. Computation of element matrices, assembly, constraint enforcement and solution 6. Convergence characteristics 7. Mechanical models and finite elements for thin-walled structures 8. Computation of thin-walled structures 9. Error estimation and hp-adaptivity 10. High-order fictitious domain methods
Literature	[1] Alexander Düster, High-Order FEM, Lecture Notes, Technische Universität Hamburg-Harburg, 164 pages, 2014 [2] Barna Szabo, Ivo Babuska, Introduction to Finite Element Analysis – Formulation, Verification and Validation, John Wiley & Sons, 2011

Course L0281: High-Order FEM	
Typ	Recitation Section (large)
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Prof. Alexander Düster
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M0605: Computational Structural Dynamics			
Courses			
Title	Typ	Hrs/wk	CP
Computational Structural Dynamics (L0282)	Lecture	3	4
Computational Structural Dynamics (L0283)	Recitation Section (small)	1	2
Module Responsible	Prof. Alexander Düster		
Admission Requirements	None		
Recommended Previous Knowledge	Knowledge of partial differential equations is recommended.		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence	<i>Knowledge</i> Students are able to + give an overview of the computational procedures for problems of structural dynamics. + explain the application of finite element programs to solve problems of structural dynamics. + specify problems of computational structural dynamics, to identify them in a given situation and to explain their mathematical and mechanical background. <i>Skills</i> Students are able to + model problems of structural dynamics. + select a suitable solution procedure for a given problem of structural dynamics. + apply computational procedures to solve problems of structural dynamics. + verify and critically judge results of computational structural dynamics. Personal Competence <i>Social Competence</i> Students are able to + solve problems in heterogeneous groups. + present and discuss their results in front of others. + give and accept professional constructive criticism. <i>Autonomy</i> Students are able to + assess their knowledge by means of exercises and E-Learning. + acquaint themselves with the necessary knowledge to solve research oriented tasks. + to transform the acquired knowledge to similar problems.		
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56		
Credit points	6		
Course achievement	None		
Examination	Written exam		
Examination duration and scale	2h		
Assignment for the Following Curricula	Civil Engineering: Specialisation Computational Engineering: Elective Compulsory Computational Engineering: Core Qualification: Elective Compulsory International Management and Engineering: Specialisation II. Mechatronics: Elective Compulsory Mechanical Engineering - Product Development and Production: Core Qualification: Elective Compulsory Materials Science and Engineering: Specialisation Modeling: Elective Compulsory Mechatronics: Technical Complementary Course: Elective Compulsory Product Development, Materials and Production: Core Qualification: Elective Compulsory Naval Architecture and Ocean Engineering: Core Qualification: Elective Compulsory Naval Architecture and Ocean Engineering: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Simulation Technology: Elective Compulsory		

Course L0282: Computational Structural Dynamics	
Typ	Lecture
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Lecturer	Prof. Alexander Düster
Language	EN
Cycle	SoSe
Content	1. Motivation 2. Basics of dynamics 3. Time integration methods 4. Modal analysis 5. Fourier transform 6. Applications
Literature	[1] K.-J. Bathe, Finite-Elemente-Methoden, Springer, 2002. [2] J.L. Humar, Dynamics of Structures, Taylor & Francis, 2012.

Course L0283: Computational Structural Dynamics	
Typ	Recitation Section (small)
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Prof. Alexander Düster
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M0673: Information Theory and Coding			
Courses			
Title	Typ	Hrs/wk	CP
Information Theory and Coding (L0436)	Lecture	3	4
Information Theory and Coding (L0438)	Recitation Section (large)	2	2
Module Responsible	Prof. Gerhard Bauch		
Admission Requirements	None		
Recommended Previous Knowledge	- Mathematics 1-3 - Probability theory and random processes - Basic knowledge of communications engineering (e.g. from lecture "Fundamentals of Communications and Random Processes")		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence	<i>Knowledge</i> The students know the basic definitions for quantification of information in the sense of information theory. They know Shannon's source coding theorem and channel coding theorem and are able to determine theoretical limits of data compression and error-free data transmission over noisy channels. They understand the principles of source coding as well as error-detecting and error-correcting channel coding. They are familiar with the principles of decoding, in particular with modern methods of iterative decoding. They know fundamental coding schemes, their properties and decoding algorithms. The students are familiar with the contents of lecture and tutorials. They can explain and apply them to new problems. <i>Skills</i> The students are able to determine the limits of data compression as well as of data transmission through noisy channels and based on those limits to design basic parameters of a transmission scheme. They can estimate the parameters of an error-detecting or error-correcting channel coding scheme for achieving certain performance targets. They are able to compare the properties of basic channel coding and decoding schemes regarding error correction capabilities, decoding delay, decoding complexity and to decide for a suitable method. They are capable of implementing basic coding and decoding schemes in software. Personal Competence <i>Social Competence</i> The students can jointly solve specific problems. <i>Autonomy</i> The students are able to acquire relevant information from appropriate literature sources. They can control their level of knowledge during the lecture period by solving tutorial problems, software tools, clicker system.		
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70		
Credit points	6		
Course achievement	None		
Examination	Written exam		
Examination duration and scale	90 min		
Assignment for the Following Curricula	Data Science: Specialisation I. Data Science & Mathematics: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Information and Communication Systems: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Wireless and Sensor Technologies: Elective Compulsory Electrical Engineering: Specialisation Information and Communication Systems: Elective Compulsory Electrical Engineering: Specialisation Wireless and Sensor Technologies: Elective Compulsory Computer Science in Engineering: Specialisation II. Engineering Science: Elective Compulsory Information and Communication Systems: Core Qualification: Compulsory International Management and Engineering: Specialisation II. Electrical Engineering: Elective Compulsory Mechatronics: Technical Complementary Course: Elective Compulsory		

Course L0436: Information Theory and Coding	
Typ	Lecture
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Lecturer	Prof. Gerhard Bauch
Language	EN
Cycle	SoSe
Content	- Introduction to information theory and coding - Definitions of information: Self information, entropy - Binary entropy function - Source coding theorem - Entropy of continuous random variables: Differential entropy, differential entropy of uniformly and Gaussian distributed random variables - Source coding - Principles of lossless source coding - Optimal source codes - Prefix codes, prefix-free codes, instantaneous codes - Morse code - Huffman code - Shannon code

- Bounds on the average codeword length
- Relative entropy, Kullback-Leibler distance, Kullback-Leibler divergence
- Cross entropy
- Lempel-Ziv algorithm
- Lempel-Ziv-Welch (LZW) algorithm
- Text compression and image compression using variants of the Lempel-Ziv algorithm
- Channel models
 - AWGN channel
 - Binary-input AWGN channel
 - Binary symmetric channel (BSC)
 - Relationship between AWGN channel and BSC
 - Binary error and erasure channel (BEEC)
 - Binary erasure channel (BEC)
 - Discrete memoryless channels (DMC)
- Definitions of information for multiple random variables
 - Mutual information and channel capacity
 - Entropy, conditional entropy
 - Chain rules for entropy and mutual information
- Channel coding theorem
- Channel capacity of fundamental channels: BSC, BEC, AWGN channel, binary-input AWGN channel etc.
- Power-limited vs. bandlimited transmission
- Capacity of parallel AWGN channels
 - Waterfilling
 - Examples: Multiple input multiple output (MIMO) channels, complex equivalent baseband channels, orthogonal frequency division multiplex (OFDM)
- Source-channel coding theorem, separation theorem
- Multiuser information theory
 - Multiple access channel (MAC)
 - Broadcast channel
 - Principles of multiple access, time division multiple access (TDMA), frequency division multiple access (FDMA), non-orthogonal multiple access (NOMA), hybrid multiple access
 - Achievable rate regions of TDMA and FDMA with power constraint, energy constraint, power spectral density constraint, respectively
 - Achievable rate region of the two-user and K-user multiple access channels
 - Achievable rate region of the two-user and K user broadcast channels
 - Multiuser diversity
- Channel coding
 - Principles and types of channel coding
 - Code rate, data rate, Hamming distance, minimum Hamming distance, Hamming weight, minimum Hamming weight
 - Error detecting and error correcting codes
 - Simple block codes: Repetition codes, single parity check codes, Hamming code, etc.
 - Syndrome decoding
 - Representations of binary data
 - Non-binary symbol alphabets and non-binary codes
 - Code and encoder, systematic and non-systematic encoders
 - Properties of Hamming distance and Hamming weight
 - Decoding spheres
 - Perfect codes
 - Linear codes
 - Decoding principles
 - Syndrome decoding
 - Maximum a posteriori probability (MAP) decoding and maximum likelihood (ML) decoding
 - Hard decision and soft decision decoding
 - Log-likelihood ratios (LLRs), boxplus operation
 - MAP and ML decoding using log-likelihood ratios
 - Soft-in soft-out decoders
 - Error rate performance comparison of codes in terms of SNR per info bit vs. SNR per code bit
 - Linear block codes
 - Generator matrix and parity check matrix, properties of generator matrix and parity check matrix
 - Dual codes
 - Low density parity check (LDPC) codes
 - Sparse parity check matrix
 - Tanner graphs, cycles and girth
 - Degree distributions
 - Code rate and degree distribution
 - Regular and irregular LDPC codes
 - Message passing decoding
 - Message passing decoding in binary erasure channels (BEC)
 - Systematic encoding using erasure message passing decoding
 - Message passing decoding in binary symmetric channels (BSC)
 - Extrinsic information
 - Bit-flipping decoding
 - Effects of short cycles in the Tanner graph
 - Alternative bit-flipping decoding
 - Soft decision message passing decoding: Sum product decoding

	▪ Bit error rate performance of LDPC codes ▪ Repeat accumulate codes and variants of repeat accumulate codes ▪ Message passing decoding and turbo decoding of repeat accumulate codes ◦ Convolutional codes ▪ Encoding using shift registers ▪ Trellis representation ▪ Hard decision and soft decision Viterbi decoding ▪ Bit error rate performance of convolutional codes ▪ Asymptotic coding gain ▪ Viterbi decoding complexity ▪ Free distance and optimum convolutional codes ▪ Generator polynomial description and octal description ▪ Catastrophic convolutional codes ▪ Non-systematic and recursive systematic convolutional (RSC) encoders ▪ Rate compatible punctured convolutional (RCPC) codes ▪ Hybrid automatic repeat request (HARQ) with incremental redundancy ▪ Unequal error protection with punctured convolutional codes ▪ Error patterns of convolutional codes ◦ Concatenated codes ▪ Serial concatenated codes ▪ Parallel concatenated codes, Turbo codes ▪ Iterative decoding, turbo decoding ▪ Bit error rate performance of turbo codes ▪ Interleaver design for turbo codes ◦ Coded modulation ▪ Principle of coded modulation ▪ Achievable rates with PSK/QAM modulation ▪ Trellis coded modulation (TCM) ▪ Set partitioning ▪ Ungerböck codes ▪ Multilevel coding ▪ Bit-interleaved coded modulation
Literature	Bossert, M.: Kanalcodierung. Oldenbourg. Friedrichs, B.: Kanalcodierung. Springer. Lin, S., Costello, D.: Error Control Coding. Prentice Hall. Roth, R.: Introduction to Coding Theory. Johnson, S.: Iterative Error Correction. Cambridge. Richardson, T., Urbanke, R.: Modern Coding Theory. Cambridge University Press. Gallager, R. G.: Information theory and reliable communication. Wiley-VCH Cover, T., Thomas, J.: Elements of information theory. Wiley.

Course L0438: Information Theory and Coding	
Typ	Recitation Section (large)
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Gerhard Bauch
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1724: Smart Monitoring				
Courses				
Title	Typ		Hrs/wk	CP
Smart Monitoring (L2762)	Integrated Lecture		2	2
Smart Monitoring (L2763)	Recitation Section (small)		2	4
Module Responsible	Prof. Kay Smarsly			
Admission Requirements	None			
Recommended Previous Knowledge	Basic knowledge or interest in object-oriented modeling, programming, and sensor technologies are helpful. Interest in modern research and teaching areas, such as Internet of Things, Industry 4.0 and cyber-physical systems, as well as the will to deepen skills of scientific working, are required. Basic knowledge in scientific writing and good English skills.			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence				
<i>Knowledge</i>	The students will become familiar with the principles and practices of smart monitoring. The students will be able to design decentralized smart systems to be applied for continuous (remote) monitoring of systems in the built and in the natural environment. In addition, the students will learn to design and to implement intelligent sensor systems using state-of-the-art data analysis techniques, modern software design concepts, and embedded computing methodologies. Besides lectures, project work is also part of this module, which will be conducted throughout the semester and will contribute to the grade. In small groups, the students will design smart monitoring systems that integrate a number of "intelligent" sensors to be implemented by the students. Specific focus will be put on the application of machine learning techniques. The smart monitoring systems will be mounted on real-world (built or natural) systems, such as bridges or slopes, or on scaled lab structures for validation purposes. The outcome of every group will be documented in a paper. All students of this module will "automatically" participate with their smart monitoring system in the annual "Smart Monitoring" competition. The written papers and oral examinations form the final grades. The module will be taught in English. Limited enrollment.			
<i>Skills</i>	The students will gain insights into operating state-of-the-art smart sensor systems, used for monitoring a wide range of physical processes relevant to engineering, such as environmental, structural, or comfort monitoring. The students will be capable of devising monitoring strategies of physical processes as part of group projects, tailored to their knowledge backgrounds, and to implement the strategies in smart wireless sensor nodes, using embedded computing and programming. Finally, the students will be able to document the findings of their projects in short reports.			
Personal Competence				
<i>Social Competence</i>	The students will be able to work in groups, share parts of the work for their projects, and develop communication skills, towards achieving the common project goals.			
<i>Autonomy</i>	The students will be able to gain a solid basis on approaching and solving problems in engineering, as well as on documenting results, through their involvement in their monitoring group projects.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	None			
Examination	Written elaboration			
Examination duration and scale	10 pages of work with 15-minute oral presentation			
Assignment for the Following Curricula	Civil Engineering: Specialisation Water and Traffic: Elective Compulsory Civil Engineering: Specialisation Geotechnical Engineering: Elective Compulsory Civil Engineering: Specialisation Coastal Engineering: Elective Compulsory Civil Engineering: Specialisation Structural Engineering: Elective Compulsory Computer Science: Specialisation II: Intelligence Engineering: Elective Compulsory Environmental Engineering: Specialisation Energy and Resources: Elective Compulsory Environmental Engineering: Specialisation Environment and Climate: Elective Compulsory Environmental Engineering: Specialisation Water Quality and Water Engineering: Elective Compulsory Mechatronics: Technical Complementary Course: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory Water and Environmental Engineering: Specialisation Cities: Elective Compulsory Water and Environmental Engineering: Specialisation Environment: Elective Compulsory Water and Environmental Engineering: Specialisation Water: Elective Compulsory			

Course L2762: Smart Monitoring	
Typ	Integrated Lecture
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Kay Smarsly
Language	EN
Cycle	SoSe
Content	In this course, principles of smart monitoring will be taught, focusing on modern concepts of data acquisition, data storage, and data analysis. Also, fundamentals of intelligent sensors and embedded computing will be illuminated. Autonomous software and decentralized data processing are further crucial parts of the course, including concepts of the Internet of Things, Industry 4.0 and cyber-physical systems. Furthermore, measuring principles, data acquisition systems, data management and data analysis algorithms will be discussed. Besides the theoretical background, numerous practical examples will be shown to demonstrate how smart monitoring may advantageously be used for assessing the condition of systems in the built or natural environment.
Literature	The course contents couples different fields, such as signal processing, sensing technologies, data analytics, environmental engineering, civil engineering, artificial intelligence, database systems, and many more. The basics will be taught in this course. However, specific literature that covers all these topics does not exist. Instead, literature will be referenced in the lectures, all of which are papers that are freely available online.

Course L2763: Smart Monitoring	
Typ	Recitation Section (small)
Hrs/wk	2
CP	4
Workload in Hours	Independent Study Time 92, Study Time in Lecture 28
Lecturer	Prof. Kay Smarsly
Language	EN
Cycle	SoSe
Content	The contents of the exercises are based on the lecture contents. In addition to the exercises, project work will be conducted throughout the semester, which will consume the majority of the workload. As part of the project work, students will design smart monitoring systems that will be tested in the laboratory or in the field. As mentioned in the module description, the students will participate in the "Smart Monitoring" competition, hosted annually by the Institute of Digital and Autonomous Construction. Students are encouraged to contribute their own ideas. The tools required to implement the smart monitoring systems will be taught in the group exercises as well as through external sources, such as video tutorials and literature.
Literature	The course contents couples different fields, such as signal processing, sensing technologies, data analytics, environmental engineering, civil engineering, artificial intelligence, database systems, and many more. The basics will be taught in this course. However, specific literature that covers all these topics does not exist. Instead, literature will be referenced in the lectures, all of which are papers that are freely available online.

Module M0769: EMC I: Coupling Mechanisms, Countermeasures and Test Procedures				
Courses				
Title	Typ		Hrs/wk	CP
EMC I: Coupling Mechanisms, Countermeasures, and Test Procedures (L0743)	Lecture		3	4
EMC I: Coupling Mechanisms, Countermeasures, and Test Procedures (L0744)	Recitation Section (small)		1	1
EMC I: Coupling Mechanisms, Countermeasures, and Test Procedures (L0745)	Practical Course		1	1
Module Responsible	Prof. Christian Schuster			
Admission Requirements	None			
Recommended Previous Knowledge	Fundamentals of Electrical Engineering			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence				
<i>Knowledge</i>	Students are able to explain the fundamental principles, inter-dependencies, and methods of Electromagnetic Compatibility of electric and electronic systems and to ensure Electromagnetic Compatibility of such systems. They are able to classify and explain the common interference sources and coupling mechanisms. They are capable of explaining the basic principles of shielding and filtering. They are able of giving an overview over measurement and simulation methods for the characterization of Electromagnetic Compatibility in electrical engineering practice.			
<i>Skills</i>	Students are able to apply a series of modeling methods for the Electromagnetic Compatibility of typical electric and electronic systems. They are able to determine the most important effects that these models are predicting in terms of Electromagnetic Compatibility. They can classify these effects and they can quantitatively analyze them. They are capable of deriving problem solving strategies from these predictions and they can adapt them to applications in electrical engineering practice. They can evaluate their problem solving strategies against each other.			
Personal Competence				
<i>Social Competence</i>	Students are able to work together on subject related tasks in small groups. They are able to present their results effectively in English, during laboratory work and exercises, e.g..			
<i>Autonomy</i>	Students are capable to gather necessary information from the references provided and relate that information to the context of the lecture. They are able to make a connection between their knowledge obtained in this lecture with the content of other lectures (e.g. Theoretical Electrical Engineering and Communication Theory). They can communicate problems and solutions in the field of Electromagnetic Compatibility in english language.			
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	Yes	None	Presentation	
Examination	Oral exam			
Examination duration and scale	45 min			
Assignment for the Following Curricula	Electrical Engineering and Information Technology: Specialisation Microwave Engineering, Optics, and Electromagnetic Compatibility: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Wireless and Sensor Technologies: Elective Compulsory Electrical Engineering: Specialisation Microwave Engineering, Optics, and Electromagnetic Compatibility: Elective Compulsory Electrical Engineering: Specialisation Wireless and Sensor Technologies: Elective Compulsory Mechatronics: Technical Complementary Course: Elective Compulsory Microelectronics and Microsystems: Specialisation Microelectronics Complements: Elective Compulsory			

Course L0743: EMC I: Coupling Mechanisms, Countermeasures, and Test Procedures	
Typ	Lecture
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Lecturer	Prof. Christian Schuster
Language	EN
Cycle	SoSe
Content	• Introduction to Electromagnetic Compatibility (EMC) • Interference sources in time an frequency domain • Coupling mechanisms • Transmission lines and coupling to electromagnetic fields • Shielding • Filters • EMC test procedures
Literature	• C.R. Paul: "Introduction to Electromagnetic Compatibility", 2nd ed., (Wiley, New Jersey, 2006). • A.J. Schwab und W. Kürner: "Elektromagnetische Verträglichkeit", 6. Auflage, (Springer, Berlin 2010). • F.M. Tesche, M.V. Ianoz, and T. Karlsson: "EMC Analysis Methods and Computational Models", (Wiley, New York, 1997).

Course L0744: EMC I: Coupling Mechanisms, Countermeasures, and Test Procedures	
Typ	Recitation Section (small)
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Prof. Christian Schuster
Language	EN
Cycle	SoSe
Content	The exercise sessions serve to deepen the understanding of the concepts of the lecture.
Literature	• C.R. Paul: "Introduction to Electromagnetic Compatibility", 2nd ed., (Wiley, New Jersey, 2006). • A.J. Schwab und W. Kürner: "Elektromagnetische Verträglichkeit", 6. Auflage, (Springer, Berlin 2010). • F.M. Tesche, M.V. Ianoz, and T. Karlsson: "EMC Analysis Methods and Computational Models", (Wiley, New York, 1997). • Scientific articles and papers

Course L0745: EMC I: Coupling Mechanisms, Countermeasures, and Test Procedures	
Typ	Practical Course
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Prof. Christian Schuster
Language	EN
Cycle	SoSe
Content	Laboratory experiments serve to practically investigate the following EMC topics: • Shielding • Conducted EMC test procedures • The GTEM-cell as an environment for radiated EMC test
Literature	Versuchsbeschreibungen und zugehörige Literatur werden innerhalb der Veranstaltung bereit gestellt.

Module M1810: Autonomous Cyber-Physical Systems				
Courses				
Title	Typ		Hrs/wk	CP
Autonomous Cyber-Physical Systems (L3000)	Lecture		2	3
Autonomous Cyber-Physical Systems (L3001)	Project/problem-based Learning		3	3
Module Responsible	Prof. Bernd-Christian Renner			
Admission Requirements	None			
Recommended Previous Knowledge	• Very good knowledge and practical experience in programming in the C/C++ language (e.g., module: Procedural Programming for Computer Scientists) • Basic knowledge in software engineering • Basic knowledge in wired and wireless communication protocols • Principal understanding of simple electronic circuits			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence <i>Knowledge</i>	This master's course centers on the advanced Internet of Things (IoT) systems, spotlighting energy sustainability and Real-Time Operating Systems (RTOS) within cyber-physical frameworks. Key focus areas include the exploration of IoT system architecture, especially leveraging RTOS for real-time responsiveness, and energy-efficient technologies for sustainable operations. Upon completion, students will be capable of: • Outlining the structure and operational principles of IoT systems and their significance in modern cyber-physical environments. • Differentiating between RTOS and conventional operating systems in the context of IoT applications. • Evaluating energy harvesting (specifically solar technologies) and storage systems for their applicability and efficiency in IoT devices • Knowing different wireless technologies (e.g., Bluetooth LE, LoRaWAN, Zigbee) and networking protocols (TCP/IP, MQTT) for IoT connectivity and functionality, and pinpointing their dis-/advantages.			
<i>Skills</i>	Students will acquire practical skills to: • Describe, plan, develop and optimize IoT applications using RTOS, focusing on multitasking, efficient memory, and power management. • Select and deploy suitable wireless and networking technologies considering device requirements and operational contexts. • Integrate energy harvesting and storage technologies into IoT solutions, enhancing their sustainability and autonomy. • Conduct comprehensive planning and evaluation of IoT projects, incorporating scientific methodologies for real-world applications.			
Personal Competence <i>Social Competence</i>	• Students are able to work goal-oriented in small mixed groups. • They learn and broaden their teamwork abilities. • They learn to define and split tasks within the team. • After completing the module, the students are able to work on similar tasks alone or in a group and to present the results in a suitable way.			
<i>Autonomy</i>	After completing the module, the students are able to independently work on sub-areas of the subject using specialist literature, to summarize and present the knowledge they have acquired and to link it to the content of other courses.			
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	No	10 %	Attestation	
Examination	Written exam			
Examination duration and scale	90 min			
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Wireless and Sensor Technologies: Elective Compulsory Electrical Engineering: Specialisation Wireless and Sensor Technologies: Elective Compulsory Computer Science in Engineering: Specialisation II. Engineering Science: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems, Focus Software and Signal Processing: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Mechatronics: Technical Complementary Course: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory			

Course L3000: Autonomous Cyber-Physical Systems	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Bernd-Christian Renner
Language	EN
Cycle	SoSe
Content	
Literature	

Course L3001: Autonomous Cyber-Physical Systems	
Typ	Project-/problem-based Learning
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Lecturer	Prof. Bernd-Christian Renner
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1725: Scientific Working in Computational Engineering				
Courses				
Title	Scientific Working in Computational Engineering (L2764)		Typ	Hrs/wk
			Project-/problem-based Learning	6
Module Responsible	Prof. Kay Smarsly			
Admission Requirements	None			
Recommended Previous Knowledge	Basic knowledge in scientific writing. String interest in topics related to computing in civil engineering.			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence				
<i>Knowledge</i>	The students will learn to apply concepts and methods of scientific working in computational engineering. In interaction with the course instructors and in collaboration with each other, the students will also learn to understand the complex process of scientific thinking, being able to accurately plan, implement and analyze scientific projects, such as prospective master theses. A project will be conducted throughout the semester, which will contribute to the grade. Since scientific writing is of particular importance in this course, a scientific paper will be developed based, which is a prerequisite for the final examination. The paper will be written based on the project conducted within this course. Project meetings in small groups, presentations, and critical discussions of scientific publications are further key activities.			
<i>Skills</i>	The students will be capable (i) of solving a scientific problem following a scientific methodology, (ii) of documenting their work effectively in the form of a paper, and (iii) of sharing their work in a presentation.			
Personal Competence				
<i>Social Competence</i>	The students will be able to work in a multidisciplinary team and develop communication skills necessary for problem solving.			
<i>Autonomy</i>	The students will be able to extend their knowledge and apply it to solve scientific problems by working independently in a project.			
Workload in Hours	Independent Study Time 96, Study Time in Lecture 84			
Credit points	6			
Course achievement	None			
Examination	Written elaboration			
Examination duration and scale	10 pages of work with 15-minute oral presentation			
Assignment for the Following Curricula	Civil Engineering: Specialisation Water and Traffic: Elective Compulsory Civil Engineering: Specialisation Geotechnical Engineering: Elective Compulsory Civil Engineering: Specialisation Coastal Engineering: Elective Compulsory Civil Engineering: Specialisation Structural Engineering: Elective Compulsory Civil Engineering: Specialisation Computational Engineering: Elective Compulsory Computer Science: Specialisation II: Intelligence Engineering: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Mechatronics: Technical Complementary Course: Elective Compulsory			

Course L2764: Scientific Working in Computational Engineering	
Typ	Project-/problem-based Learning
Hrs/wk	6
CP	6
Workload in Hours	Independent Study Time 96, Study Time in Lecture 84
Lecturer	Prof. Kay Smarsly
Language	EN
Cycle	WiSe/SoSe
Content	In the course, a scientific problem of practical relevance will first be defined, taking into account the interests of the students participating in the course. The scientific problem will then systematically be solved within the framework of a comprehensive project. The principles of scientific working will be taught based on the scientific problem defined previously. As an integral part of scientific working, fundamentals of scientific writing will be presented and applied to a scientific paper to be written during the course. Topics related to scientific writing include structuring in scientific writing (structuring the abstract, the introduction, the main part, the summary and conclusions, and the acknowledgments and references) and recommendations on effective scientific writing (principles of composition, use of English in scientific writing, useful tips, creating figures, writing in mathematics, referencing, and formal email correspondence). A final paper and a final presentation will be assembled by the students.
Literature	Smarsly, K. & Dragos, K., 2019. Scientific Writing in Engineering. Tredition, Hamburg, Germany.

Module M2058: Design optimization in structural mechanics and dynamics			
Courses			
Title	Typ	Hrs/wk	CP
Design optimization in structural mechanics and dynamics (L1873)	Lecture	2	3
Design optimization in structural mechanics and dynamics (L1874)	Recitation Section (large)	2	3
Module Responsible	Prof. Benedikt Kriegesmann		
Admission Requirements	None		
Recommended Previous Knowledge	• Technical mechanics • Higher math		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence <i>Knowledge</i> • Design optimization ◦ Gradient based methods ◦ Genetic algorithms ◦ Optimization with constraints ◦ Topology optimization • Reliability analysis ◦ Stochastic basics ◦ Monte Carlo methods ◦ Semi-analytic approaches • robust design optimization ◦ Robustness measures ◦ Coupling of design optimization and reliability analysis <i>Skills</i> • Application of optimization algorithms and probabilistic methods in the design of structures • Programming with Matlab • Implementation of algorithms • Debugging Personal Competence <i>Social Competence</i> • Team work • Oral explanation of the the work <i>Autonomy</i> • Application of methods learned in the framework of a home work • Familiarizing with source code provided • Description of approaches and results			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56		
Credit points	6		
Course achievement	None		
Examination	Written elaboration		
Examination duration and scale	10 pages		
Assignment for the Following Curricula	Aircraft Systems Engineering: Core Qualification: Elective Compulsory Aeronautics: Core Qualification: Elective Compulsory Mechatronics: Technical Complementary Course: Elective Compulsory Product Development, Materials and Production: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Core Qualification: Elective Compulsory		

Course L1873: Design optimization in structural mechanics and dynamics	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Benedikt Kriegesmann, Dr. Alexander Held, Prof. Robert Seifried
Language	DE
Cycle	SoSe
Content	In the course the theoretic basics for design optimization and reliability analysis are taught, where the focus is on the application of such methods. The lectures will consist of presentations as well as computer exercises. In the computer exercises, the methods learned will be implemented in Matlab for understanding the practical realization. The following contents will be considered: • Design optimization ◦ Gradient based methods ◦ Genetic algorithms ◦ Optimization with constraints ◦ Topology optimization • Reliability analysis ◦ Stochastic basics ◦ Monte Carlo methods ◦ Semi-analytic approaches • robust design optimization ◦ Robustness measures ◦ Coupling of design optimization and reliability analysis
Literature	[1] Arora, Jasbir. Introduction to Optimum Design. 3rd ed. Boston, MA: Academic Press, 2011. [2] Haldar, A., and S. Mahadevan. Probability, Reliability, and Statistical Methods in Engineering Design. John Wiley & Sons New York/Chichester, UK, 2000.

Course L1874: Design optimization in structural mechanics and dynamics	
Typ	Recitation Section (large)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Benedikt Kriegesmann, Dr. Alexander Held, Prof. Robert Seifried
Language	DE
Cycle	SoSe
Content	Matlab exercises complementing the lecture
Literature	siehe Vorlesung

Module M0603: Nonlinear Structural Analysis				
Courses				
Title	Typ		Hrs/wk	CP
Nonlinear Structural Analysis (L0277)	Lecture		3	4
Nonlinear Structural Analysis (L0279)	Recitation Section (small)		1	2
Module Responsible	Prof. Alexander Düster			
Admission Requirements	None			
Recommended Previous Knowledge	Knowledge of partial differential equations is recommended.			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence				
<i>Knowledge</i>	Students are able to + give an overview of the different nonlinear phenomena in structural mechanics. + explain the mechanical background of nonlinear phenomena in structural mechanics. + to specify problems of nonlinear structural analysis, to identify them in a given situation and to explain their mathematical and mechanical background.			
<i>Skills</i>	Students are able to + model nonlinear structural problems. + select for a given nonlinear structural problem a suitable computational procedure. + apply finite element procedures for nonlinear structural analysis. + critically verify and judge results of nonlinear finite elements. + to transfer their knowledge of nonlinear solution procedures to new problems.			
Personal Competence				
<i>Social Competence</i>	Students are able to + solve problems in heterogeneous groups. + present and discuss their results in front of others. + give and accept professional constructive criticism.			
<i>Autonomy</i>	Students are able to + assess their knowledge by means of exercises and E-Learning. + acquaint themselves with the necessary knowledge to solve research oriented tasks. + to transform the acquired knowledge to similar problems.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	None			
Examination	Written exam			
Examination duration and scale	120 min			
Assignment for the Following Curricula	Civil Engineering: Specialisation Structural Engineering: Elective Compulsory Civil Engineering: Specialisation Computational Engineering: Compulsory Computational Engineering: Core Qualification: Elective Compulsory International Management and Engineering: Specialisation II. Civil Engineering: Elective Compulsory Mechanical Engineering - Product Development and Production: Core Qualification: Elective Compulsory Materials Science and Engineering: Specialisation Modeling: Elective Compulsory Materials Science: Specialisation Modeling: Elective Compulsory Mechatronics: Technical Complementary Course: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Product Development, Materials and Production: Core Qualification: Elective Compulsory Naval Architecture and Ocean Engineering: Core Qualification: Elective Compulsory Naval Architecture and Ocean Engineering: Core Qualification: Elective Compulsory Ship and Offshore Technology: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Simulation Technology: Elective Compulsory			

Course L0277: Nonlinear Structural Analysis	
Typ	Lecture
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Lecturer	Prof. Alexander Düster
Language	EN
Cycle	WiSe
Content	1. Introduction 2. Nonlinear phenomena 3. Mathematical preliminaries 4. Basic equations of continuum mechanics 5. Spatial discretization with finite elements 6. Solution of nonlinear systems of equations 7. Solution of elastoplastic problems 8. Stability problems 9. Contact problems
Literature	[1] Alexander Düster, Nonlinear Structural Analysis, Lecture Notes, Technische Universität Hamburg-Harburg, 2014. [2] Peter Wriggers, Nonlinear Finite Element Methods, Springer 2008. [3] Peter Wriggers, Nichtlineare Finite-Elemente-Methoden, Springer 2001. [4] Javier Bonet and Richard D. Wood, Nonlinear Continuum Mechanics for Finite Element Analysis, Cambridge University Press, 2008.

Course L0279: Nonlinear Structural Analysis	
Typ	Recitation Section (small)
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Prof. Alexander Düster
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M0781: EMC II: Signal Integrity and Power Supply of Electronic Systems				
Courses				
Title	Typ		Hrs/wk	CP
EMC II: Signal Integrity and Power Supply of Electronic Systems (L0770)	Lecture		3	4
EMC II: Signal Integrity and Power Supply of Electronic Systems (L0771)	Recitation Section (small)		1	1
EMC II: Signal Integrity and Power Supply of Electronic Systems (L0774)	Practical Course		1	1
Module Responsible	Prof. Christian Schuster			
Admission Requirements	None			
Recommended Previous Knowledge	Fundamentals of electrical engineering			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence				
<i>Knowledge</i>	Students are able to explain the fundamental principles, inter-dependencies, and methods of signal and power integrity of electronic systems. They are able to relate signal and power integrity to the context of interference-free design of such systems, i.e. their electromagnetic compatibility. They are capable of explaining the basic behavior of signals and power supply in typical packages and interconnects. They are able to propose and describe problem solving strategies for signal and power integrity issues. They are capable of giving an overview over measurement and simulation methods for characterization of signal and power integrity in electrical engineering practice.			
<i>Skills</i>	Students are able to apply a series of modeling methods for characterization of electromagnetic field behavior in packages and interconnect structure of electronic systems. They are able to determine the most important effects that these models are predicting in terms of signal and power integrity. They can classify these effects and they can quantitatively analyze them. They are capable of deriving problem solving strategies from these predictions and they can adapt them to applications in electrical engineering practice. They can evaluate their problem solving strategies against each other.			
Personal Competence				
<i>Social Competence</i>	Students are able to work together on subject related tasks in small groups. They are able to present their results effectively in English (e.g. during CAD exercises).			
<i>Autonomy</i>	Students are capable to gather necessary information from the references provided and relate that information to the context of the lecture. They are able to make a connection between their knowledge obtained in this lecture with the content of other lectures (e.g. theory of electromagnetic fields, communications, and semiconductor circuit design). They can communicate problems and solutions in the field of signal integrity and power supply of interconnect and packages in English.			
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	Yes	None	Presentation	
Examination	Oral exam			
Examination duration and scale	45 min			
Assignment for the Following Curricula	Electrical Engineering and Information Technology: Specialisation Microwave Engineering, Optics, and Electromagnetic Compatibility: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Nanoelectronics and Microsystems Technology: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Wireless and Sensor Technologies: Elective Compulsory Electrical Engineering: Specialisation Microwave Engineering, Optics, and Electromagnetic Compatibility: Elective Compulsory Electrical Engineering: Specialisation Wireless and Sensor Technologies: Elective Compulsory Mechatronics: Technical Complementary Course: Elective Compulsory Microelectronics and Microsystems: Specialisation Microelectronics Complements: Elective Compulsory			

Course L0770: EMC II: Signal Integrity and Power Supply of Electronic Systems	
Typ	Lecture
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Lecturer	Prof. Christian Schuster
Language	EN
Cycle	WiSe
Content	- The role of packages and interconnects in electronic systems - Components of packages and interconnects in electronic systems - Main goals and concepts of signal and power integrity of electronic systems - Repeat of relevant concepts from the theory electromagnetic fields - Properties of digital signals and systems - Design and characterization of signal integrity - Design and characterization of power supply - Techniques and devices for measurements in time- and frequency-domain - CAD tools for electrical analysis and design of packages and interconnects - Connection to overall electromagnetic compatibility of electronic systems
Literature	- J. Franz, "EMV: Störungssicherer Aufbau elektronischer Schaltungen", Springer (2012) - R. Tummala, "Fundamentals of Microsystems Packaging", McGraw-Hill (2001) - S. Ramo, J. Whinnery, T. Van Duzer, "Fields and Waves in Communication Electronics", Wiley (1994) - S. Thierauf, "Understanding Signal Integrity", Artech House (2010) - M. Swaminathan, A. Engin, "Power Integrity Modeling and Design for Semiconductors and Systems", Prentice-Hall (2007)

Course L0771: EMC II: Signal Integrity and Power Supply of Electronic Systems	
Typ	Recitation Section (small)
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Prof. Christian Schuster
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Course L0774: EMC II: Signal Integrity and Power Supply of Electronic Systems	
Typ	Practical Course
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Prof. Christian Schuster
Language	EN
Cycle	WiSe
Content	- The role of packages and interconnects in electronic systems - Components of packages and interconnects in electronic systems - Main goals and concepts of signal and power integrity of electronic systems - Repeat of relevant concepts from the theory electromagnetic fields - Properties of digital signals and systems - Design and characterization of signal integrity - Design and characterization of power supply - Techniques and devices for measurements in time- and frequency-domain - CAD tools for electrical analysis and design of packages and interconnects - Connection to overall electromagnetic compatibility of electronic systems
Literature	- J. Franz, "EMV: Störungssicherer Aufbau elektronischer Schaltungen", Springer (2012) - R. Tummala, "Fundamentals of Microsystems Packaging", McGraw-Hill (2001) - S. Ramo, J. Whinnery, T. Van Duzer, "Fields and Waves in Communication Electronics", Wiley (1994) - S. Thierauf, "Understanding Signal Integrity", Artech House (2010) - M. Swaminathan, A. Engin, "Power Integrity Modeling and Design for Semiconductors and Systems", Prentice-Hall (2007)

Module M0720: Matrix Algorithms				
Courses				
Title	Typ		Hrs/wk	CP
Matrix Algorithms (L0984)	Lecture		2	3
Matrix Algorithms (L0985)	Recitation Section (small)		2	3
Module Responsible	Dr. Jens-Peter Zemke			
Admission Requirements	None			
Recommended Previous Knowledge	- Mathematics I - III - Numerical Mathematics 1/ Numerics - Basic knowledge of the programming language Python			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> Students are able to - name, state and classify state-of-the-art Krylov subspace methods for the solution of the core problems of the engineering sciences, namely, eigenvalue problems, solution of linear systems, and model reduction; - state approaches for the solution of matrix equations (Sylvester, Lyapunov, Riccati). <i>Skills</i> Students are capable to - implement and assess basic Krylov subspace methods for the solution of eigenvalue problems, linear systems, and model reduction; - assess methods used in modern software with respect to computing time, stability, and domain of applicability; - adapt the approaches learned to new, unknown types of problem. Personal Competence <i>Social Competence</i> Students can - develop and document joint solutions in small teams; - form groups to further develop the ideas and transfer them to other areas of applicability; - form a team to develop, build, and advance a software library. <i>Autonomy</i> Students are able to - correctly assess the time and effort of self-defined work; - assess whether the supporting theoretical and practical exercises are better solved individually or in a team; - define test problems for testing and expanding the methods; - assess their individual progress and, if necessary, to ask questions and seek help.			
<i>Knowledge</i>				
<i>Skills</i>				
Personal Competence				
<i>Social Competence</i>				
<i>Autonomy</i>				
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	None			
Examination	Oral exam			
Examination duration and scale	25 min			
Assignment for the Following Curricula	Computational Engineering: Core Qualification: Elective Compulsory Data Science: Specialisation IV. Special Focus Area: Elective Compulsory Data Science: Specialisation I. Data Science & Mathematics: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Mechatronics: Technical Complementary Course: Elective Compulsory Technomathematics: Specialisation I. Mathematics: Elective Compulsory Technomathematics: Specialisation II. Informatics: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Simulation Technology: Elective Compulsory			

Course L0984: Matrix Algorithms	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Dr. Jens-Peter Zemke
Language	EN
Cycle	WiSe
Content	- Part A: Krylov Subspace Methods: - Basics (derivation, basis, Ritz, OR, MR) - Arnoldi-based methods (Arnoldi, GMRes) - Lanczos-based methods (Lanczos, CG, BiCG, QMR, SymmLQ, PVL) - Sonneveld-based methods (IDR, BiCGStab, TFQMR, IDR(s)) - Part B: Matrix Equations: - Sylvester Equation - Lyapunov Equation - Algebraic Riccati Equation
Literature	Skript (224 Seiten) Ergänzend können die folgenden Lehrbücher herangezogen werden: - Saad, Yousef. Numerical methods for large eigenvalue problems: revised edition. Society for Industrial and Applied Mathematics, 2011. - Saad, Yousef. Iterative methods for sparse linear systems. Society for Industrial and Applied Mathematics, 2003. - Van der Vorst, Henk A. Iterative Krylov methods for large linear systems. No. 13. Cambridge University Press, 2003. - Liesen, Jörg, and Zdenek Strakos. Krylov subspace methods: principles and analysis. Oxford University Press, 2013.

Course L0985: Matrix Algorithms	
Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Dr. Jens-Peter Zemke
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M2036: Advanced Topics in Production Engineering				
Courses				
Title		Typ	Hrs/wk	CP
Advanced Topics in Production Engineering (L3288)		Project-/problem-based Learning	4	6
Module Responsible	Dr. Martin Gomse			
Admission Requirements	None			
Recommended Previous Knowledge	Depending on the type of project: e.g. Robotics, computer science, measurement technology, control engineering, machine learning			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence <i>Knowledge</i> <i>Skills</i> <				

Course L3288: Advanced Topics in Production Engineering	
Typ	Project-/problem-based Learning
Hrs/wk	4
CP	6
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56
Lecturer	Dr. Martin Gomse
Language	DE/EN
Cycle	WiSe/SoSe
Content	Advanced and Research Topics in Production Engineering
Literature	Aktuelle Veröffentlichungen / Recent publications

Thesis

Module M-002: Master Thesis

Courses

Title	Typ	Hrs/wk	CP
Module Responsible	Professoren der TUHH		
Admission Requirements	According to General Regulations §21 (1): At least 60 credit points have to be achieved in study programme. The examinations board decides on exceptions.		
Recommended Previous Knowledge			
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence <i>Knowledge</i>	- The students can use specialized knowledge (facts, theories, and methods) of their subject competently on specialized issues. - The students can explain in depth the relevant approaches and terminologies in one or more areas of their subject, describing current developments and taking up a critical position on them. - The students can place a research task in their subject area in its context and describe and critically assess the state of research.		
Skills	The students are able: - To select, apply and, if necessary, develop further methods that are suitable for solving the specialized problem in question. - To apply knowledge they have acquired and methods they have learnt in the course of their studies to complex and/or incompletely defined problems in a solution-oriented way. - To develop new scientific findings in their subject area and subject them to a critical assessment.		
Personal Competence <i>Social Competence</i>	Students can - Both in writing and orally outline a scientific issue for an expert audience accurately, understandably and in a structured way. - Deal with issues competently in an expert discussion and answer them in a manner that is appropriate to the addressees while upholding their own assessments and viewpoints convincingly.		
<i>Autonomy</i>	Students are able: - To structure a project of their own in work packages and to work them off accordingly. - To work their way in depth into a largely unknown subject and to access the information required for them to do so. - To apply the techniques of scientific work comprehensively in research of their own.		
Workload in Hours	Independent Study Time 900, Study Time in Lecture 0		
Credit points	30		
Course achievement	None		
Examination	Thesis		
Examination duration and scale	According to General Regulations		
Assignment for the Following Curricula	Civil Engineering: Thesis: Compulsory Bioprocess Engineering: Thesis: Compulsory Chemical and Bioprocess Engineering: Thesis: Compulsory Chemical and Bioprocess Engineering: Thesis: Compulsory Computational Engineering: Thesis: Compulsory Computer Science: Thesis: Compulsory Data Science: Thesis: Compulsory Electrical Engineering and Information Technology: Thesis: Compulsory Electrical Engineering: Thesis: Compulsory Energy Systems: Thesis: Compulsory Environmental Engineering: Thesis: Compulsory Aircraft Systems Engineering: Thesis: Compulsory Global Innovation Management: Thesis: Compulsory Computer Science in Engineering: Thesis: Compulsory Information and Communication Systems: Thesis: Compulsory Interdisciplinary Mathematics: Thesis: Compulsory International Production Management: Thesis: Compulsory International Management and Engineering: Thesis: Compulsory Joint European Master in Environmental Studies - Cities and Sustainability: Thesis: Compulsory Logistics, Infrastructure and Mobility: Thesis: Compulsory Aeronautics: Thesis: Compulsory Mechanical Engineering - Product Development and Production: Thesis: Compulsory Materials Science and Engineering: Thesis: Compulsory		

	Materials Science: Thesis: Compulsory
	Mechanical Engineering and Management: Thesis: Compulsory
	Mechatronics: Thesis: Compulsory
	Biomedical Engineering: Thesis: Compulsory
	Microelectronics and Microsystems: Thesis: Compulsory
	Product Development, Materials and Production: Thesis: Compulsory
	Renewable Energies: Thesis: Compulsory
	Naval Architecture and Ocean Engineering: Thesis: Compulsory
	Naval Architecture and Ocean Engineering: Thesis: Compulsory
	Ship and Offshore Technology: Thesis: Compulsory
	Theoretical Mechanical Engineering: Thesis: Compulsory
	Process Engineering: Thesis: Compulsory
	Water and Environmental Engineering: Thesis: Compulsory
	Certification in Engineering & Advisory in Aviation: Thesis: Compulsory